

TOSHIBA TSL3000 COMMUNICATIONS MANUAL



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INSTRUCTION MANUAL

COMMUNICATION MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of Toshiba Machine's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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Preface

This manual describes serial communication and ethernet communication between the robot controller and peripheral devices. It covers such subjects as connecting communication channels and setting communication modes. It also describes communication protocols, communication commands and data format, and presents information on how to operate the robot with data communication and how to handle communication dialogue with controller programs.

Before reading this manual, we ask that you first read and understand the contents of the following user manuals.

- Operator's Manual
- Robot Language Manual
- Interface Manual

This manual is divided into six (6) sections:

Section 1 Introduction

This section presents an introduction to the communication functions provided by the TS3000 Series robot system.

Section 2 COM1 Port and HOST Port Specifications and Settings

This section describes the RS-232C port hardware interfaces, communication modes, and other information.

Section 3 Ethernet Specifications and Settings

This section describes the Ethernet hardware interfaces, communication modes, and other information.

Section 4 Non-Protocol Communication

This section describes the protocol for communication between an external device and the robot program only.

Section 5 Simple Protocol Communication

This section describes the communication protocol, commands, and robot program files for communication with the host computer.

Section 6 Robot Operation Sequence Using External Communication

This section shows examples of methods for operating the robot.

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Section 1

Introduction

This manual describes serial communication and ethernet communication between the robot controller and an external computer. There are two procedures for data transmission: non-protocol communication and simple protocol communication, each of which can achieve the following functions.

Non-protocol communication

- (1) Input variables into the program from the external device.
- (2) Output variables or messages in the program.

Simple protocol communication

- (1) Start and stop operation of the robot
- (2) Program reset, step reset, cycle reset and output signal reset
- (3) Servo ON/OFF
- (4) Upload and download program files
- (5) Erase program files
- (6) Select programs
- (7) Monitor the status of the robot
- (8) Monitor for robot errors

The communication interface configuration is as shown below.

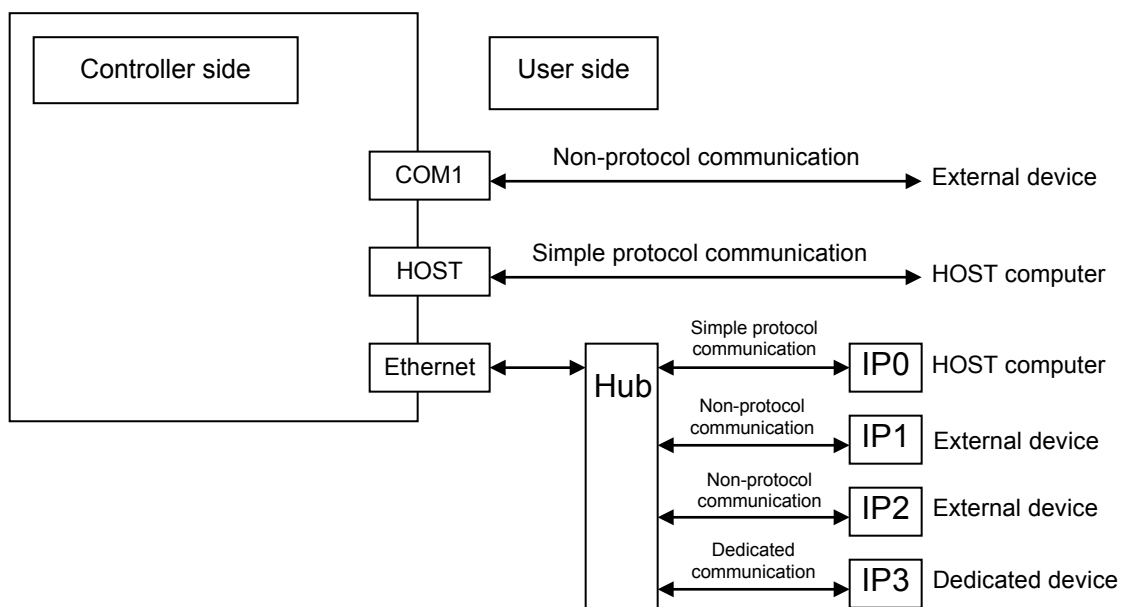


Fig. 1.1 Communication interface configuration

Section 2

COM1 Port and HOST Port Specifications and Settings

The TS3000/TSL3000 controller has two (2) serial communication ports. The COM1 port is used for non-protocol communication, being connected to external devices, and the HOST port is used for protocol communication and communicates with the HOST computer.

2.1 Connections

The COM1 port and HOST port should be connected, using an RS-232C cross connect cable (D-SUB: 9-pin). (For details, see the Interface Manual.)

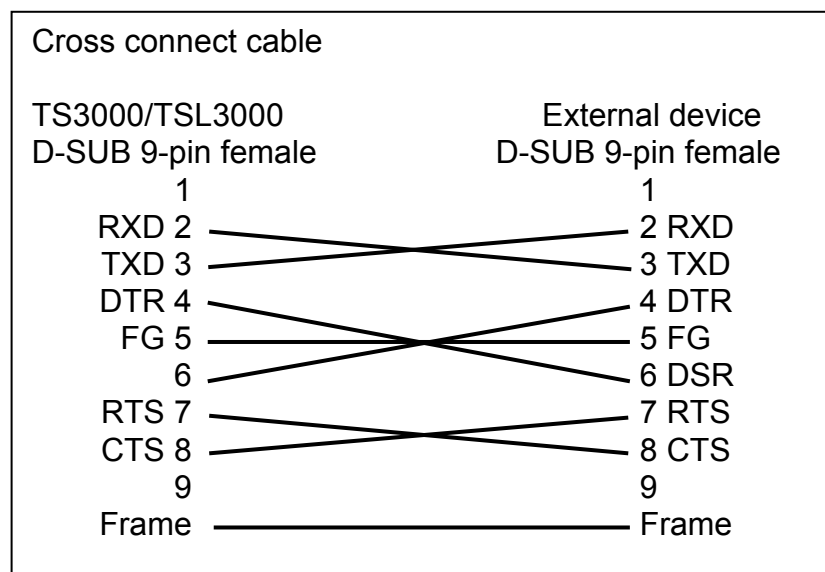


Fig. 2.1 Pin arrangement for cross connect cable

2.2 Communication Specifications

2.2.1 COM1 Port

Table 2.1 COM1 port communication specifications

Item	Specifications
Interface	RS-232C
Synchronous system	Start-stop synchronization system
Communication system	Full duplex system
Communication rate	300, 600, 1200, 2400, 4800, 9600, 19200, 38400 bps
Data format	ASCII code
No. of transfer bytes	Max. 256 bytes
Transfer data structure	Data length: 7 or 8 bits Parity: None, odd, even Stop bit: 1 or 2 bits
Protocol	Non-protocol communication
Timeout	No timeout
Fault recovery	No special protocols are used.

Note: The communication rate and transfer data structure are specified in the user parameter file. (For details, see Para. 2.3.)

2.2.2 HOST Port

Table 2.2 HOST port communication specifications

Item	Specifications
Interface	RS-232C
Synchronous system	Start-stop synchronization system
Communication system	Half-duplex system
Communication rate	300, 600, 1200, 2400, 4800, 9600, 19200, 38400, 115200 bps
Data format	ASCII code
No. of transfer bytes	Max. 255 bytes
Transfer data structure	Data length: 7 or 8 bits Parity: None, odd, even Stop bit: 1 or 2 bits
Protocol	Simple protocol communication
Timeout	10 sec. (character receiving intervals)
Fault recovery	Command is re-sent from the host computer. No special protocols are used.

Note: The communication rate and transfer data structure are specified in the user parameter file. (For details, see Para. 2.3.)

2.3 Communication Parameters

The communication rate and transfer data structure of the COM1 port and HOST port are defined in the user parameter file (USER. PAR).

Communication is performed according to the contents defined under [U06] of the user parameter file (USER.PAR) in the RAM drive.

[U06] Serial port setting

Specify the communication ports [COM1 port, HOST port].

TS3000

```
[U06] Serial port setting
{Speed } (38400, 19200, 9600, 4800, 2400, 1200)
{Character} (7, 8)
{Parity } (0:Without, 1:Odd, 2:Even)
{Stop bit } (1, 2)
{COM1}
= 9600 8 0 1
{HOST}
= 9600 8 0 1
```

{COM1}

= (Speed) (Character length) (Parity) (Stop bit length) ← COM1 port setting

{HOST}

= (Speed) (Character length) (Parity) (Stop bit length) ← HOST port setting

(Speed) : Select the speed for data communication from the following six (6) rates.

115200	:	115200 bps	* Host port only
38400	:	38400 bps	
19200	:	19200 bps	
9600	:	9600 bps	
4800	:	4800 bps	
2400	:	2400 bps	
1200	:	1200 bps	

(Character length) : Specify the length of characters to be transferred.

- 8 : 8 bits
- 7 : 7 bits

(Parity) : Specify the parity of characters to be transferred.

- 0 : Without parity
- 1 : Odd-number parity
- 2 : Even-number parity

(Stop bit length) : Specify the stop bit length of characters to be transferred.

- 1 : Stop bit 1
- 2 : Stop bit 2

Set value

Example: TS3000

= 9600 8 0 1

= 38400 8 1 1

"Speed 9,600 bps, character length 8 bits, without parity and stop bit 1" are specified for the COM1 port.

"Speed 38,400 bps, character length 8 bits, even-number parity and stop bit 1" are specified for the HOST port.

Section 3

Ethernet Specifications and Settings

The TS3000/TSL3000 controller has one Ethernet connector.

IP0 can communicate with the host computer using simple protocol, IP1 and IP2 can communicate with an external device using non-protocol communication, and IP3 can communicate with a dedicated device using a dedicated protocol.

This function uses TCP/IP protocol to enable communication on the same local area network with computers and visual equipment on the Ethernet. This allows program creation and editing, debugging start support, and maintenance using the computer programmer TSPC.

Data can be sent and received by the robot program for exchanging numerical and positional data.

INPUT and PRINT statements from SCOL language can be used to perform data input and output.

Example: INPUT IP1, INDATA, PRINT IP1, OUTDATA, CR, etc.

3.1 Connections

If a hub is used to connect to an external device, connect using a straight cable. Use a cross connect cable when connecting a computer directly to the controller. Some devices have the automatic recognition function so that either a straight cable or cross connect cable can be used, and so check the specifications of the device that is used.

3.2 Communication Specifications

Table 3.1 Communication specifications

Item	Specification
Interface	10BASE-T
Communication system	TCP
Communication rate	10 Mbps
Maximum packet size	512 bits/packet
Number of simultaneous connectable ports	4
Applications	IP0: Simple protocol communication IP1: Non-protocol communication IP2: Non-protocol communication IP3: Dedicated communication

3.3 Communication Parameters

This section describes the setting of the Ethernet parameters. These settings are defined in the Ethernet Parameter File (ETHERNET.PAR).

[E00] Open mode IP0 IP1 IP2 IP3

[E00] Open mode IP0 IP1 IP2 IP3 { 0: non 1: Robot is TCP server 2: Robot is TCP client } = 1 0 0 0

IP0 is for simple protocol only, and so the server mode must be specified.

If the external device is a server, the controller uses client settings, and if the external device is a client, the controller uses server settings.

=(IP0) (IP1) (IP2) (IP3)

This enables or disables the Ethernet function.

0: Not used

1: Operates as a server

2: Operates as a client

[E01] Robot controller IP address

[E01] Robot controller IP address = 192.168.0.124
--

This sets the IP address of the robot controller itself.

The IP address is an address for identifying the controller. It is expressed as four (4) numbers from 0 to 255 with a period (.) in between. If the controller is connected directly to a computer on the network, the IP address may remain at the initial value (any setting), but if it is connected to a LAN, the IP address must be set by following the instructions of the user's LAN system administrator. This function will not work properly if any IP addresses are duplicated. The robot controller and computer to communicate with it must be connected on the same network.

When sending data to devices on the network, the controller can be operated as a client or a server.

If the external device is a server, use the client setting for the controller, and if the external device is a client, use the server setting for the controller.

= (Robot controller IP address)

[E02] Robot controller name

(Planned for use in function expansion)

[E02] Robot controller name = "NO1-TS3K"

This parameter specifies the name of the robot controller.
This is planned for use when the Ethernet function is expanded.

[E03] Subnet mask

[E03] Subnet mask = 255.255.255.0

This parameter specifies the subnet mask of the connected network.
When connecting to a LAN, set the subnet mask by following the instructions of the user's LAN system administrator.

= (Subnet mask)

[E04] Default gateway

[E04] Default gateway = 192.168.0.1
--

This parameter specifies the default gateway address.

When connecting to a LAN, set the default gateway by following the instructions of the user's LAN system administrator.

= (Default gateway address)

[05] Own port no

[E05] Own port no { IP0 } = 1000 { IP1 } = 1001 { IP2 } = 1002 { IP3 } = 1003
--

This parameter specifies each port number that is used by the controller. A port number is similar to a radio frequency. The numbers IP0 to IP3 indicate the port numbers. Be careful that no port numbers are duplicated.

{IP0}
=(Port number used)
{IP1}
=(Port number used)
{IP2}
=(Port number used)
{IP3}
=(Port number used)

[06] Port number of destination

[E06] Port number of destination { IP0 } = 1000 { IP1 } = 1001 { IP2 } = 1002 { IP3 } = 1003
--

This parameter specifies each port number that is used by the destination. This is used in client mode only. This must match the port number used by the server at the destination.

{IP0}
=(Port number of destination)
{IP1}
=(Port number of destination)
{IP2}
=(Port number of destination)
{IP3}
=(Port number of destination)

[E07] IP address of destination

[E07] IP address of destination {IP0} = 192.168.0.150 {IP1} = 192.168.0.151 {IP2} = 192.168.0.152 {IP3} = 192.168.0.153

This parameter specifies the IP address of the destination where each port is connected. This is used in client mode only.

{IP0}
=(IP address of destination)
{IP1}
=(IP address of destination)
{IP2}
=(IP address of destination)
{IP3}
=(IP address of destination)

[E08] Robot community name

(Planned for use in function expansion)

[E08] Robot community name = "NO1-TS3K"
--

This parameter specifies the community name of the network.
This is planned for usage when the Ethernet function is expanded.

[E09] DHCP

(Planned for use in function expansion)

[E09] DHCP = 0

This parameter specifies if an address is obtained from the DHCP server.
This is planned for usage when the Ethernet function is expanded.

[E10] TCP status alarm IP0 IP1 IP2 IP3

[E10] TCP status alarm IP0 IP1 IP2 IP3 {0:NON 1:Lv_1 2:Lv_2 4:LV_4 8:Lv_8} = 0 0 0 0
--

For the port being used, the alarm level (1Lv, 2Lv, 4Lv, 8Lv) generated when the Ethernet connection is cut off during running of a program can be set.

When the value is 0 (zero), no alarm is generated.

*Parameter (ETHERNET.PAR) setting example

```
[E00] OPEN MODE IP0 IP1 IP2 IP3
{0:NON 1:ROBOT IS TCP SERVER 2:ROBOT IS CLIENT}
= 1 0 0 0
[E01] ROBOT CONTROLLER IP ADDRESS
= 192.168.0.124
[E02] ROBOT CONTROLLER NAME
= "NO1-TS3K"
[E03] SUBNET MASK
= 255.255.255.0
[E04] DEFAULT GATEWAY
= 192.168.0.1
[E05] OWN PORT NO
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003
[E06] PORT NUMBER OF DESTINATION
{IP0}
= 1000
{IP1}
= 1001
{IP2}
= 1002
{IP3}
= 1003
[E07] IP ADDRESS OF DESTINATION
{IP0}
= 192.168.0.150
{IP1}
= 192.168.0.151
{IP2}
= 192.168.0.152
{IP3}
= 192.168.0.153
[E08] ROBOT COMMUNITY NAME
= "NO1-TS3K"
[E09] DHCP
= 0
[E10] TCP status alarm IP0 IP1 IP2 IP3
{0:NON 1:Lv_1 2:Lv_2 4:Lv_4 8:Lv_8}
= 0 0 0 0
[END]
```

Section 4

Non-Protocol Communication

Non-protocol communication is used for exchanging data with a robot language program and for printing out files.

The COM1 (RS-232C) port, and Ethernet IP1 and IP2 ports are used for non-protocol communication.

The data format for character strings to be transferred between the robot controller and the TCP/IP device is as shown below.

	These character strings are based on ASCII format.
Character string + CR (0x0D)	ASCII 0x0D (carriage return) is added in the character string data. When a plurality of data is transferred, the character string is separated by commas.
Character string, character string + CR (0x0D)	

The functions as described in the simple protocol cannot be used for the non-protocol data communication. Also, response to data received by the controller and timeout check are not processed at all.

4.1 Communication with a Robot Language Program

Data communication with a robot language program can be performed using the PRINT and INPUT commands. For more information on the robot language, see "Robot Language Manual".

4.1.1 PRINT Command

The PRINT command allows you to send out a specified character string from the controller. The controller transfers character strings or values of variable specified by the PRINT command. The format of the PRINT command is shown below:

PRINT □ [<channel>,{<character string>|<expression>|}[,<character string>|<expression>}][,CR]

<channel>:

Specifies the communication port over which the data is to be transmitted. One of the following should be specified as the channel.

COM1	: COM1 port
IP1	: Ethernet IP1 port
IP2	: Ethernet IP2 port
TP	: Screen output to teach pendant

Unless <channel> is specified, data is transmitted to the teach pendant.

<character string>:

The character string to be transmitted is specified by enclosing that string in double quotation marks ("").

<expression>:

Expressions made up of constants, variables, arithmetic operands and functions may be specified.

CR:

CR is used when the record end code (0x0D) is added to the last of sending data.

Ex.: PRINT COM1, "INPUT DATA =", -1000.0/3, CR
PRINT IP1, "INPUT DATA =", -1000.0/3, CR

Commas are used to separate any multiple character strings or expressions specified in the PRINT command. Character strings are enclosed inside double quotation marks, and everything inside of those double quotation marks is transmitted in ASCII code. Expressions are first solved, and the result is sent as a 12-character block of fixed length with the result pushed over to the right of that block. Should the result of the expression be an integer, that result is sent as a Base 10 number having a maximum of ten places (digits).

Should the result of the expression be a real number, the result is sent as a number having an integral part with a maximum of four (4) digits and a decimal part with a maximum of three (3) digits (for a maximum of eight (8) places counting the decimal point). One space in front of the number is allocated for the sign (+ or -) of that number, although the sign is omitted if it is plus (+). The number is sent in a 12-character block, with the number pushed over to the right. The remaining spaces are filled with space codes (0x20). The number itself is sent in ASCII code. The commas used to separate character strings or expressions in a PRINT command are themselves not transmitted. When multiple character strings and/or expressions are specified with the PRINT command, the controller will send out these character strings and/or expressions as a single text.

Ex.: When the above example program is executed, the following data are transmitted.

INPUT DATA = -333.333 (CR)

Note: CR signifies the record end code (0x0D).

*To left-justify the output display by the PRINT command without space, set 1 in the value of (FUNCTION 14), the 2nd column in 3rd row of user parameter [U25] FUNCTION SELECT SWITCH. Data can be left-justified without displaying space. For details, refer to the Instruction Manual "User Parameter Manual."

*To use the PRINT command to output multiple character strings and/or expressions delimited by a comma ',', add a comma enclosed by double quotation marks "" separately from delimiter ','. They can be output delimited by a comma.

Ex.: PRINT COM1, A,"", B, CR

4.1.2 INPUT Command

As opposed to the PRINT command, which is used for transmitting data, the INPUT command is used for receiving data. The only data which may be received by the controller are integers and real numbers. Data received by the controller is put into a variable (in a robot language program) specified by the INPUT command. This data can be referred to later in the program to operate the robot.

The format of the INPUT command is shown below:

INPUT □ [<channel>,] <variable>, [<variable>] ...

<channel>:

Specifies the communication channel over which the data is to be received. One of the following should be specified as the channel.

COM1	:	COM1 port
IP1	:	Ethernet IP1 port
IP2	:	Ethernet IP2 port
TP	:	Key input from teach pendant

Unless <channel> is specified, the controller receives data from the teach pendant.

<variable>:

Specifies the variable in the robot program into which the data is to be entered.

Ex.: INPUT COM1, N1, N2 INPUT IP1, N1, N2
--

Commas are used to separate two (2) or more variables specified in the INPUT command. The controller waits until data comes in over the communication channel specified by the INPUT command. Add a record end code (0x0D) to the end of data which is sent to the controller. When multiple units of data are to be sent to the controller, the individual data units should be separated with commas before being transmitted. When more units of data are received by the controller than was specified with the INPUT command, the surplus data are ignored and used for the next INPUT command. If fewer units of data are received by the controller

than was specified with the INPUT command, the controller waits until the short data reach.

Note 1)

The controller starts reading any data sent to it only after an INPUT command is executed. Data are set in the ring buffer, and the input data are picked up according to the request of the INPUT statement.

Any data received before the INPUT command is executed may be ignored. Take careful precautions when determining the data transmission timing.

Note 2)

Should program execution be suspended while the controller is waiting for data to come in (as directed by an INPUT command), the execution of the INPUT command will be cancelled. Should the program be resumed, program execution will start from the step following the INPUT command. Any variables for which data was not yet received when the program was suspended will be treated as 0.

You should keep this in mind when writing your robot language program, i.e., you should arrange your program in such a way that it will still function properly even should the controller (mistakenly) treat input values as 0. One way to do this is to have the controller ask the host computer for confirmation (retransmission) whenever the controller receives a 0. Another way to do this is to add on check-sum data (to the data to be transmitted) and check the validity of the received data. (For programming examples, see Para. 4.2, "Programming examples for communication with a robot language program.")

Note 3)

In the step operation mode, which is one of the test operation modes, when the INPUT command is executed, the program enters the wait state until data are received. The same holds true when an INPUT command is executed directly.

Note 4)

When an INPUT command is to be used to receive multiple variables transmitted as one text from the host computer, individual numerals should be separated in the text with commas.

4.1.3 Clearing Communication Buffer

If the number of data the controller has received is larger than the number of data specified by the INPUT command, such data are stored in the communication buffer and used at request of the next INPUT command. If data is left in the communication buffer, unexpected data will reach by the INPUT command. To clear the data left in the communication buffer, output character string "BUFFRESET" to relevant communication port.

Example)

When clearing the communication buffer of the COM1 port:

```
PRINT □ COM1, "BUFFRESET"
```

Example)

When clearing the communication buffer of the IP1 port:

```
PRINT □ IP1, "BUFFRESET"
```

4.2 Programming Examples for Communication with a Robot Language Program

It is possible to utilize communication functions in a robot language program in order to do such things as specify program branches, correct the position of the robot to be performed, etc. Listed below are several programming examples showing how to do this.

4.2.1 Program Branching

The INPUT command can be utilized to specify the number of times a certain action is to be repeated, to select a task for execution, etc.

Specifying the number of times an action is to be repeated (COM1)

```
PROGRAM REPEAT
  N=0
START:
  PRINT COM1, "REQ"
  INPUT COM1, N
  IF N == 0 THEN GOTO START
  FOR K = 1 TO N
    MOVE A1
    MOVE A2
    . . .
  NEXT K
  PRINT COM1, "END"
END
```

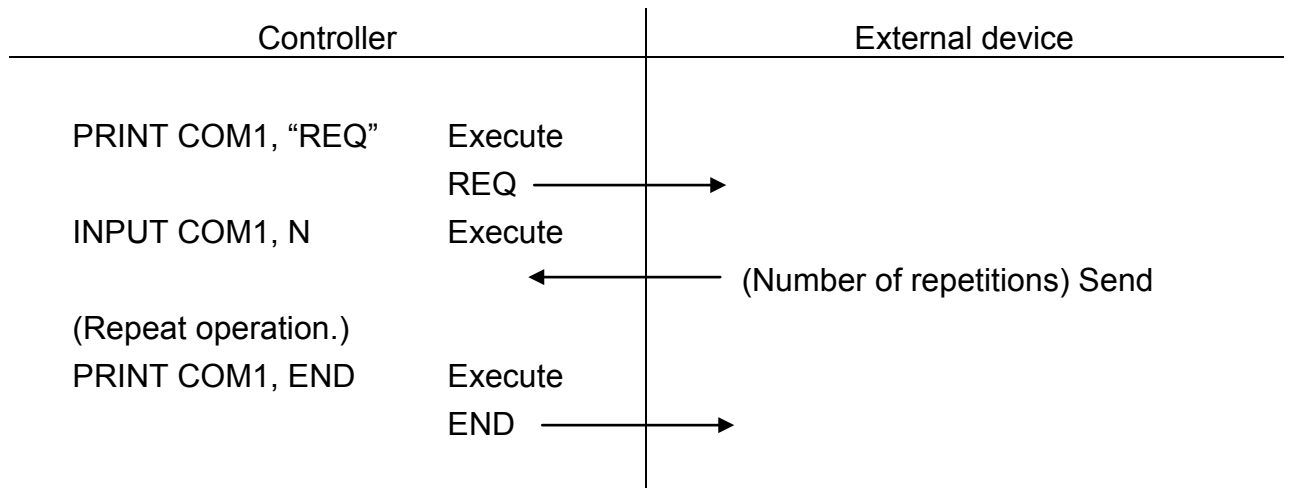
In this program, the robot will repeat a certain series of motions as many times as the external device tells it to.

First, the controller will send the character string REQ to the external device. The external device will reply by sending back the number of times the operation is to be repeated.

The controller will read in this number as the variable N, and will use this variable in the FOR statement.

Should the program execution have been suspended while the controller was waiting receiving data in the INPUT command, the variable N would be (mistakenly) taken as 0 when the program is resumed. In such a case, branch is executed to the top address of the program by asking the external device for a retransmission. When the task is completed, the controller will send the character string END to the external device.

Data exchange



Selecting a task to be performed (COM1)

```
PROGRAM SELECT
K = 0
START:
  PRINT COM1, "STR"
SELECT:
  INPUT COM1, K
  GOTO(K) L1, L2, L3
  PRINT COM1, "NG"
  GOTO FIN
L1:
  (Task 1)
  GOTO FIN
L2:
  (Task 2)
  GOTO FIN
L3:
  (Task 3)
  GOTO FIN
FIN:
END
```

In this program, the external device tells the controller what task is to be performed.

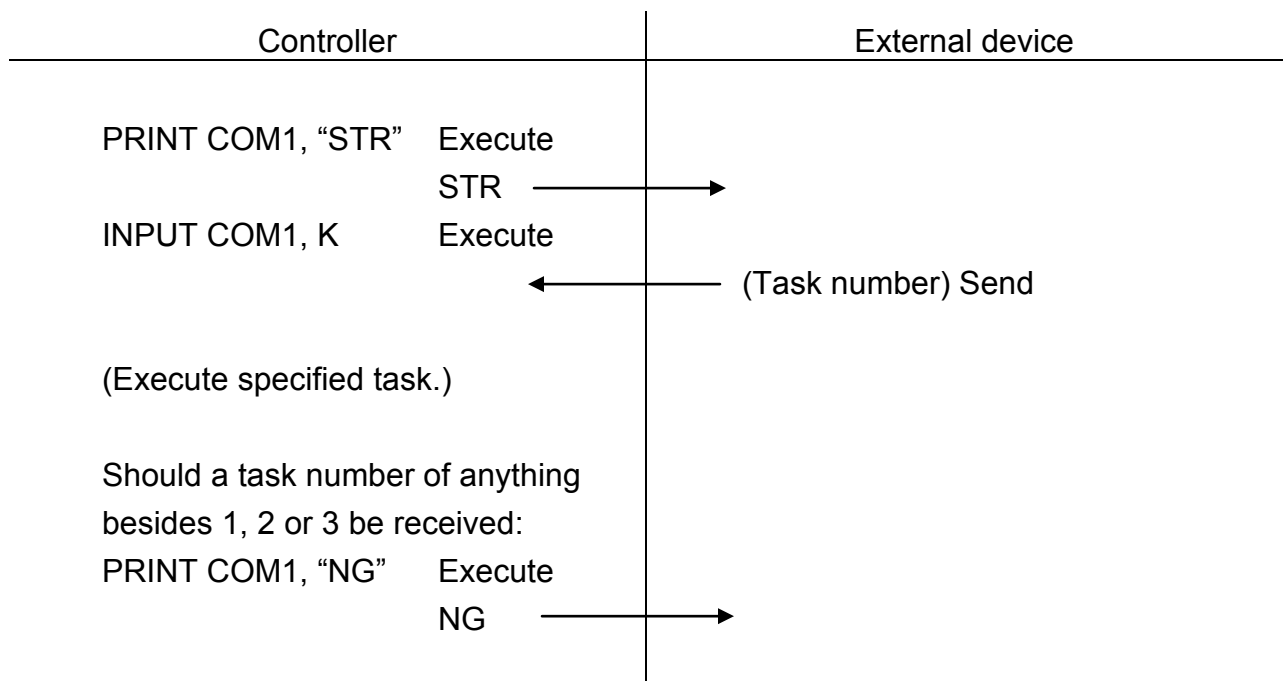
First, the controller will send the character string STR to the external device. The external device will send back a number which specifying task the controller (robot) is to perform. The controller reads in this number as variable K, which is then used in the GOTO () command to branch the program to the appropriate task.

Task 1 will be performed if the value of K is 1, Task 2 if the value is 2, and Task 3 if the value is 3. If the value of K is anything else, the controller will send the character string NG

(non-acknowledge) back to the external device.

Should program execution have been suspended while the controller was waiting for the number to come in, variable K will be taken as 0 when the program is resumed. In this case also, the controller will send the character string NG back to the host computer.

Data exchange



4.2.2 Correcting the Position of the Robot

Data received from the host computer can be used to correct the position of the robot.

Directly specifying position (COJM1)

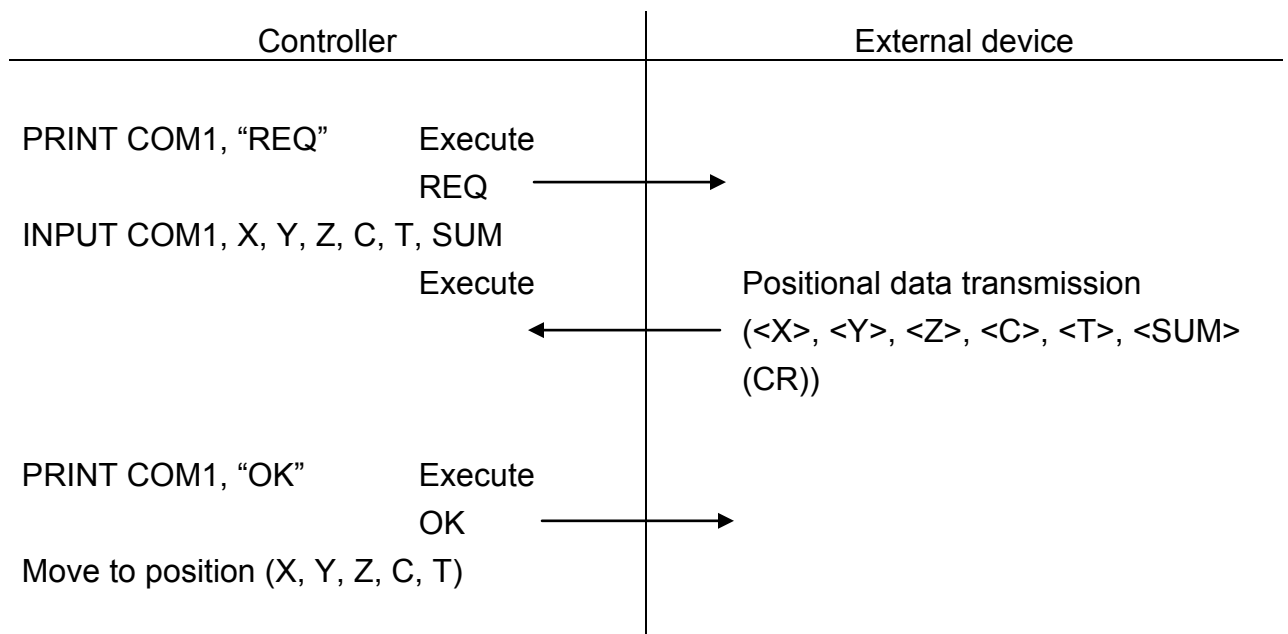
```
PROGRAM DIRECT
X=0.0
Y=0.0
Z=0.0
C=0.0
T=0.0
PRINT COM1,"REQ"
INPUT X,Y,Z,C,T,SUM
IF SUM==X+Y+Z+C+T+1
    THEN GOTO ACTION
PRINT COM1,"NG"
GOTO FIN
ACTION:
PRINT COM1,"OK"
P1=POINT(X,Y,Z,C,T)
MOVE P1
FIN:
END
```

In this program, the controller moves the robot to a position specified by the external device.

The controller sends the character string REQ to the external device. The external device will respond by sending back the values for positional data X, Y, Z, C, and T, and also the sum of these values plus 1 (as check-sum data). The controller will then see if the received data is correct by adding all the positional data, adding 1 to the result, and determining if check-sum value agrees with the value. If it does agree, the controller will send the character string OK.

If the received data is wrong, the controller will send the character string NG back to the external device and quit the program. The controller will create positional data P1 using that data and then tell the robot to move to that point.

Data exchange



Specifying a relative position (COM1)

PROGRAM RELATIV

```
X = 0.0
Y = 0.0
SUM = 0.0
MOVE P1
PRINT COM1, "REQ"
INPUT COM1, X, Y, SUM
IF SUM == X + Y + 1 THEN GOTO
ACTION
PRINT COM1, "NG"
GOTO FIN
ACTION:
PRINT COM1, "OK"
MOVE P1 + POINT (X, Y)
FIN:
END
```

In this program, the controller moves the robot to a position specified by the external device.

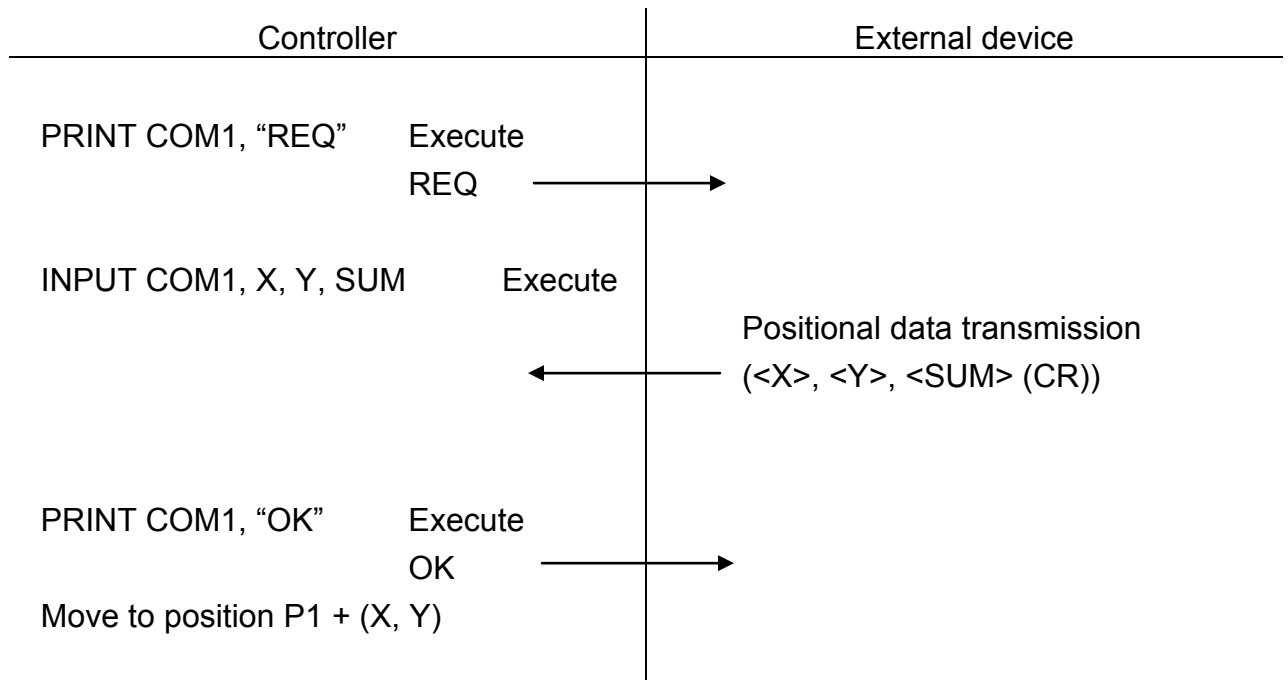
The controller sends the character string REQ to the external device.

The external device will respond by sending back the values for coordinate positions X and Y, and also the sum of these values plus 1 (as check-sum data). The controller will then see if the received data is correct by adding X and Y, adding 1 to the result, value agrees with the check-sum value. If it does agree, the controller will send back the character string OK to the external device.

If the received data is wrong, the controller will send the character string NG back to the external device and quit the program.

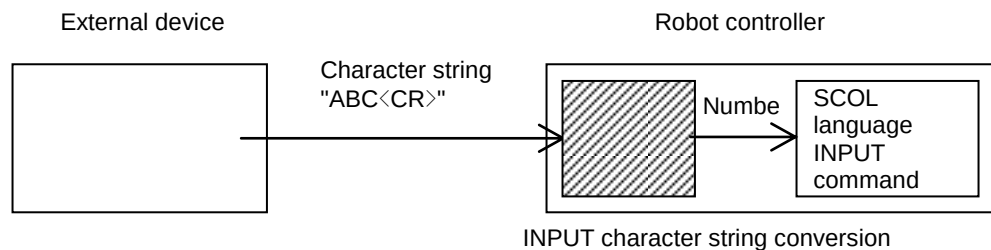
The controller will add the X and Y values of the transmission to the X and Y values of positional data P1 and make the robot move to the new position thus created.

Data exchange



4.3 INPUT Character String Conversion Function

The INPUT command in SCOL language cannot receive a character string. If a nonnumeric character string is received, 2-046 Invalid Channel error occurs. A character string may be sent depending on the specification of the external device. This function converts the character string received by the robot into a numeric value.



Note: <CR> signifies the record end code (0x0D).

The value of (FUNCTION 1), the 1st column in 1st row of user parameter [U25] FUNCTION SELECT SWITCH is set to the following to switch the processing if a nonnumeric character string is received.

- 0: Function disabled (if a nonnumeric character string is received, Invalid Channel error occurs.)
- 1: A nonnumeric character string is converted to 0 if received.
- 2: A nonnumeric character string is converted to any numeric value if received.

4.3.1 Converting a Character String to 0

The value of (FUNCTION 1), the 1st column in 1st row of user parameter [U25] FUNCTION SELECT SWITCH is set to 1.

[U25] FUNCTION SELECT SWITCH

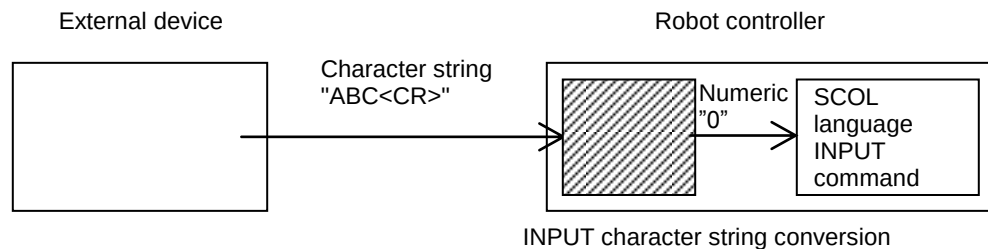
= 1 0 0 0 0 0

= 0 0 0 0 0 0

= 0 0 0 0 0 0

= 0 0 0 0 0 0

When the value is set to 1, a nonnumeric character string is converted to 0 if received.



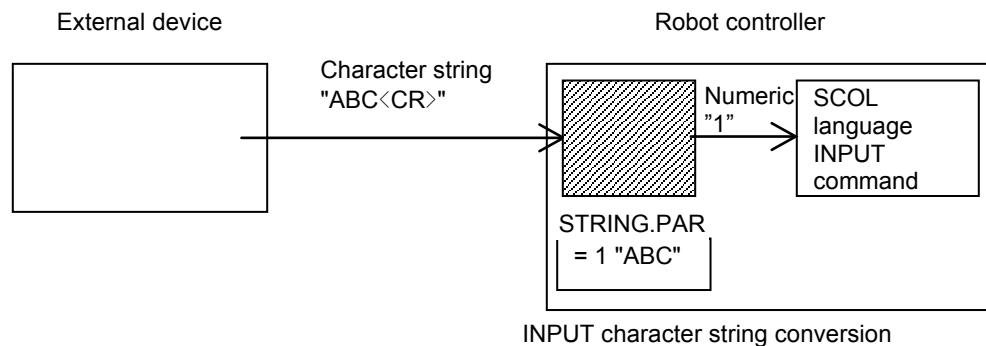
Note: <CR> signifies the record end code (0x0D).

4.3.2 Converting a Character String to Any Numeric Value

The value of (FUNCTION 1), the 1st column in 1st row of user parameter [U25] FUNCTION SELECT SWITCH is set to 2.

[U25] FUNCTION SELECT SWITCH						
=	<u>2</u>	0	0	0	0	0
=	0	0	0	0	0	0
=	0	0	0	0	0	0
=	0	0	0	0	0	0

When the value is set to 2, a nonnumeric character string is compared with the character string in the previously created parameter file "STRING.PAR" and the conversion number of the matching character string is returned if received.



Note: <CR> signifies the record end code (0x0D).

STRING.PAR should be created in the following format.

	①	②
=	<u>1</u>	"▲▲▲▲"

- ① Conversion number (0 to 32767)
- ② Character string to be received (up to 32 characters)

Example of STRING.PAR:

=	0	"OK000"
=	1	"OK001"
=	2	"OK002"
	:	
=	127	"OK127"

The number of data rows can be specified to be up to 128. If 128 rows are exceeded, data is truncated by 128 rows.

If the first data is other than a number (integer), the second data has 33 characters or more, and the number of data elements is other than 2, 8-015 Parameter error occurs.

*If a matching character string is not found, Invalid Channel error occurs without numeric conversion.

*If the received character string can be determined to be numeric, conversion using this function is not performed.

*STRING.PAR becomes valid when the power is turned OFF and then turned ON.

Section 5

Simple Protocol Communication

The simple protocol communication issues a command to the robot controller and performs start/stop operation of the robot, transfer of program files, and monitoring of the status.

5.1 Transmission Protocol

The RS-232C HOST port and Ethernet IP0 port wait for requests from the host computer after the power is supplied to the robot controller.

Basically, the host computer is a master station and the robot controller is a slave station. The robot controller sends back necessary data to the host computer in reply to commands reached from the host computer.

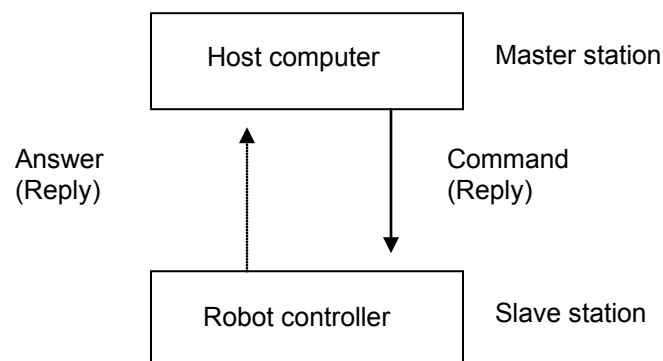


Fig. 5.1 Basic communication

Should the robot controller return an NG (Non-acknowledge) code in response to a command from the host computer, or should the robot controller give no response at all, resend the command from the host computer. Even should an error have occurred while transmitting a file, resend the file upload (or download) command from the host computer.

Furthermore, after receiving text from the controller, insert a delay of about 50 msec before beginning to transmit subsequent text from the host computer.

5.2 Transmission Format

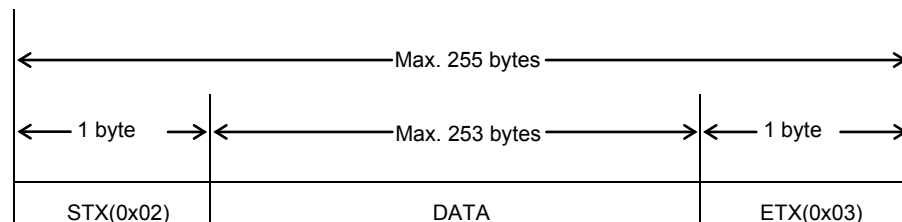
Transmissions are made with the text unit shown below. A maximum of 253 bytes of actual data can be transmitted as a single text. Actual data in amounts over 253 bytes will be transmitted in one of the following two ways.

- (1) Files (robot programs, positional data, parameters) will be broken down into multiple texts and transmitted as described in Para. 5.3.3. The receiving station will send an answer signal back for each block of text transmitted. File upload (or download) commands are not necessary to transmit the second and following blocks of text.

- (2) Messages to be transmitted

Messages will be sent with multiple transmissions. In other words, the data will be broken down into multiple texts and each text will be transmitted independently. The station that received the data will reconstruct the multiple texts back into a single message.

5.2.1 Text



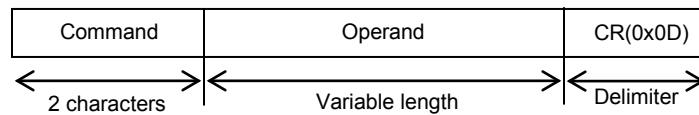
Text length: Max. 255 byte (from STX through ETX)

Text content:

- STX Start of text code (0x02) – 1 byte
- DATA Data section (Max. 253 bytes)
- ETX End of text code (0x03) – 1 byte

5.2.2 Data Section Format

The basic format of data is shown below.



(1) Command

Command consists of two alphabetical letters which signify the type of command. See Table 5.1 for command types and descriptions.

(2) Operand

The form of the operand varies depending on the type of command. For more information, read the description for the command in question.

(3) Delimiter

Add CR as delimiter at the end of the data. In the description of commands in Para 5.3.3, CR is used.

(4) Characters

The characters to be used should be ASCII code alphanumeric characters and symbols.

Alphanumeric characters:

a b c d e f g h i j k l m n o p q r s t u v w x y z
 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
 0 1 2 3 4 5 6 7 8 9

Special symbols:

“ ‘ () + - * / , . < > = ! [] { } % ^ & ? ;

5.2.3 Ethernet Status Display [ETHER]

(1) Function

Displays the Ethernet status.

(2) Procedures

1. Press the NEXT key repeatedly until [ETHER] is displayed in the menu, and then press [ETHER].

The screen shown below is displayed.

E T H E R N E T S T A T U S				[1 / 3]
I P A D D R E S S	1 9 2 . 1 6 8 .	0 . 1 2 4		
S U B N E T M A S K	2 5 5 . 2 5 5 . 2 5 5 .	0		
M A C A D D R E S S	0 0 0 0 4 8 :	1 4 0 2 0 6		
F I R M W A R E	S 1 S 6 0 0 0 0 / 0 0 0 1 :	0 0 : 2 0		

2. The display items can be changed by pressing ALT + [↓] and the NEXT key (or ALT + [↑] and the ALT + NEXT key).

The information displayed on each page is shown below.

(Page 1)

E T H E R N E T S T A T U S				[1 / 3]
I P A D D R E S S	1 9 2 . 1 6 8 .	0 . 1 2 4		
S U B N E T M A S K	2 5 5 . 2 5 5 . 2 5 5 .	0		
M A C A D D R E S S	0 0 0 0 4 8 :	1 4 0 2 0 6		
F I R M W A R E	S 1 S 6 0 0 0 0 / 0 0 0 1 :	0 0 : 2 0		

- ◆ IP ADDRESS: IP address of the robot controller itself that is written in the "ETHERNET.PAR" [E01] file
- ◆ SUBNET MASK: Subnet mask of the network that is written in the "ETHERNET.PAR" [E03] file
- ◆ MAC ADDRESS: MAC address of the robot controller itself
- ◆ FIRMWARE: Version of the Ethernet unit firmware

(Page 2)

E T H E R N E T S T A T U S								[2 / 3]
	O w n P o r t	C o n n e c t	I P				P o r t	
I P 0	1 0 0 0	0 0 0 . 0 0 0 . 0 0 0 . 0 0 0	0	0	0	0	0 0 0 0 0	
I P 1	0 0 0 0	0 0 0 . 0 0 0 . 0 0 0 . 0 0 0	0	0	0	0	0 0 0 0 0	
I P 2	0 0 0 0	0 0 0 . 0 0 0 . 0 0 0 . 0 0 0	0	0	0	0	0 0 0 0 0	
I P 3	0 0 0 0	0 0 0 . 0 0 0 . 0 0 0 . 0 0 0	0	0	0	0	0 0 0 0 0	

- ◆ IP0 to IP3: Ethernet channel number
- ◆ OwnPort: Robot port number
- ◆ Port: Port number of the destination
- ◆ Connect IP: IP address of the destination

(Page 3)

E T H E R N E T S T A T U S								[3 / 3]
	A p p .	M o d e	S t a t u s					
I P 0	H O S T	T C P / S	L	I	S	E	N	
I P 1	S C O L	T C P / C	C	L	O	S	E	D
I P 2	S C O L	T C P / S	C	L	O	S	E	D
I P 3	S Y S T E M	T C P / S	C	L	O	S	E	D

- ◆ IP0 to IP3: Ethernet channel number
- ◆ App.: Name of application using the channel
The connection in the control status from robot language uses simple protocol when HOST is shown and uses non-protocol communication when SCOL is shown.
- ◆ Mode: Protocol type, open status
TCP/S indicates a server connection using TCP protocol, and TCP/C indicates a client connection using TCP protocol.
- ◆ Status: Ethernet port status
The statuses below are displayed. For the specific meaning of each status, refer to any commercially-available reference that provides explanations of TCP/IP.
Closed
Listen
Syn-Sent
Syn-Received

Established
Fin-Wait1
Fin-Wait2
Close wait
Closing
Last ACK
Time Wait

(3) Cautions

- Check that the Ethernet cable between the controller and external device is connected correctly.
- Use a cross connect cable if connecting the robot controller directly to an external device, and use a straight cable if the connection passes through a hub.
- After making any parameter settings, turn the power off and then on again to enable the new settings.

5.2.4 Confirmation with a Windows Computer

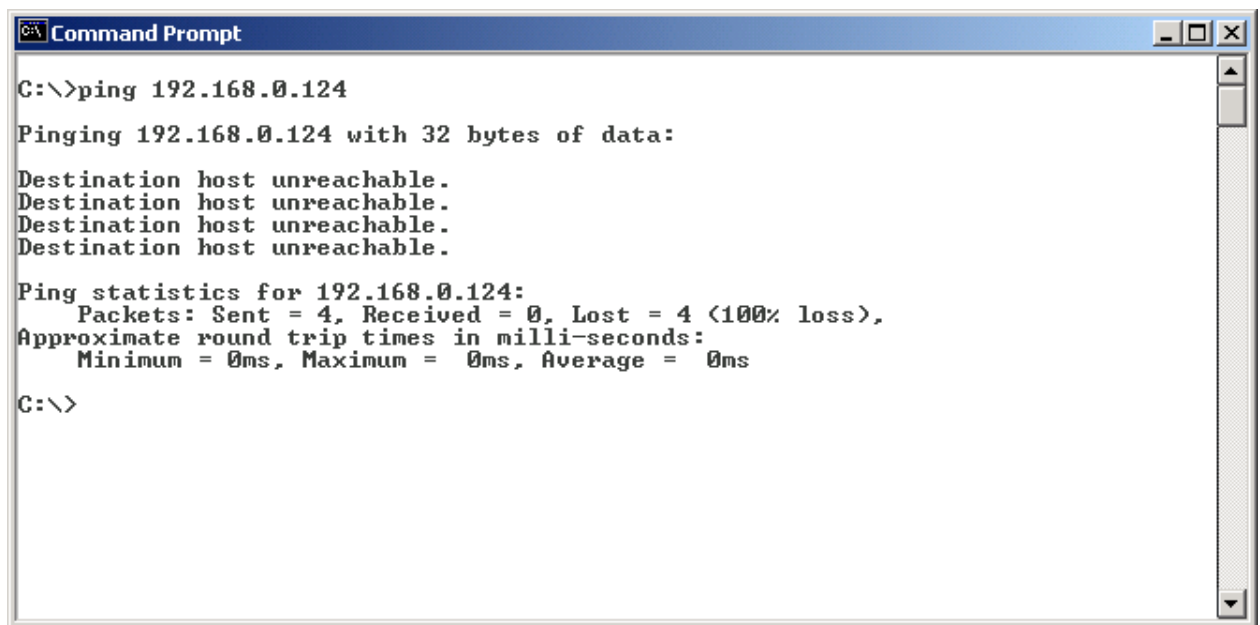
This describes how to confirm the connection using the DOS command "ping".

From the Start menu in Windows, go to the Command Prompt, and specify the IP address of the robot controller as shown below.

C:¥> ping Robot controller IP address

If there is a transmission problem, the result "unreachable" or "Request Time out" is displayed as shown in Figure 5.2.

If transmission is performed normally, the result "Reply from ..." is displayed as shown in Figure 5.3.



```
C:\>ping 192.168.0.124

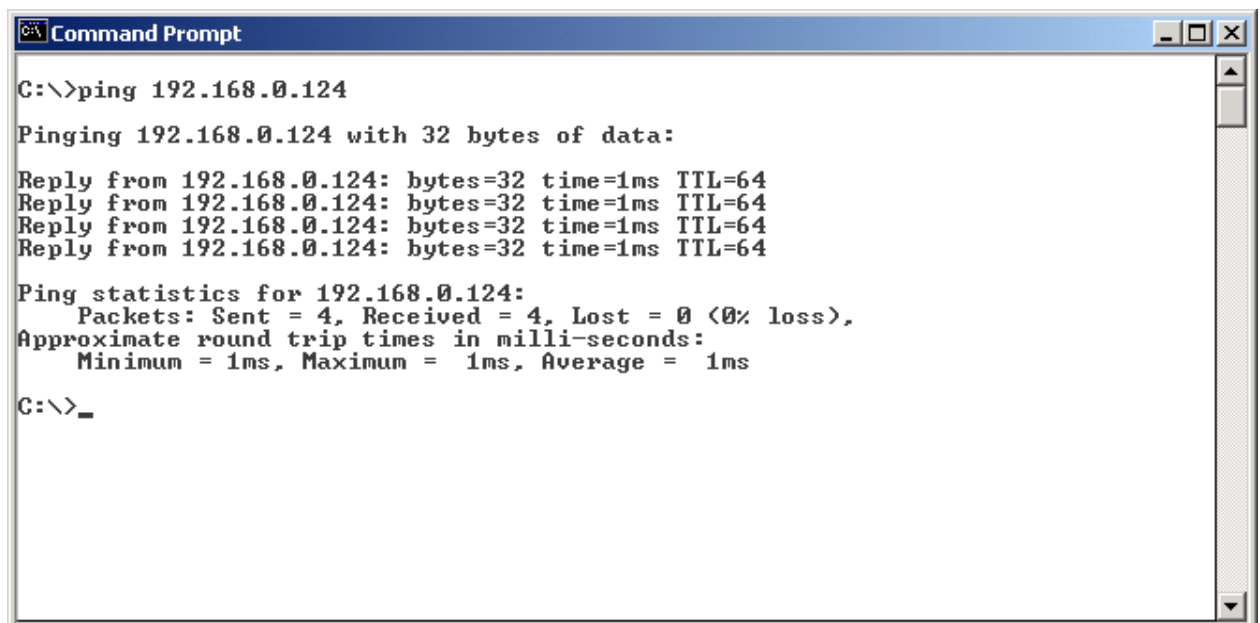
Pinging 192.168.0.124 with 32 bytes of data:

Destination host unreachable.
Destination host unreachable.
Destination host unreachable.
Destination host unreachable.

Ping statistics for 192.168.0.124:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Fig. 5.2 Ping execution screen (Problem result)



```
C:\>ping 192.168.0.124

Pinging 192.168.0.124 with 32 bytes of data:

Reply from 192.168.0.124: bytes=32 time=1ms TTL=64
Reply from 192.168.0.124: bytes=32 time=1ms TTL=64
Reply from 192.168.0.124: bytes=32 time=1ms TTL=64
Reply from 192.168.0.124: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.0.124:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>_
```

Fig. 5.3 Ping execution screen (Normal result)

5.3 Commands

5.3.1 List of Commands

Commands which may be used with this communication protocol are shown in Tables 5.1 and 5.2.

Table 5.1 List of commands (Host computer → Controller)

No.	Command	Descriptions	Details	Text
1	RN	Start	Automatic operation	RN
2	SP	Stop		SP
3	BR	Servo OFF		BR
4	SO	Servo ON		SO
5	RS	Reset	Program reset	RS, PRG
			Step reset	RS, STP
			Cycle reset	RS, CYC
			Output signal reset	RS, SIG
			Select reset	RS, SEL
			Alarm reset	RS, ERR
6	SL	Program selection		SL, <u>File name</u>
7	UL	File name	RAM file upload	UL, <u>File name</u>
8	DL	File name	RAM file download	DL, <u>File name</u>
9	CA	Directory	Directory request	CA
10	SU	Status	Status request	SU
11	EU	Error information	Error history upload	EU
12	ER	File erase		ER, <u>File name</u>
13	FL	File		FL, <u>File contents</u>
14	OK	Acknowledge		OK
15	NG	Non-acknowledge		NG
16	EC*	Internal command	Internal command transmit	EC, <u>Internal command</u>
17	MW	Variable write	Global-defined variable write	MW, <u>Flag Name, Type Data</u>
18	FD	Feed hold	Feed hold	FD
19	SF	System total status	Request for fast-speed status information	SF
20	MR	Variable read	Global-defined variable read	MR, <u>Name, Type</u>
21	IW	I/O write	I/O forced-writing	IW, <u>Line No., Status</u>
22	VR	Version read	Robot and software version	VR
23	DO	Execution of DO statement	Execution of DO statement	DO, <u>Statement</u>
24	AC	Currently occurring alarm	Currently occurring alarm acquisition with message	AC

No.	Command	Descriptions	Details	Text
25	MD	Guide mode	JOG setting request	MD, 0
			INCHING setting request	MD, 1
26	RT	Guide rate	Slow speed setting request	RT, 0
			Mid speed setting request	RT, 1
			Fast speed setting request	RT, 2
27	SC	Guidance coordinate	Joint coordinate setting request	SC, 0
			Tool coordinate setting request	SC, 1
			Work coordinate setting request	SC, 2
			World coordinate setting request	SC, 3
28	JG	JOG execution	Plus direction JOG execution request	JG, 1 +
			Minus direction JOG execution request	JG, 1 -
			JOG stop request	JG, 1 !
29	ZS	Coordinate setting	Work coordinate setting request	ZS, 1 coordinate name coordinate
			Tool coordinate setting request	ZS, 2 coordinate name coordinate
30	MP	Teaching point movement	Normal teaching point movement execution request	MP, 0 movement position
			Z axis current position teaching point movement execution request	MP, 1 movement position
31	PS	Current position	Current position request	PS
32	PR	Individual current position	Joint coordinate request	PR, 0
			World coordinate request	PR, 1
			Work coordinate request	PR, 2
			Joint feedback coordinate request	PR, 3
			World feedback coordinate request	PR, 4
			Work feedback coordinate request	PR, 5

No.	Command	Descriptions	Details	Text
33	MN	Maintenance information	Information request for total rotational amount of motor	MN, 1
			Information request for movement amount of joint axis	MN, 2
			Information request for integrated amount of motor torque	MN, 3
			Information request for integrated amount of motor torque exceeding rating	MN, 4
			Peak torque information request *Cleared by communication request.	MN, 5
			3-second load factor information request	MN, 6
			1-minute load factor information request	MN, 7
			15-minute load factor information request	MN, 8
			2-hour load factor information request	MN, 9
			Total time information request	MN, 10
			Energizing time information request	MN, 11
			Power OFF time information request	MN, 12
			Motion time information request	MN, 13
			Fault time information request	MN, 14
			Servo information request (servo OFF count)	MN, 15
			Servo ON time information request	MN, 16
			Peak torque information request *Not cleared by communication request.	MN, 17

No.	Command	Descriptions	Details	Text
34	RM	Maintenance information reset	Reset request for total rotational amount of motor	RM, 1
			Reset request for movement amount of joint axis	RM, 2
			Reset request for integrated amount of motor torque	RM, 3
			Reset request for integrated amount of motor torque exceeding rating	RM, 4
			Peak torque reset request	RM, 5
			3-second load factor reset request	RM, 6
			1-minute load factor reset request	RM, 7
			15-minute load factor reset request	RM, 8
			2-hour load factor reset request	RM, 9
			Total time reset request	RM, 10
			Energizing time reset request	RM, 11
			Power OFF time reset request	RM, 12
			Motor time reset request	RM, 13
			Fault time reset request	RM, 14
			Servo information reset request (servo OFF count)	RM, 15
			Servo ON time reset request	RM, 16
35	IO	IO information	IO information request *Field bus area extension	IO, 1
36	WD	Watchdog	Watchdog timer setting	WD, <u>time alarm level</u>
37	AH	Alarm history	Alarm history message acquisition	AH
38	HO	Output signal write	Hexadecimal-coding and forced writing of output signal	HO, signal name signal length expression
39	VL	Robot tip speed information	Robot tip speed information request (joint)	VL, 0
			Robot tip speed information request (world)	VL, 1

No.	Command	Descriptions	Details	Text
40	SM	Motion status	Motion status information acquisition	SM, version
41	HI	Input signal read	Hexadecimal-coding and reading of Input signal	HI, signal name signal length
42	ZC	3-axis origin position change information	3-axis origin position change information request	ZC
43	EM	Controller mode change	EXT.SIG change request	EM, 0
			EXT.RS232C change request	EM, 1
			EXT.ETHER change request	EM, 2

* The internal commands executable by the EC command are as follows:

- (1) MODE
- (2) OVRD
- (3) BREAK

Table 5.2 List of commands (Controller → Host computer)

No.	Command	Descriptions	Details	Text
1	FL*	File		FL, <u>File contents</u>
2	OK	Acknowledge		OK
3	NG	Non-acknowledge		NG

* The following files can be sent with the FL command.

- (1) RAM files
 - User file (program and positional data)
 - Parameter file
- (2) File directories
- (3) Status files
- (4) Error history files
- (5) System total status files
- (6) Version information files
- (7) Variable read data files
- (8) Currently occurring alarm

5.3.2 Commands and Operation Modes

Tables 5.3 and 5.4 show the operation modes in which each command is operative. The communications through the HOST port are effective irrespective of master modes, but their function is limited by a master mode. For details, see the following table.

Table 5.3 HOST port (RS232C) commands and operation modes

• HOST port commands and operation modes

Master mode (Key switch)		TEACHING	INTERNAL (TS3000)	EXT		
Parameter (EXTERNAL mode control right)				SIG	RS-232C	ETHER
Maximum speed of robot		250 mm/sec	100%	100%	100%	100%
Command	Description					
RN	Automatic operation start	X	X	X	△	X
SP	Stop	○	○	○	○	○
BR	Servo OFF	○	○	○	○	○
SO	Servo ON	X	X	X	○	X
RS	Program reset	X	X	X	○	X
SL	Program selection	△	△	△	△	△
UL	File upload	△	△	△	△	△
DL	File download	△	△	△	△	△
CA	Directory request	○	○	○	○	○
SU	Status request	○	○	○	○	○
ER	File erase	△	△	△	△	△
EU	Error history upload	○	○	○	○	○
EC	Internal command	X	X	X	○	X
MW	Variable write	○	○	○	○	○
FD	Feed hold	○	○	○	○	○
SF	System total status	○	○	○	○	○
MR	Variable read	○	○	○	○	○
IW	I/O write	X	X	X	○	X
VR	Version read	○	○	○	○	○
DO	Execution of DO statement	X	X	X	○	X
AC	Currently occurring alarm acquisition	○	○	○	○	○
MD	Guide mode setting	X	X	X	△	X
RT	Guide rate setting	X	X	X	△	X
SC	Guidance coordinate setting	X	X	X	△	X
JG	JOG execution	X	X	X	△	X
ZS	Coordinate setting	X	X	X	△	X

MP	Teaching point movement execution	X	X	X	△	X
PS	Current position acquisition	○	○	○	○	○
PR	Individual current information acquisition	○	○	○	○	○
MN	Maintenance information acquisition	○	○	○	○	○
RM	Maintenance information reset execution	○	○	○	○	○
IO	IO information acquisition	○	○	○	○	○
WD	Watchdog execution	○	○	○	○	○
AH	Alarm history acquisition	○	○	○	○	○
HO	Output signal write	X	X	X	○	X
VL	Robot tip speed information acquisition	○	○	○	○	○
SM	Motion status acquisition	○	○	○	○	○
HI	Input signal read	X	X	X	○	X
ZC	3-axis origin position change information acquisition	○	○	○	○	○
EM	Controller mode change execution	X	X	X	△	X

○: Valid, X: Invalid, △: Valid when the robot is stopped

The communications through the IP0 port are effective irrespective of master modes, but their function is limited by a master mode. For details, see the following table.

Table 5.4 IPO port (Ethernet) commands and operation modes

Master mode (Key switch)		TEACHING	INTERNAL (TS3000)	EXT		
Parameter (EXTERNAL mode control right)				SIG	RS-232C	ETHER
Maximum speed of robot		250 mm/sec	100%	100%	100%	100%
Command	Description					
RN	Automatic operation start	X	X	X	X	△
SP	Stop	○	○	○	○	○
BR	Servo OFF	○	○	○	○	○
SO	Servo ON	X	X	X	X	○
RS	Program reset	X	X	X	X	○
SL	Program selection	△	△	△	△	△
UL	File upload	△	△	△	△	△
DL	File download	△	△	△	△	△
CA	Directory request	○	○	○	○	○
SU	Status request	○	○	○	○	○
ER	File erase	△	△	△	△	△
EU	Error history upload	○	○	○	○	○
EC	Internal command	X	X	X	X	○
MW	Variable write	○	○	○	○	○
FD	Feed hold	○	○	○	○	○
SF	System total status	○	○	○	○	○
MR	Variable read	○	○	○	○	○
IW	I/O write	X	X	X	X	○
VR	Version read	○	○	○	○	○
DO	Execution of DO statement	X	X	X	X	○
AC	Currently occurring alarm acquisition	○	○	○	○	○
MD	Guide mode setting	X	X	X	X	△
RT	Guide rate setting	X	X	X	X	△
SC	Guidance coordinate setting	X	X	X	X	△
JG	JOG execution	X	X	X	X	△
ZS	Coordinate setting	X	X	X	X	△
MP	Teaching point movement execution	X	X	X	X	△
PS	Current position acquisition	○	○	○	○	○
PR	Individual current position acquisition	○	○	○	○	○
MN	Maintenance information acquisition	○	○	○	○	○
RM	Maintenance information reset execution	○	○	○	○	○

IO	IO information acquisition	○	○	○	○	○
WD	Watchdog execution	○	○	○	○	○
AH	Alarm history acquisition	○	○	○	○	○
HO	Output signal write	X	X	X	X	○
VL	Robot tip speed information acquisition	○	○	○	○	○
SM	Motion status acquisition	○	○	○	○	○
HI	Input signal read	X	X	X	X	○
ZC	3-axis origin position change information acquisition	○	○	○	○	○
EM	Controller mode change execution	X	X	X	X	△

○: Valid, X: Invalid, △: Valid when the robot is stopped

Commands transmitted from the host computer to the controller can be received by the controller in the modes marked "○".

Commands transmitted from the controller to the host computer can be transmitted by the controller in the mode marked "○".

Note: All external control input signals are also limited by a mater mode.
For details, see the Interface Manual.

5.3.3 Details of Commands

: Automatic Operation Start (Host computer → Controller) RN

Format

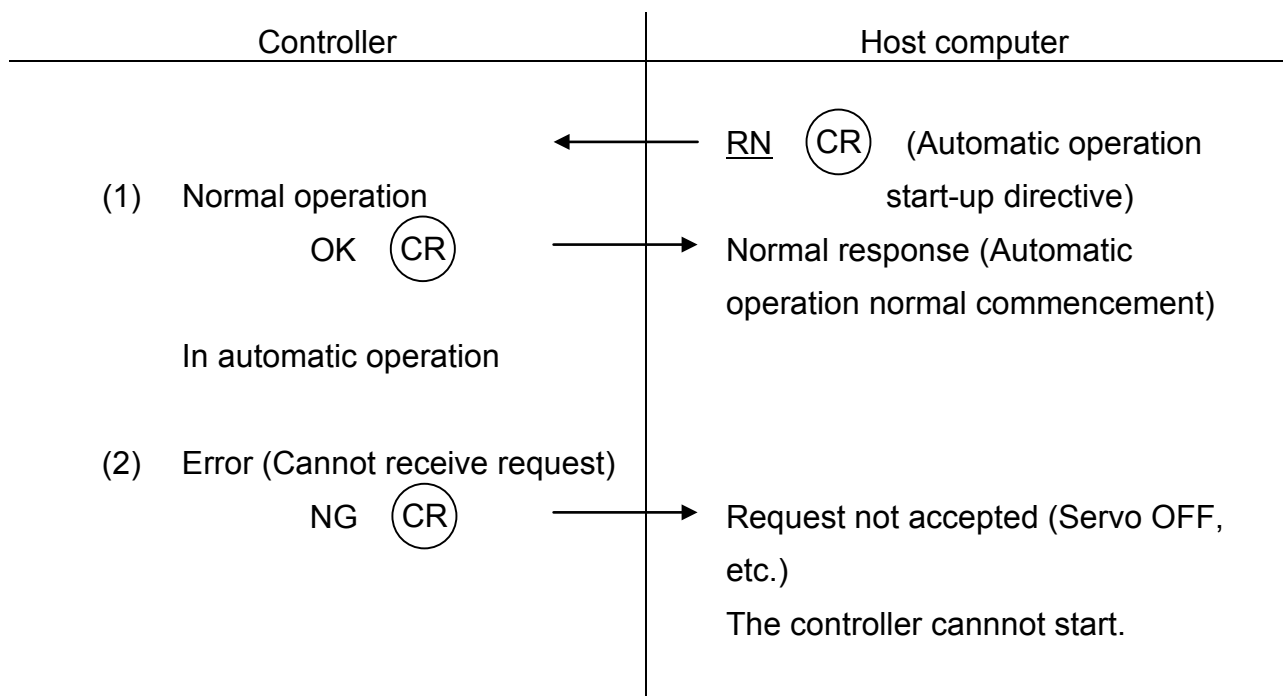
RN



Description

The RN (Automatic Operation Start) command is a command given by the host computer to the controller telling the controller to start up the program. If, after stopping the program with the SP (Automatic Operation Stop) command, one sends the RN command again, the robot will start up from the step immediately following the step at which it was stopped.

Protocol



Note

If the power is turned on while the controller is set for the host mode, the operation mode of the system will be the cycle operation mode. If the controller is changed over to the host mode from the internal automatic mode, the operation mode of the system will be that in effect beforehand.

: Automatic Operation Stop (Host computer → Controller) SP

Format

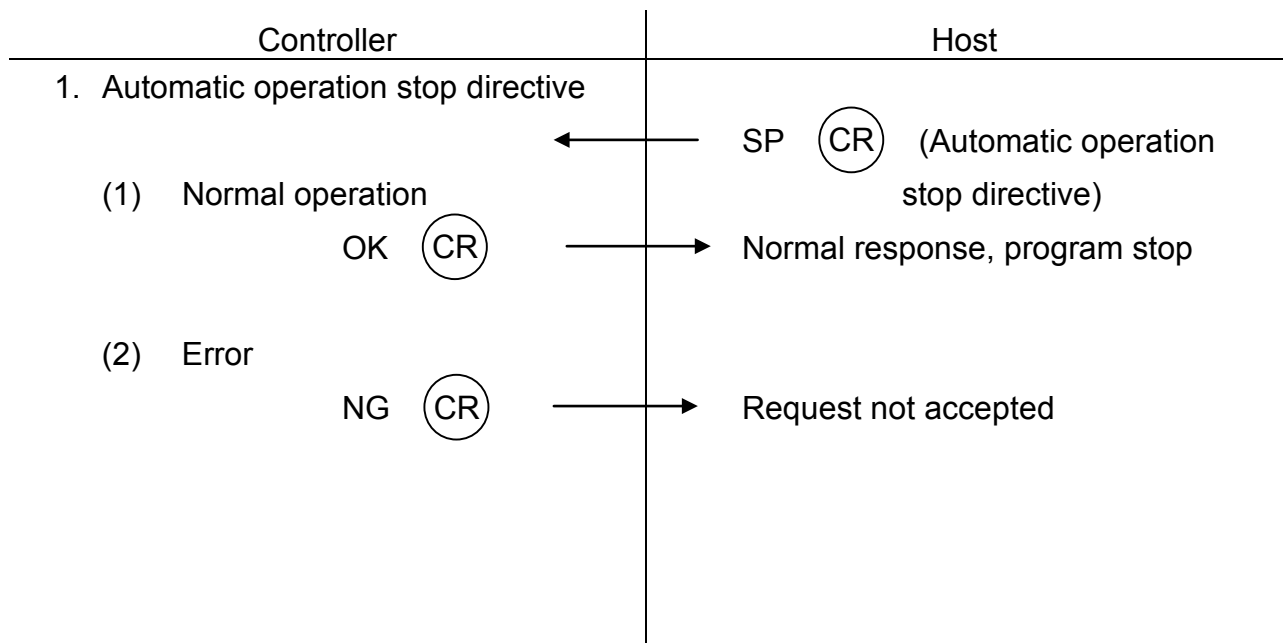
SP

(CR)

Description

The SP (Automatic Operation Stop) command is given by the host computer to the controller telling the controller to stop automatic operation. Also, this command informs the host computer when the controller is stopped. Furthermore, should the robot be stopped in accordance with any request other than that made by the host computer, this command will automatically be sent from the controller to the host computer.

Protocol



: Servo OFF (Host computer → Controller) BR

Format

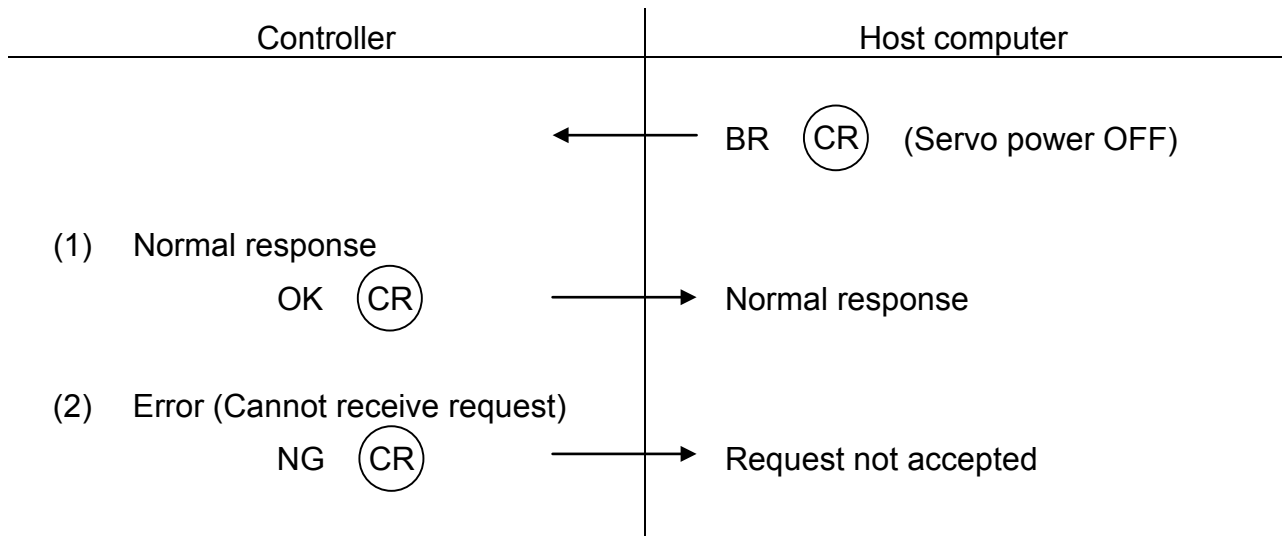
BR

CR

Description

The BR (Servo OFF) command is given by the host computer to the controller telling the controller to turn off the servo power.

Protocol



: Servo ON

(Host computer → Controller) SO

Format

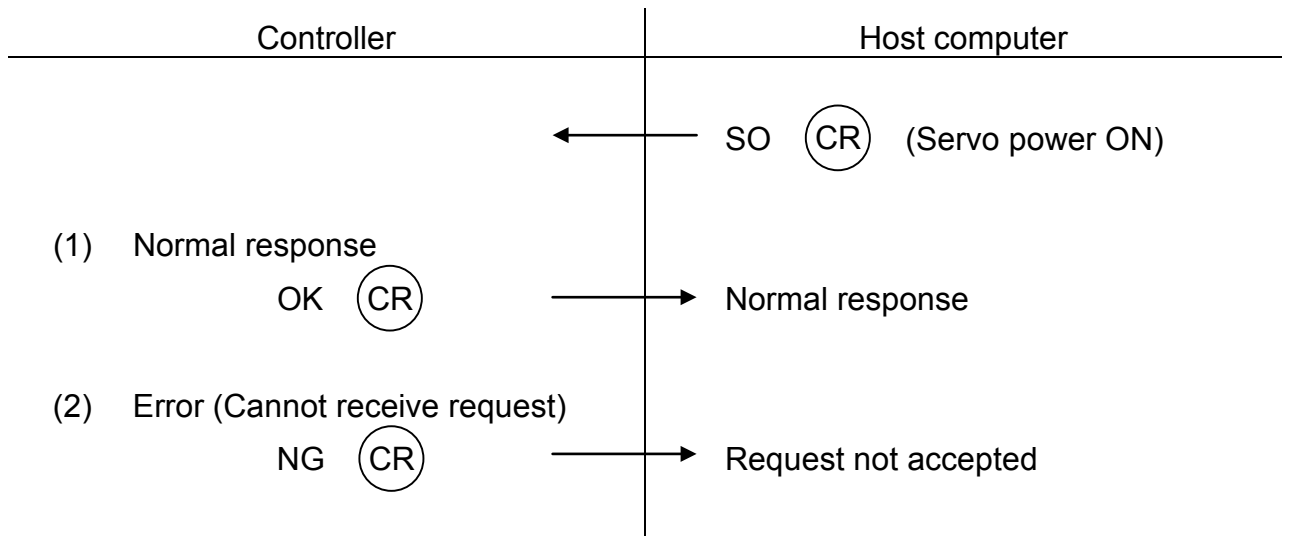
SO

(CR)

Description

The servo ON command is given from the host computer to the controller to turn the power on.

Protocol



: Reset (Host computer → Controller) RS

Format

RS, XXX



XXX : PRG Reset all program data (internal data)
 STP Reset the program back to Step 1. All program data remains unchanged.
 CYC Reset the program back to the step marked by the label RCYCLE.
 SIG Turns off the digital output signals (DO1 to DO24) available to the user.
 SEL Resets the execution file.
 ERR Rests an alarm.

Description

The RS command will not be accepted unless the system is in a stop mode.

(1) PRG

PRG will reset the program to Step 1. All program data will be initialized.

(2) STP

STP will reset the program to Step 1. All program data (such as variables) will remain unchanged.

(3) CYC

CYC will specify step number to the specified label (RCYCLE). All program data will remain unchanged.

(4) SIG

SIG will put the digital outputs in the non-active state.

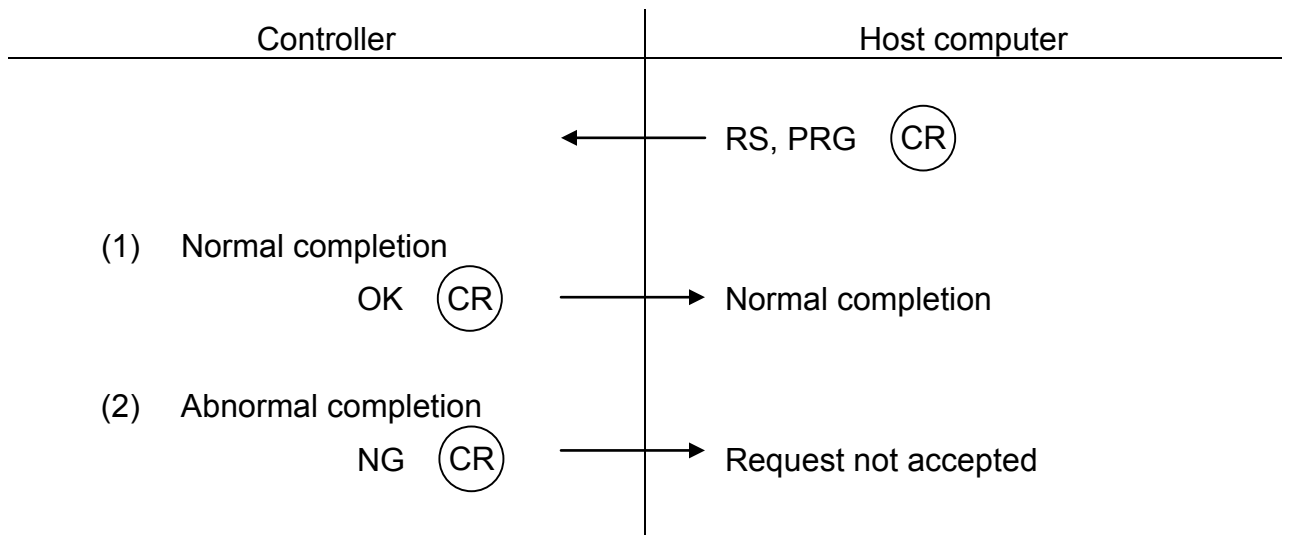
(5) SEL

SEL will reset the execution file.

(6) ERR

ERR will reset an alarm.

Protocol



: Program Selection (Host computer → Controller) SL

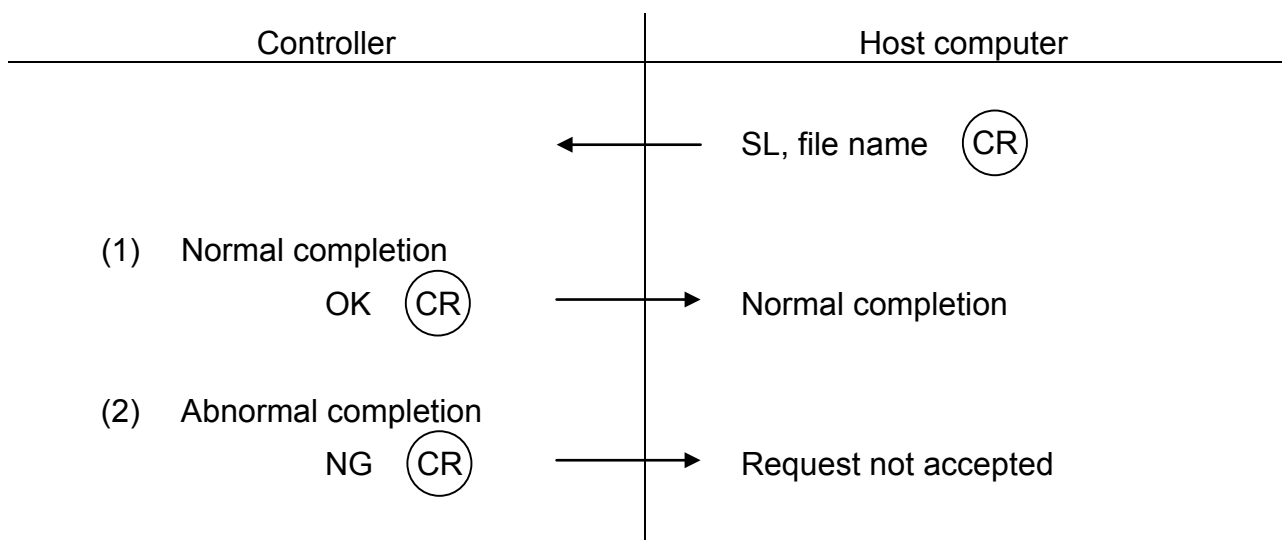
Format

SL, file name (CR)

Description

The SL command selects a program to be executed under automatic operation.

Protocol



: File Upload Request (Host computer → Controller) UL

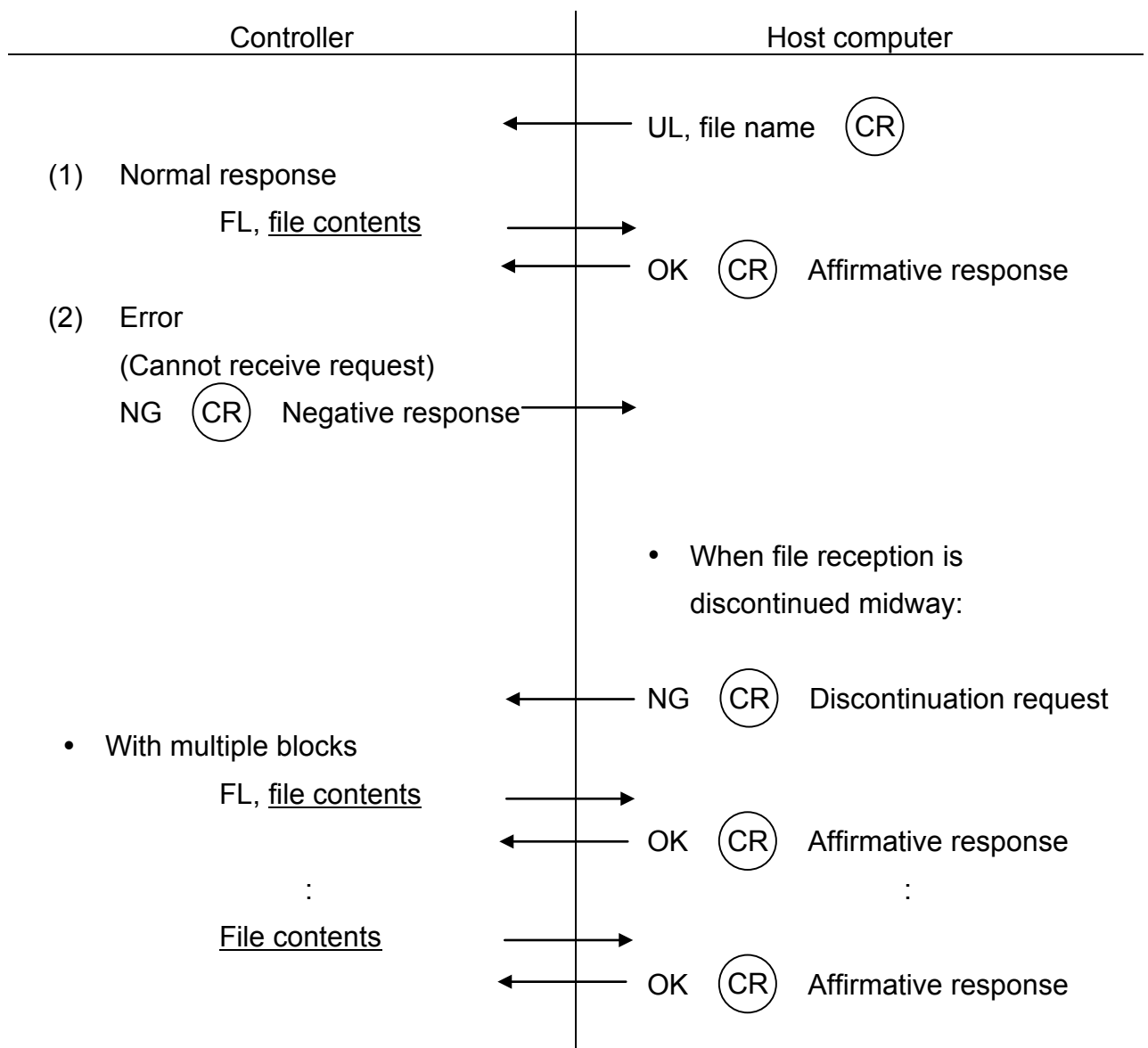
Format

UL, file name (CR)

Function

The UL command is used to upload (transmit) the specified RAM file in the controller to the host computer. For information on the contents of files to be uploaded, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: File Download Request (Host computer → Controller) DL

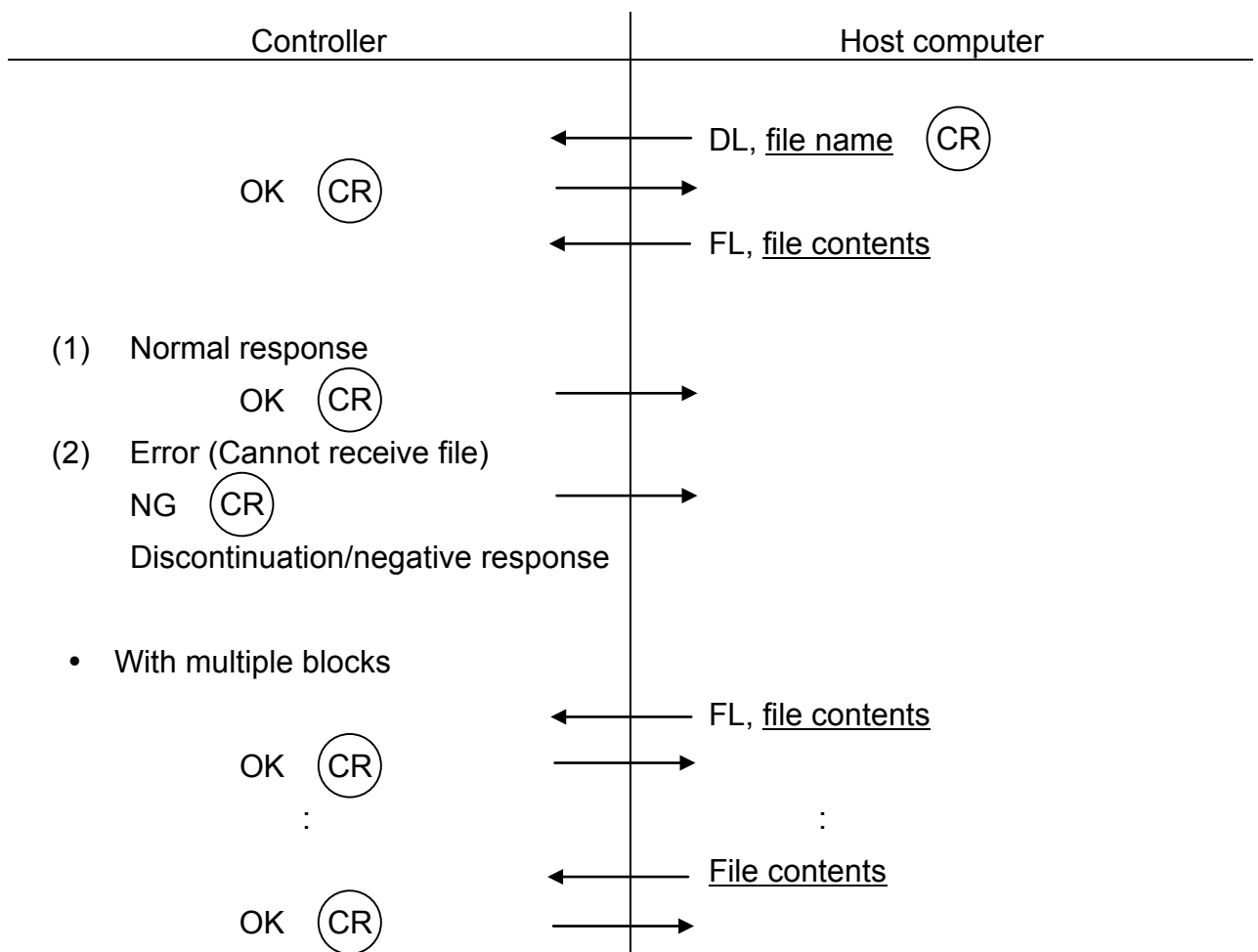
Format

DL, file name (CR)

Function

The DL command is used to download (transmit) a specified file from the host computer to the controller RAM drive. For information on the contents of files to be downloaded, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends the file contents to the controller after the host computer has received OK.

: File Directory Request (Host computer → Controller) CA

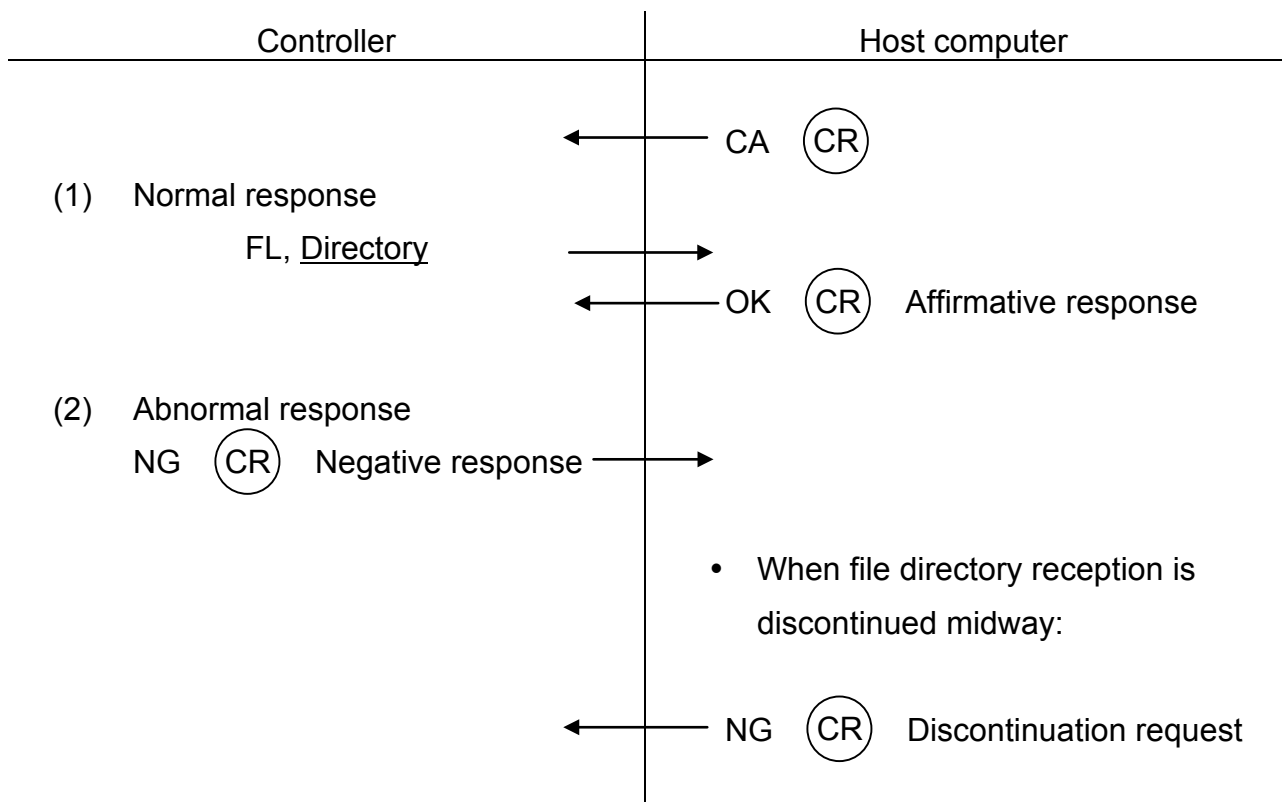
Format

CA (CR)

Function

The CA command is used to transmit the directory of files in the RAM drive from the controller to the host computer. For information on the contents of directories to be transmitted, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the directory.

: Status Request	(Host computer → Controller)	SU
: System Total Status Request	(Host computer → Controller)	SF

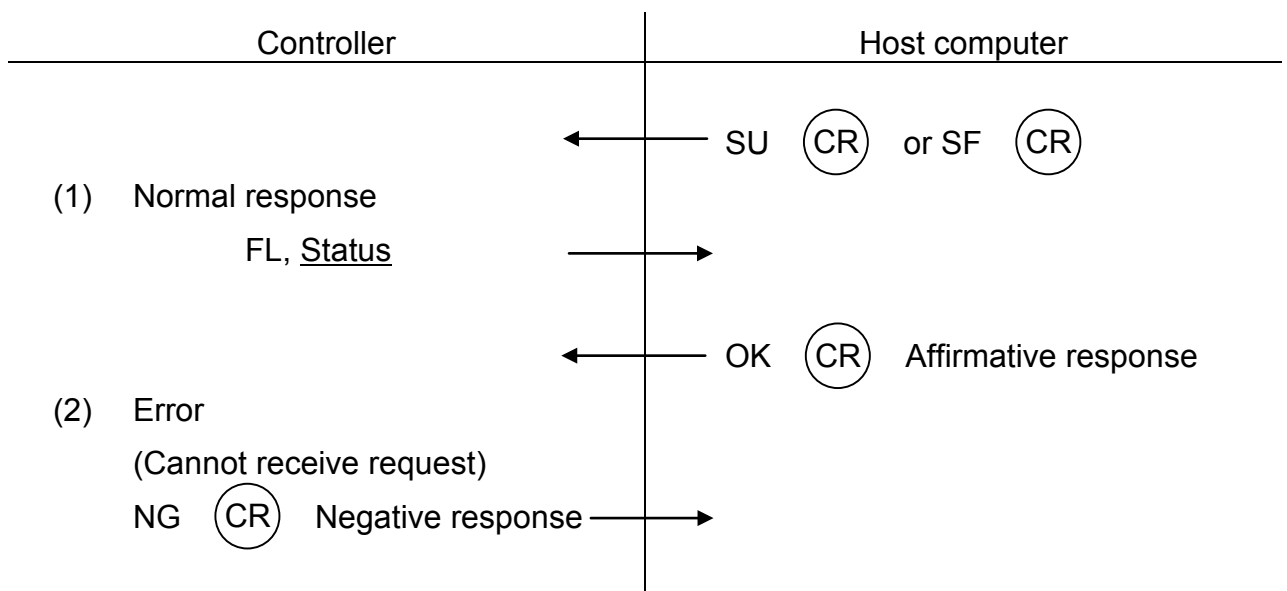
Format

SU	(CR)	Status information
SF	(CR)	System total status information

Function

The SU command is used to send the internal status (state) of the controller to the host computer. For information on the contents of status file to be transmitted, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the status command.

: Error History Request (Host computer → Controller) EU

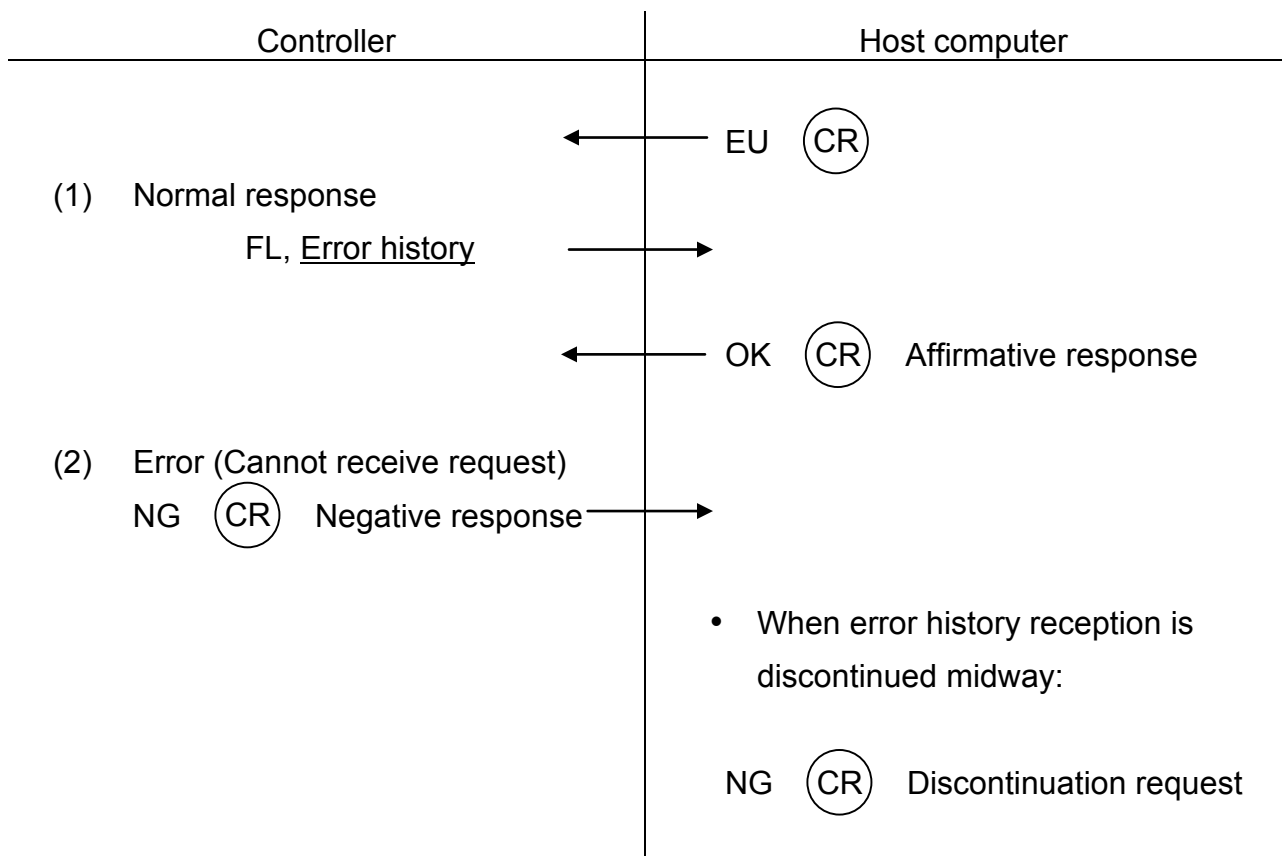
Format

EU (CR)

Function

The EU command is used to send the controller error history data to the host computer. For information on the contents of error history files to be transmitted, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the error history.

: File Erase

(Host computer → Controller) ER

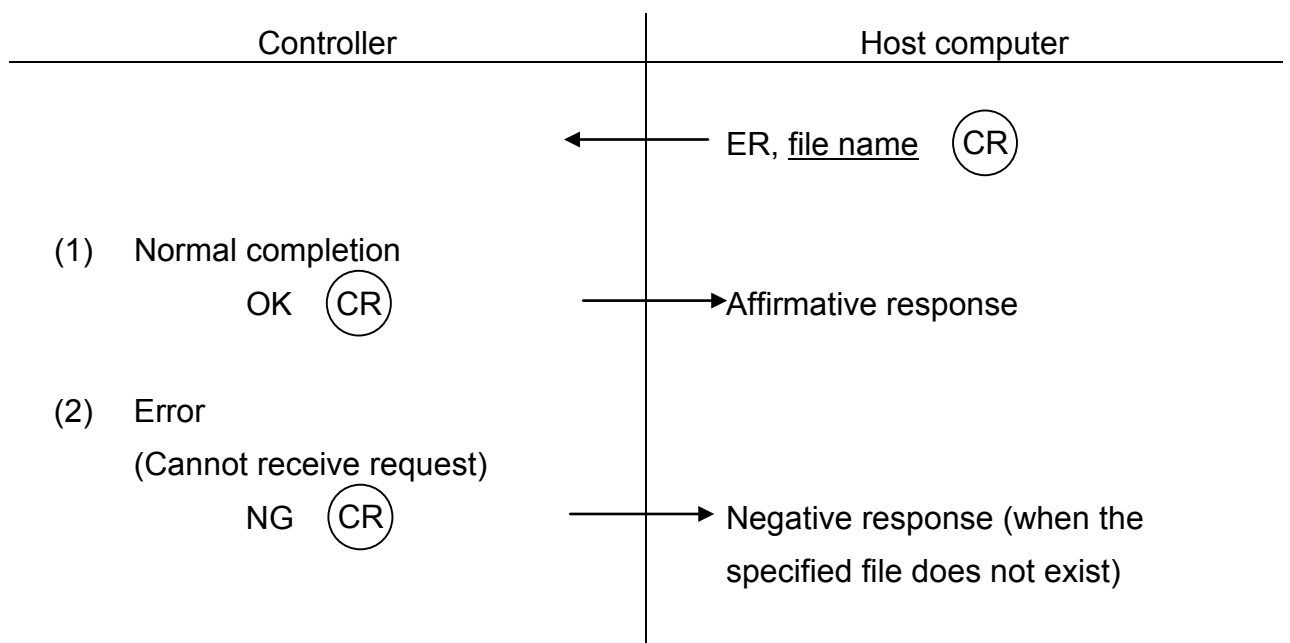
Format

ER, file name (CR)

Description

The ER command is used to erase a specified file from the controller RAM drive.

Protocol



: I/O Write

(Host computer → Controller) IW

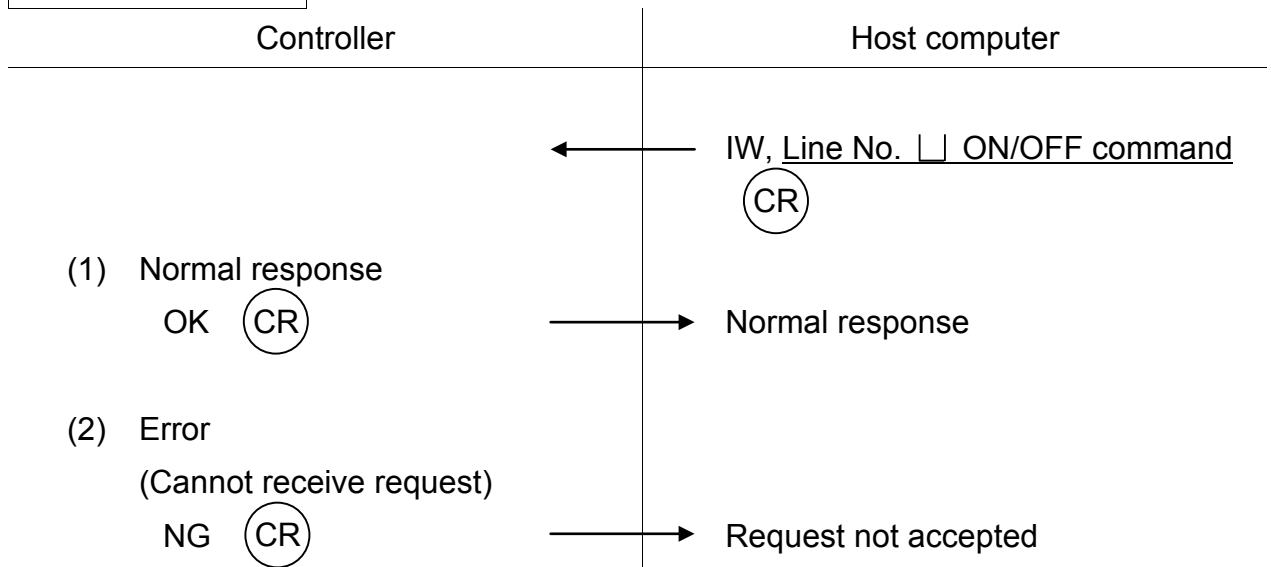
Format

IW, Line No. [] ON/OFF command (CR)

Description

The IW command is used to instruct forced writing of up to 32 I/Os from the host computer to the controller. For the ON/OFF command, specify "0" for OFF and "1" for ON.

Protocol



: Version read

(Host computer → Controller) VR

Format

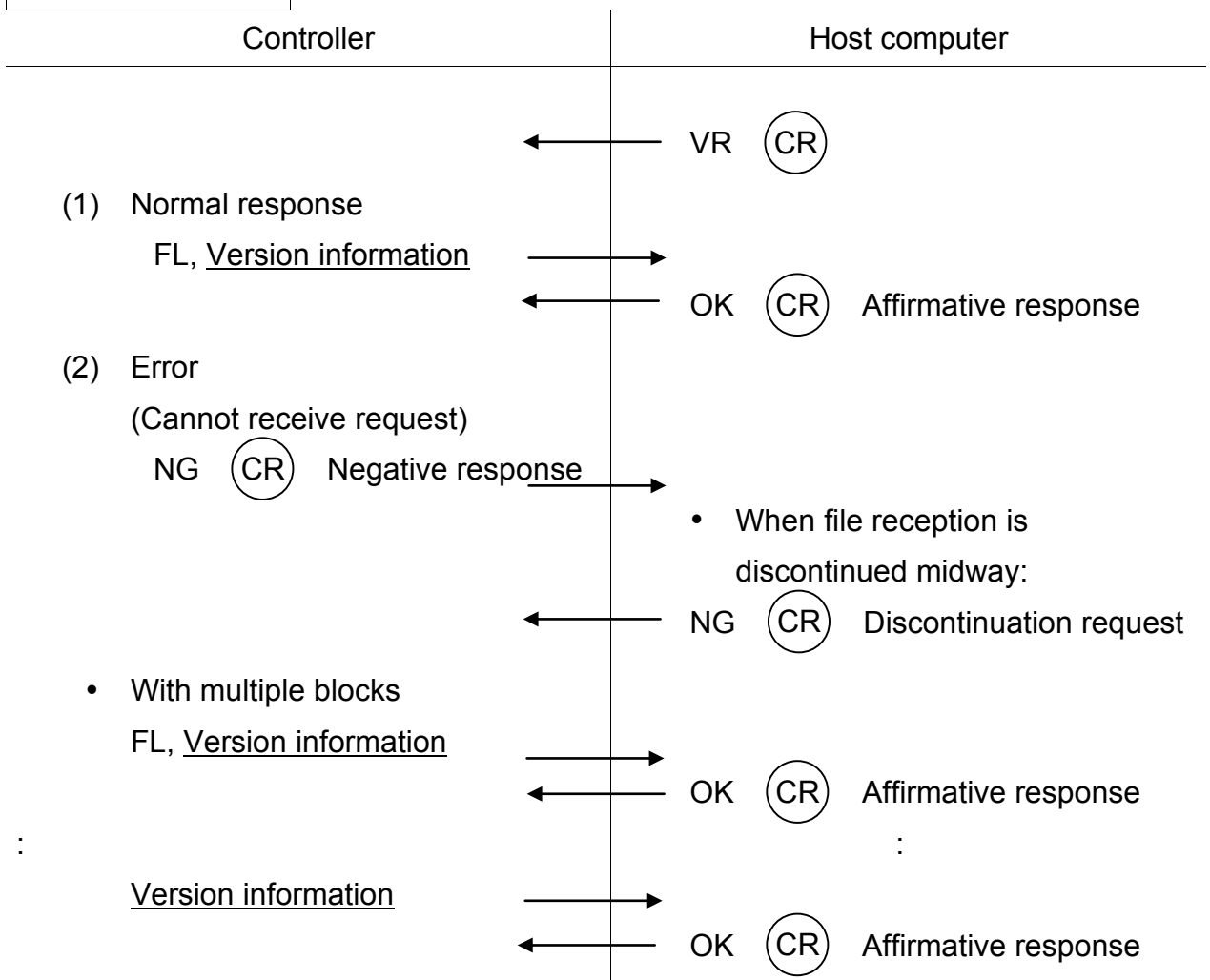
VR (CR)

Description

The VR command is used to send the system version information to the host computer.

For details on the version information, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Execution of DO statement (Host computer → Controller) DO

Format

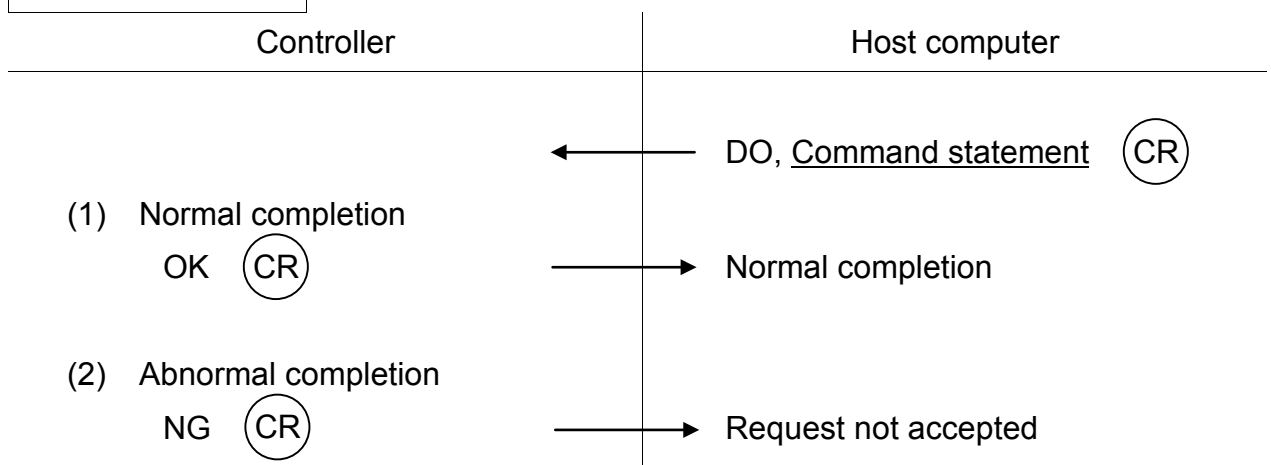
DO, Command statement (CR)

Description

The DO command is used to directly execute a command from the host computer. For details on the command, see the Robot Language Manual.

In the feed hold status or servo power OFF status, negative response NG is sent back from the controller.

Protocol



: Variable read (Host computer → Controller) MR

Format

MR, Variable name ☐ Variable type (CR)

Description

The MR command is used to transfer the read data of global-defined variable to the host computer. Each variable name should consist of up to ten (10) characters.

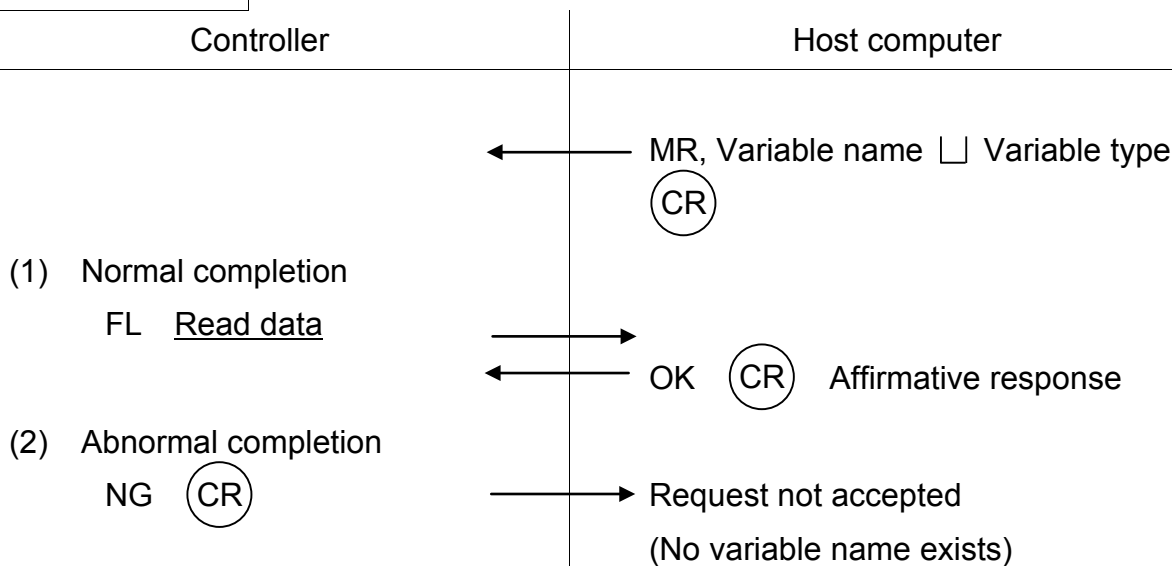
For the array variable, specify the array elements ("Variable name (*, *, ...)").

Specify one of the following numbers for the variable type data.

- | | |
|----------------------|----------------------------|
| 0 : Integer type | 5 : Array integer type |
| 1 : Real number type | 6 : Array real number type |
| 2 : Load type | 7 : Array load type |
| 3 : Coordinate type | 8 : Array coordinate type |
| 4 : Position type | 9 : Array position type |
| -1 : Unclarified | |

For details of the read data, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Write global variable (Host computer → Controller) MW

Format

MW, Flag ☐ Variable name ☐ Variable type ☐ Write data ... (CR)

Descriptions

This function requests writing of data into the user's defined variable.

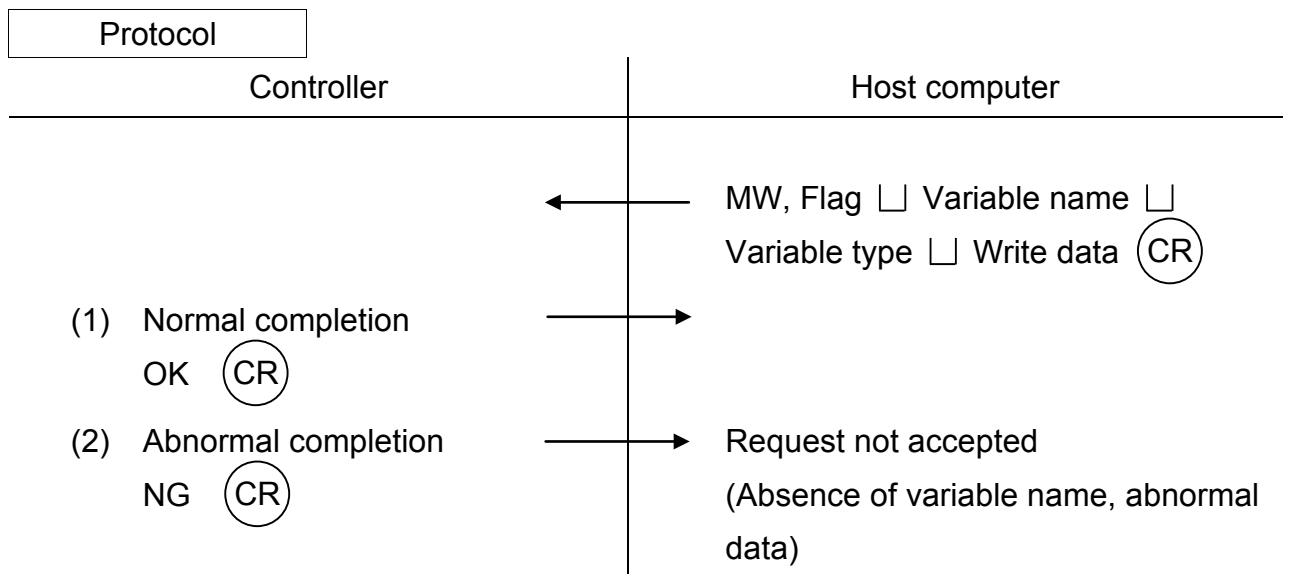
The flag is a control flag for restoring or non-restoring of data in the program file, as shown below.

Flag	Description
0	Restoring of data in the program file is not executed.
1	Restoring of data in the program file is executed.

The variable name should consist of up to ten (10) characters. Specify the array element ("Variable name (*, *, ...)") for the array variable.

Specify the variable type data and write data in the following manner.

Variable type	Variable type data	Write data
Integer type	0	1 pc. (long)
Real number type	1	1 pc. (float)
Load type	2	2 pcs. (float↵float)
Coordinate type	3	4 pcs. (float↵float↵float↵float)
Position type	4	6 pcs. (float↵float↵float↵float↵float↵float)
Array integer type	5	1 pc. (long)
Array real number type	6	1 pc. (float)
Array load type	7	2 pcs. (float↵float)
Array coordinate type	8	4 pcs. (float↵float↵float↵float)
Array position type	9	6 pcs. (float↵float↵float↵float↵float↵float)



: Feed hold

(Host computer → Controller) FD

Format

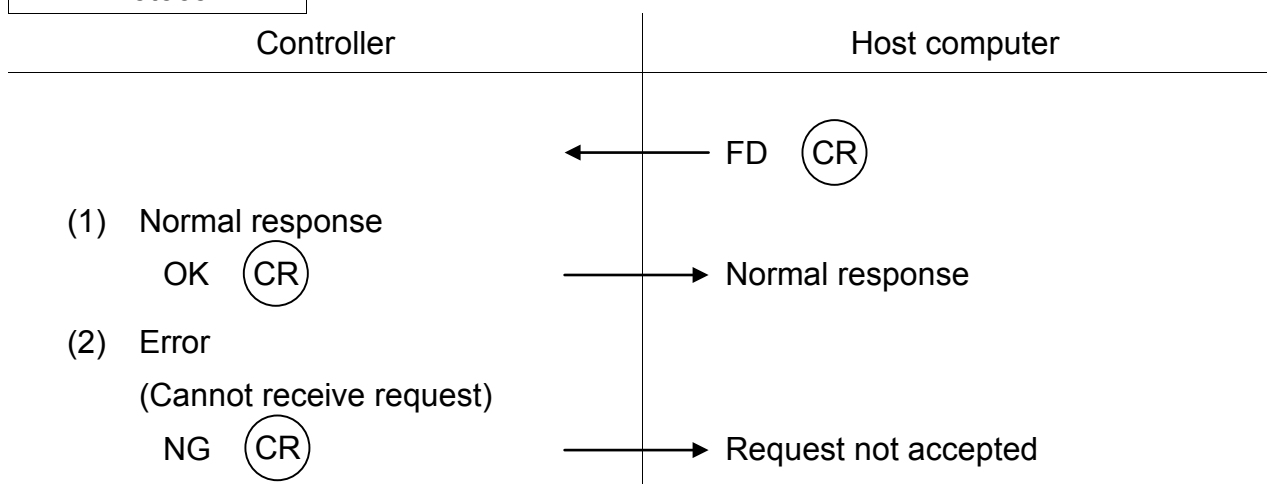
FD

(CR)

Description

The FD command is used to inform from the host computer to the controller that the FEED HOLD pushbutton switch has been pressed.

Protocol



: Acknowledge

(Host computer ↔ Controller)

OK

Format

OK



Description

The OK (Acknowledge) command indicates an affirmative response.

: Non-acknowledge (Host computer ↔ Controller) NG

Format

NG



Description

The NG (Non-acknowledge) command indicates a negative response.

: Internal Command (Host computer → Controller) EC

Format

EC, Command (CR)

Description

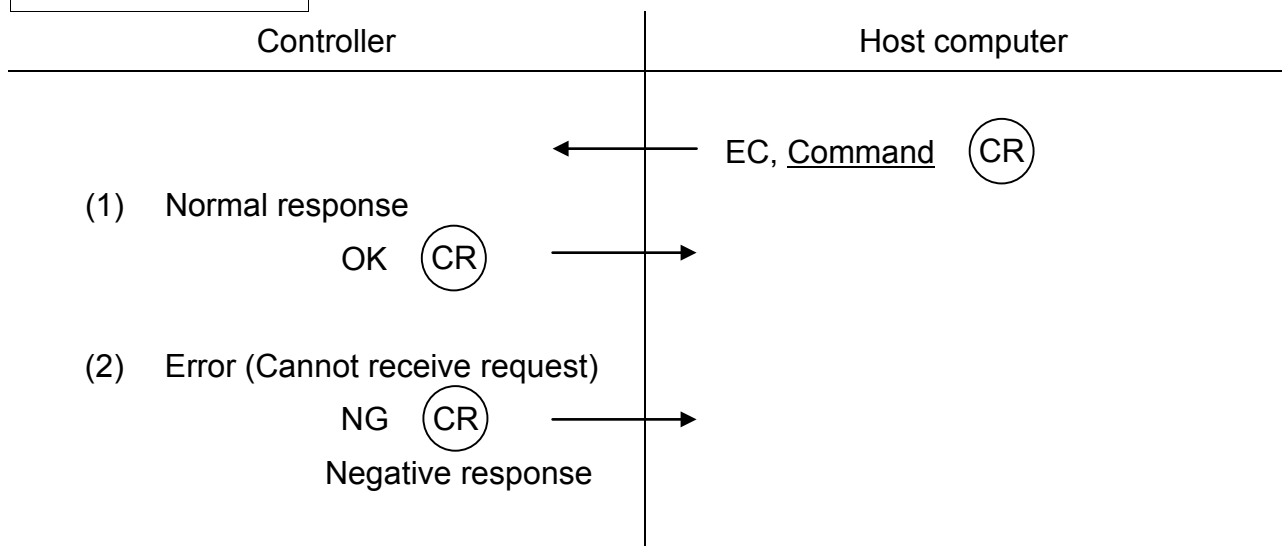
The internal command can be executed from the host computer.

The executable internal commands are as follows.

MODE□CONT
 MODE□CYCLE
 MODE□STEP
 MODE□SEG
 OVRD□Set value (1 - 100)
 BREAK

Note: "□" signifies a space.

Protocol



: Currently occurring alarm acquisition with message (Host computer→ Controller)

AC

Format

AC

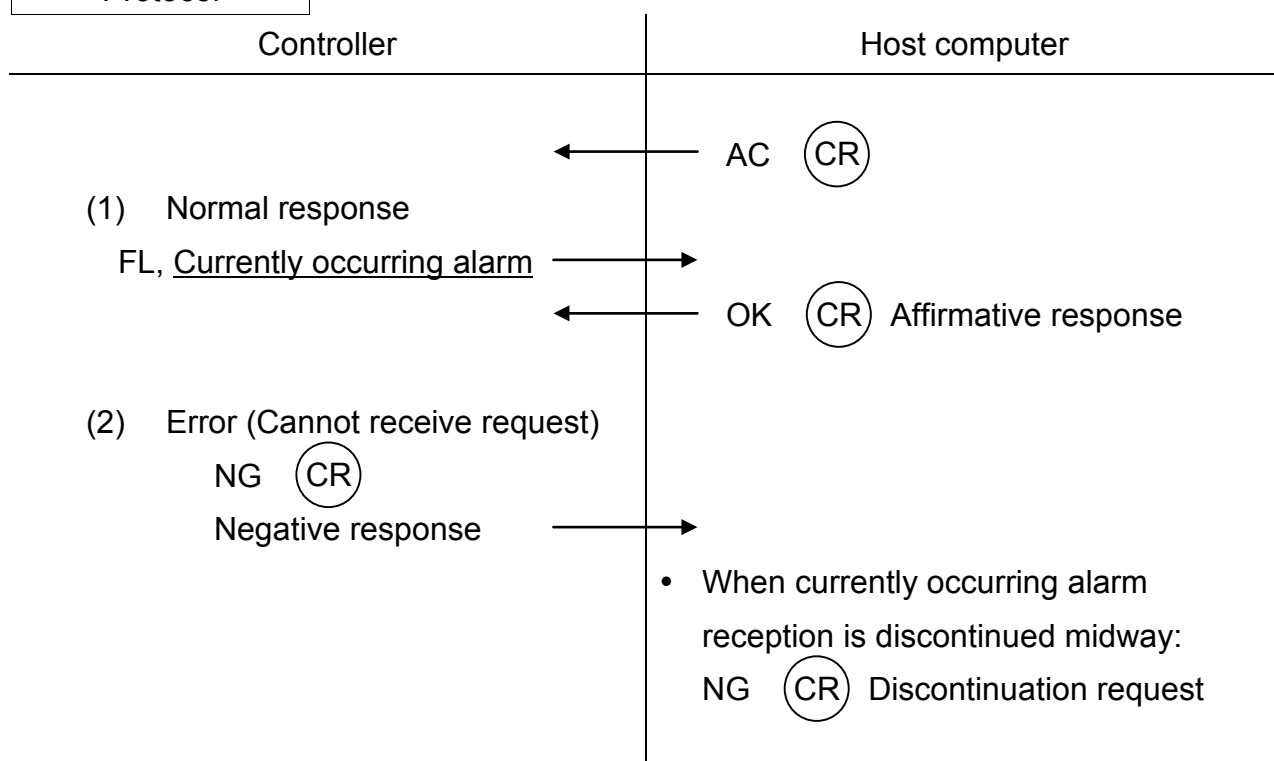
(CR)

Function

The AC command is used to transfer the alarm that currently occurs in the controller with message to the host computer.

For the details on the currently occurring alarm, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the currently occurring alarm.

If many alarms occur, all alarms may not be able to be received by one communication. To receive continuing alarms, send an affirmative response OK during each reception.

: Guide mode setting (Host computer → Controller) MD

Format

MD, Guide mode (CR)

Description

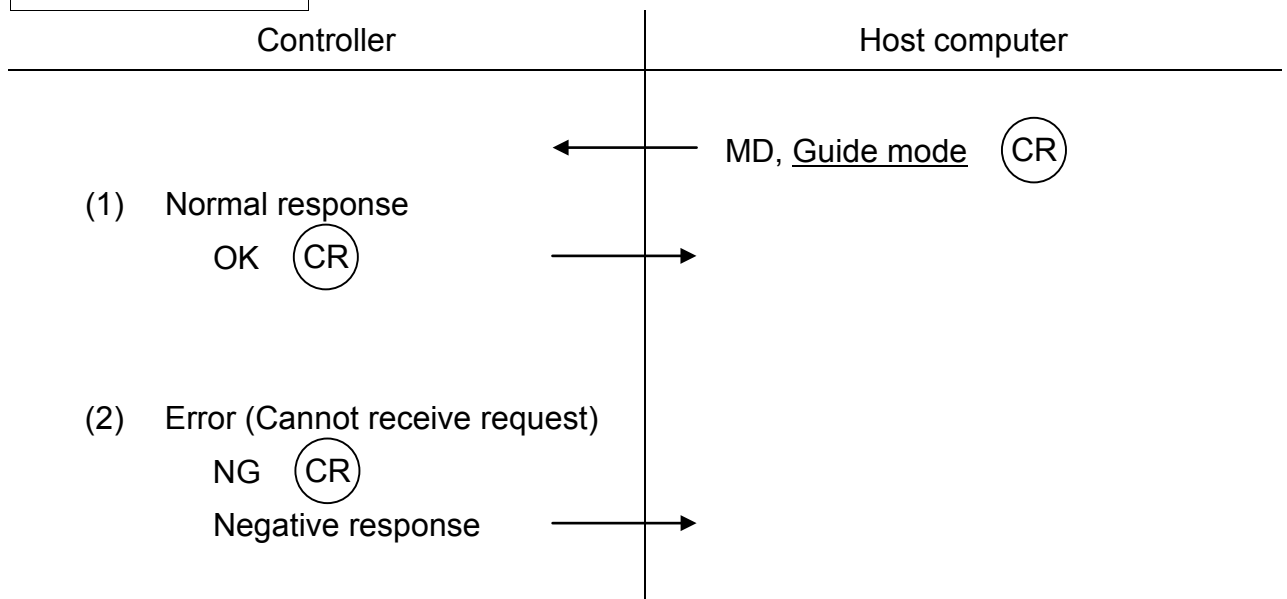
The MD command is used to set the guide mode of the controller from the host computer.

Specify one of the following numbers for the guide mode.

0: JOG

1: INCHING

Protocol



: Guide rate setting (Host computer → Controller) RT

Format

RT, Guide rate (CR)

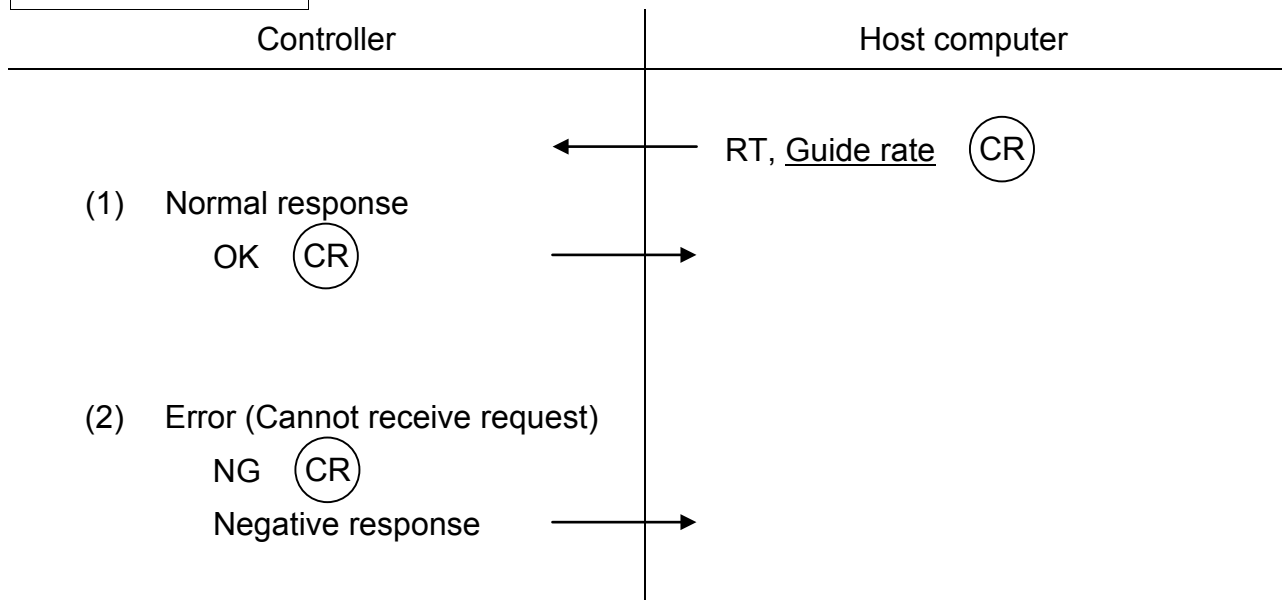
Description

The RT command is used to set the guide rate of the controller from the host computer.

Specify one of the following numbers for the guide rate.

- 0: Slow speed
- 1: Mid speed
- 2: Fast speed

Protocol



: Guidance coordinate setting (Host computer → Controller) SC

Format

SC, Guidance coordinate (CR)

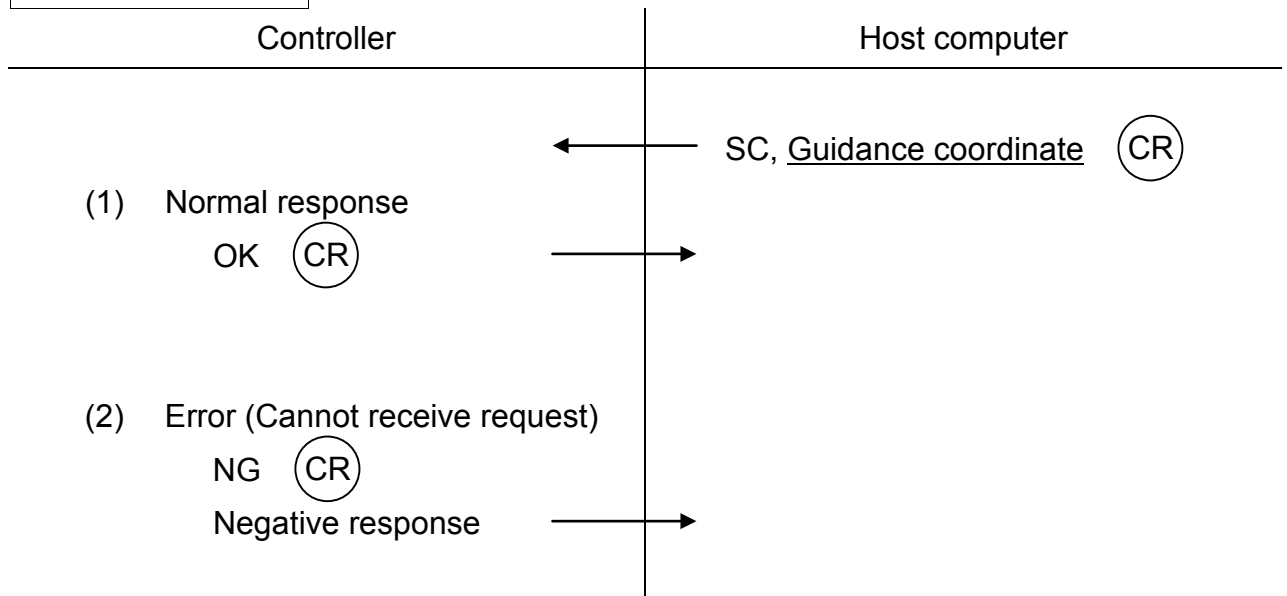
Description

The SC command is used to set the guidance coordinate of the controller from the host computer.

Specify one of the following numbers for the guidance coordinate.

- 0: Joint coordinate
- 1: Tool coordinate
- 2: Work coordinate
- 3: World coordinate

Protocol



: JOG execution

(Host computer → Controller)

JG

Format

JG, Axisdirection (CR)

Description

The JG command is used to direct the controller to start jog from the host computer. Specify one of the following numbers for the axis.

1 to 5

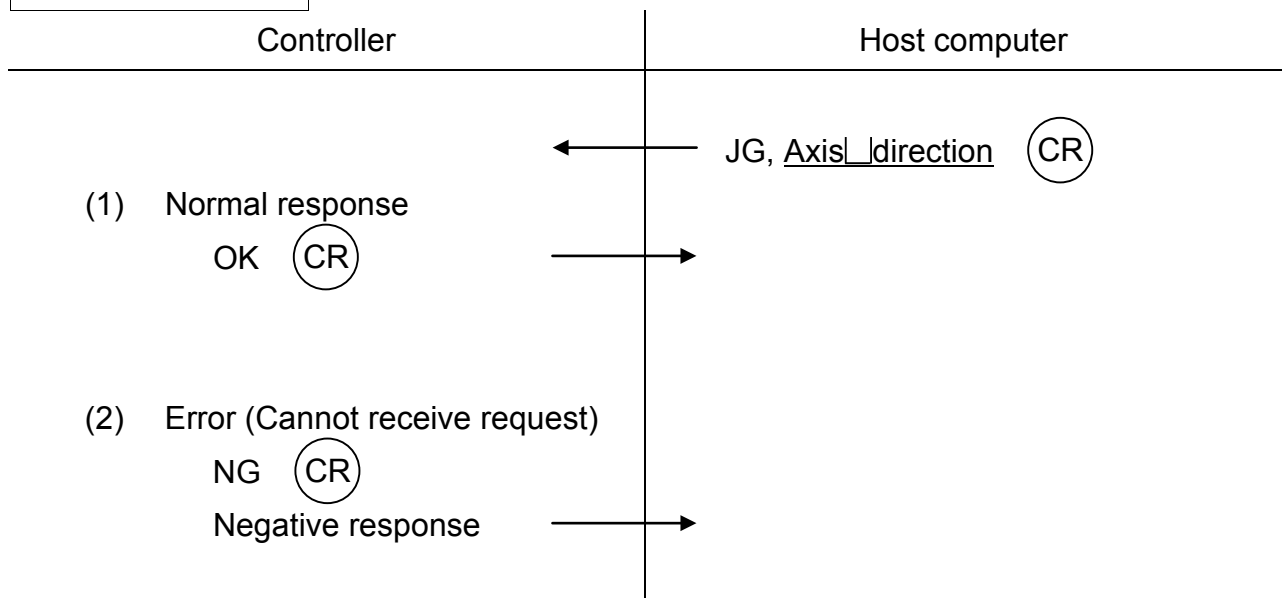
Specify one of the following items for the direction

+: Plus direction jog

-: Minus direction jog

!: Jog motion stop

Protocol



: Coordinate setting (Host computer → Controller) ZS

Format

ZS, Coordinate identification_coordinate name_coordinate value (CR)

Description

The ZS command is used to set the work and tool coordinates of the controller from the host computer.

Specify one of the following numbers for the coordinate identification.

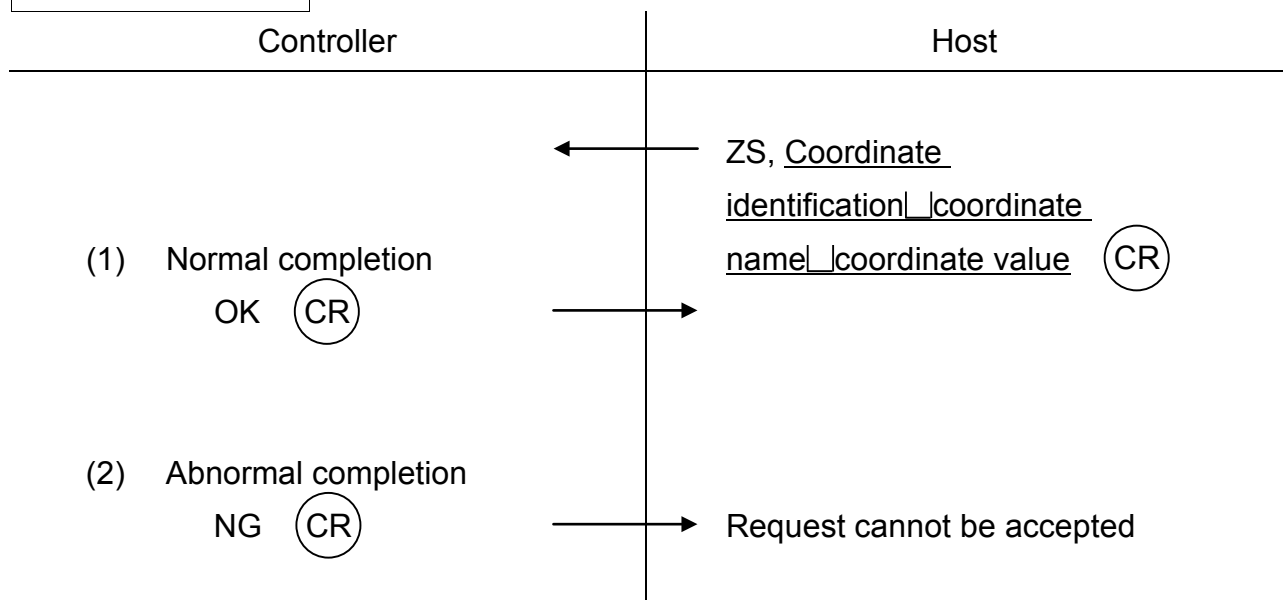
- 1: Tool coordinate
- 2: Work coordinate

A coordinate name consists of up to 20 characters.

Specify the coordinate value in the following format.

Xcoordinate value_Ycoordinate value_Z coordinate value_C coordinate value

Protocol



: Teaching point movement (Host computer → Controller) MP

Format

MP, coordinate identification codemovement position (CR)

Description

The MP command is used to direct the controller to start teaching point movement from the host computer.

Specify one of the following numbers for the coordinate identification code.

0: Normal movement motion

1: Move the Z axis at the current position

Specify the movement position in the following format.

Xcoordinate valueY coordinate valueZ coordinate valueC coordinate valueT coordinate valueconfiguration

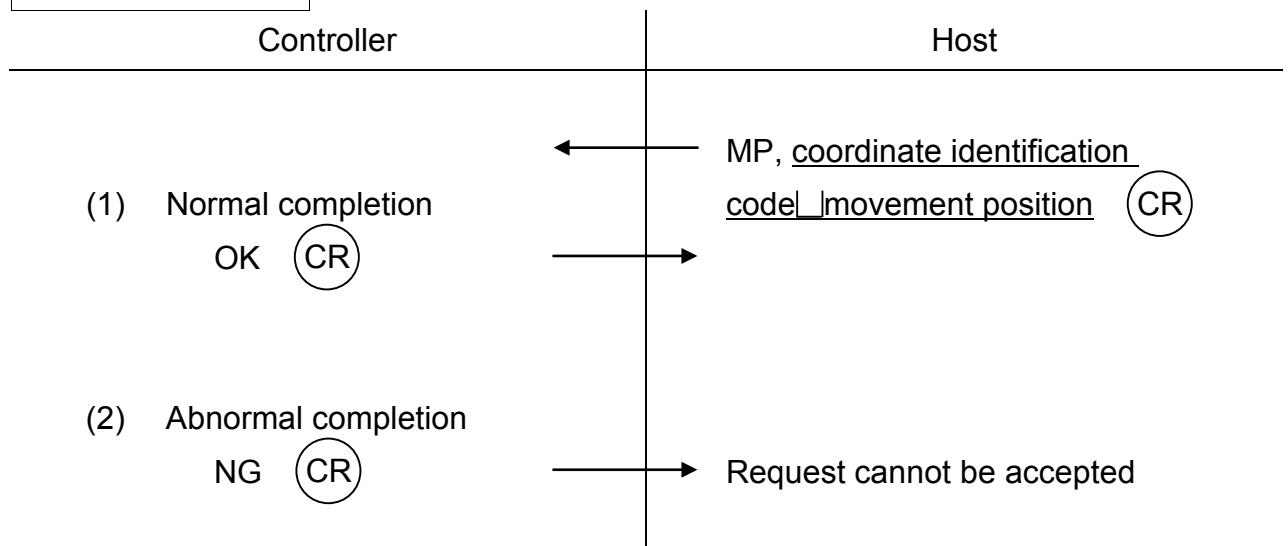
Specify the configuration in the following format.

FREE

LEFTY

RIGHTY

Protocol



: Current position information acquisition (Host computer → Controller) PS

Format

PS

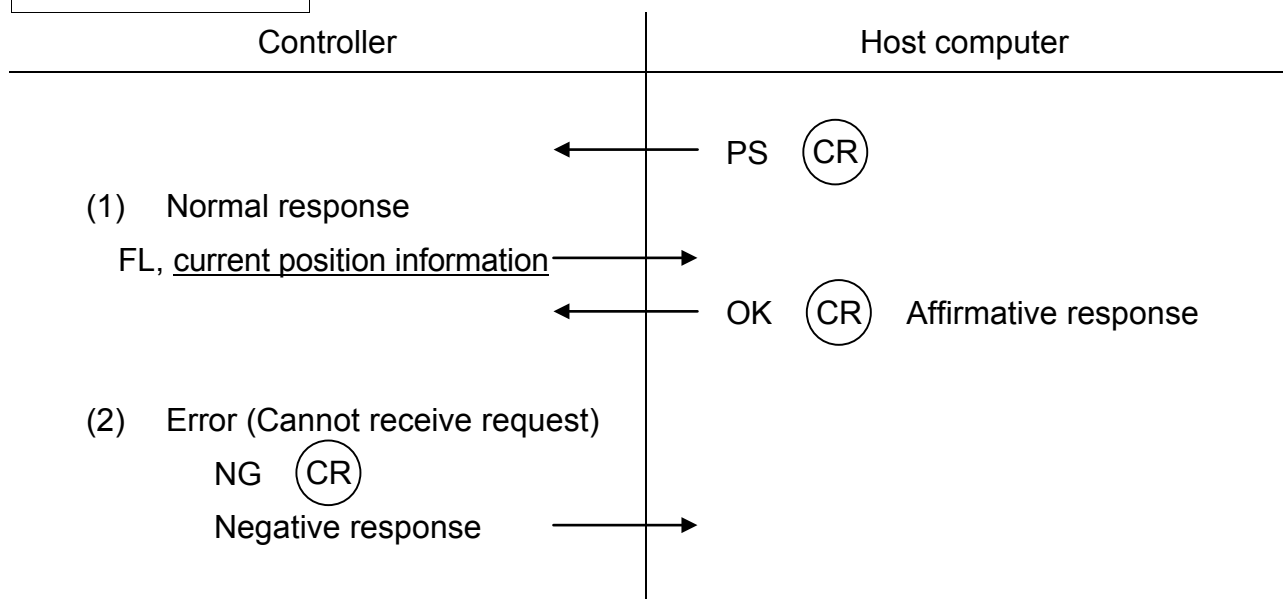
(CR)

Description

The PS command is used to transfer current position information (motion status, execution line number, current position) to the host computer.

For details on the contents of current position information, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Individual current position information acquisition (Host computer → Controller)
PR

Format

PR, coordinate system (CR)

Description

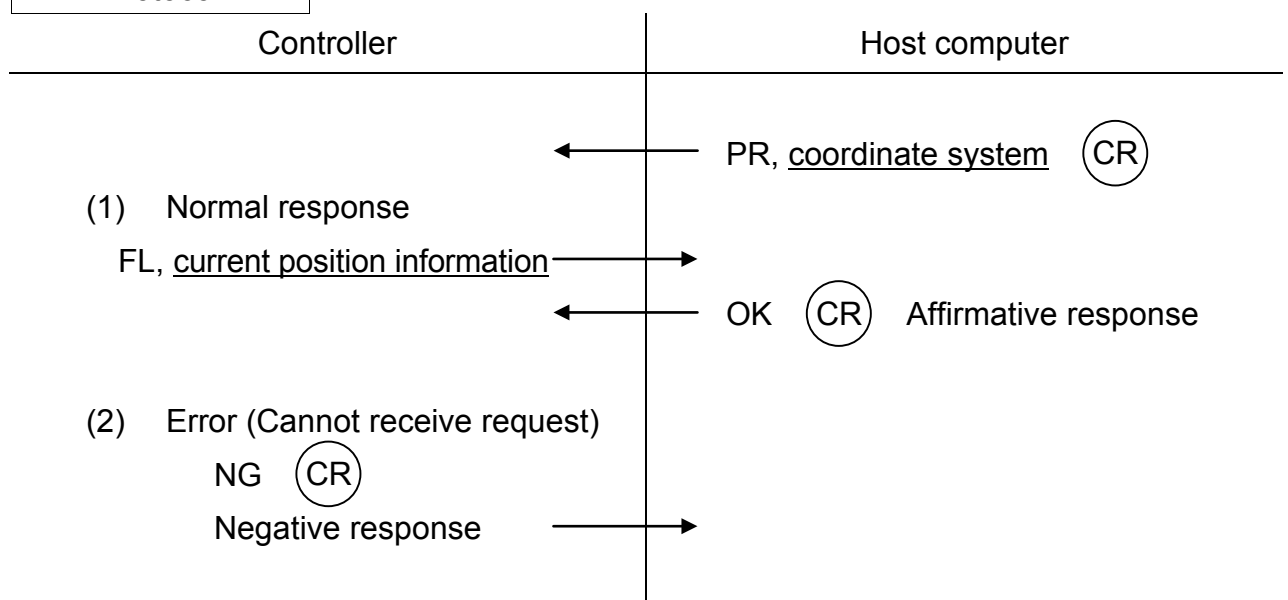
The PR command is used to transfer the current position information for the specified coordinate system to the host computer.

For details on the contents of the current position information for the specified coordinate system, see the FL (File) command.

Specify one of the following numbers for the coordinate system.

- 0: Joint coordinate
- 1: World coordinate
- 2: Work coordinate
- 3: Joint feedback coordinate
- 4: World feedback coordinate
- 5: Work feedback coordinate

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Maintenance information acquisition (Host computer → Controller) MN

Format

MN, maintenance information number



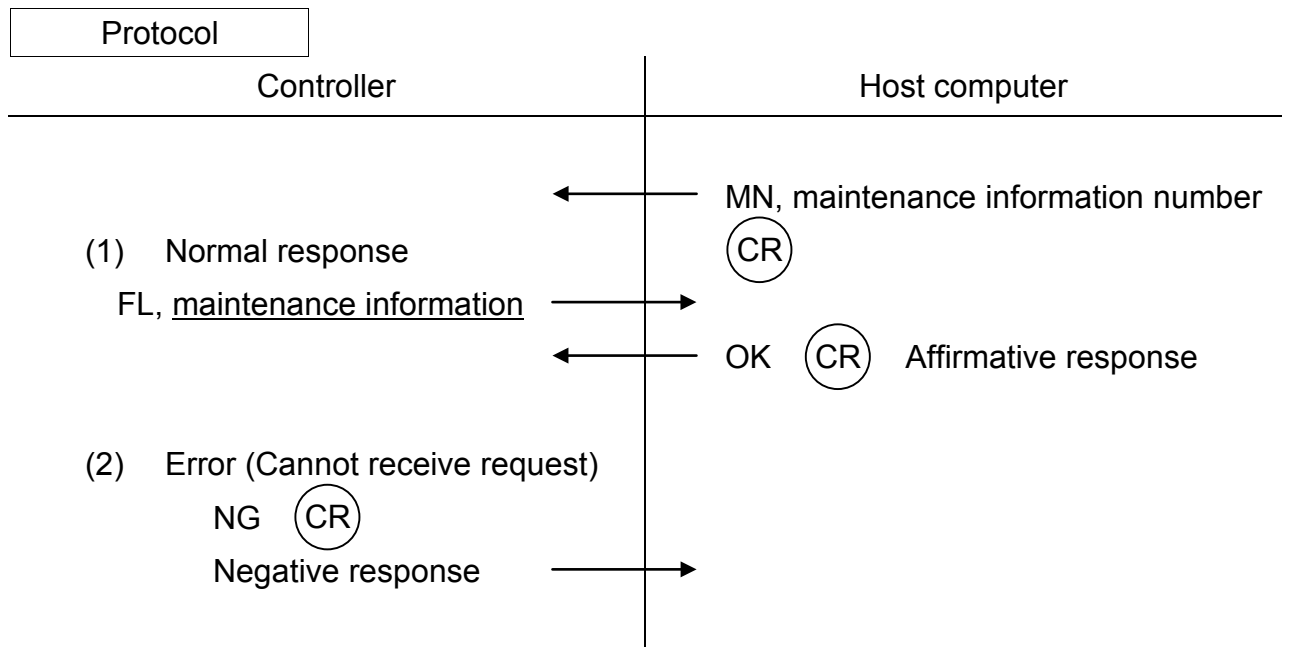
Description

The MN command is used to transfer the specified maintenance information to the host computer.

For details on the contents of the specified maintenance information, see the FL (File) command.

Specify one of the following numbers for the maintenance information number.

- 1: Total rotational amount of motor
- 2: Movement amount of joint axis
- 3: Integrated amount of motor torque
- 4: Integrated amount of motor torque exceeding rating
- 5: Peak torque (cleared by communication request)
- 6: 3-second load factor
- 7: 1-minute load factor
- 8: 15-minute load factor
- 9: 2-hour load factor
- 10: Total time
- 11: Energizing time
- 12: Power OFF time
- 13: Motion time
- 14: Fault time
- 15: Servo information (servo OFF count)
- 16: Total servo ON time
- 17: Peak torque


Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Maintenance information reset (Host computer → Controller) RM

Format

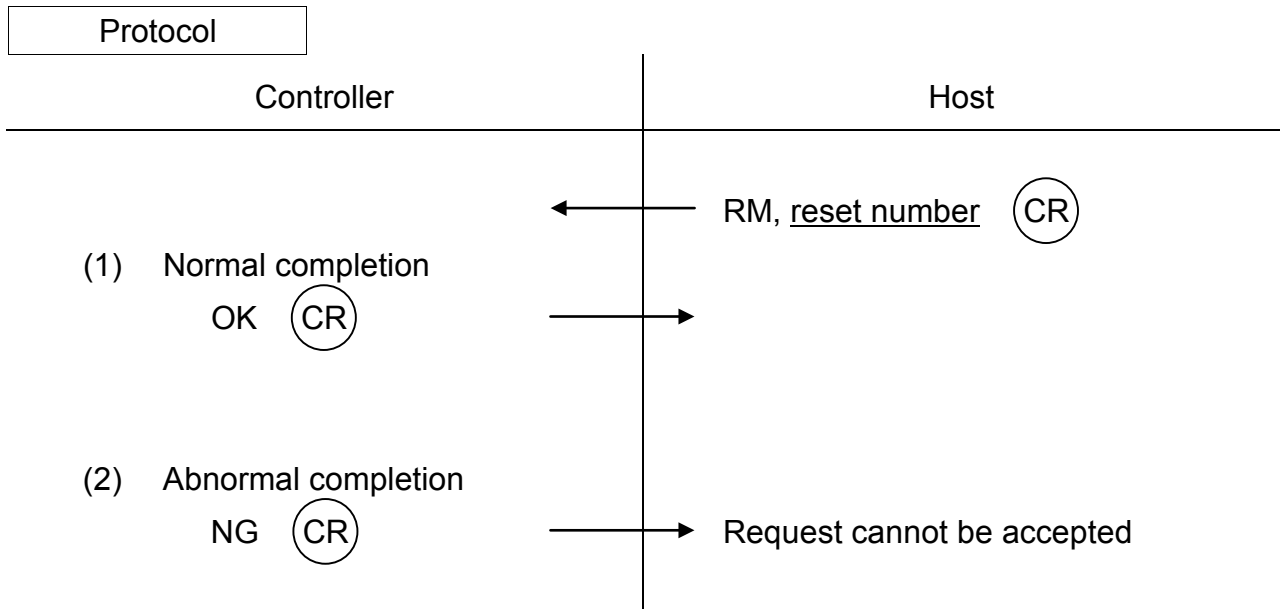
RM, reset number (CR)

Description

The RM command is used to direct the controller to reset the specified maintenance information from the host computer.

Specify one of the following numbers for the reset number.

- 1: Total rotational amount of motor
- 2: Movement amount of joint axis
- 3: Integrated amount of motor torque
- 4: Integrated amount of motor torque exceeding rating
- 5: Peak torque (cleared by communication request)
- 6: 3-second load factor
- 7: 1-minute load factor
- 8: 15-minute load factor
- 9: 2-hour load factor
- 10: Total time
- 11: Energizing time
- 12: Power OFF time
- 13: Motion time
- 14: Fault time
- 15: Servo information (servo OFF count)
- 16: Total servo ON time



: IO information acquisition (Host computer → Controller) IO

Format

IO, version (CR)

Description

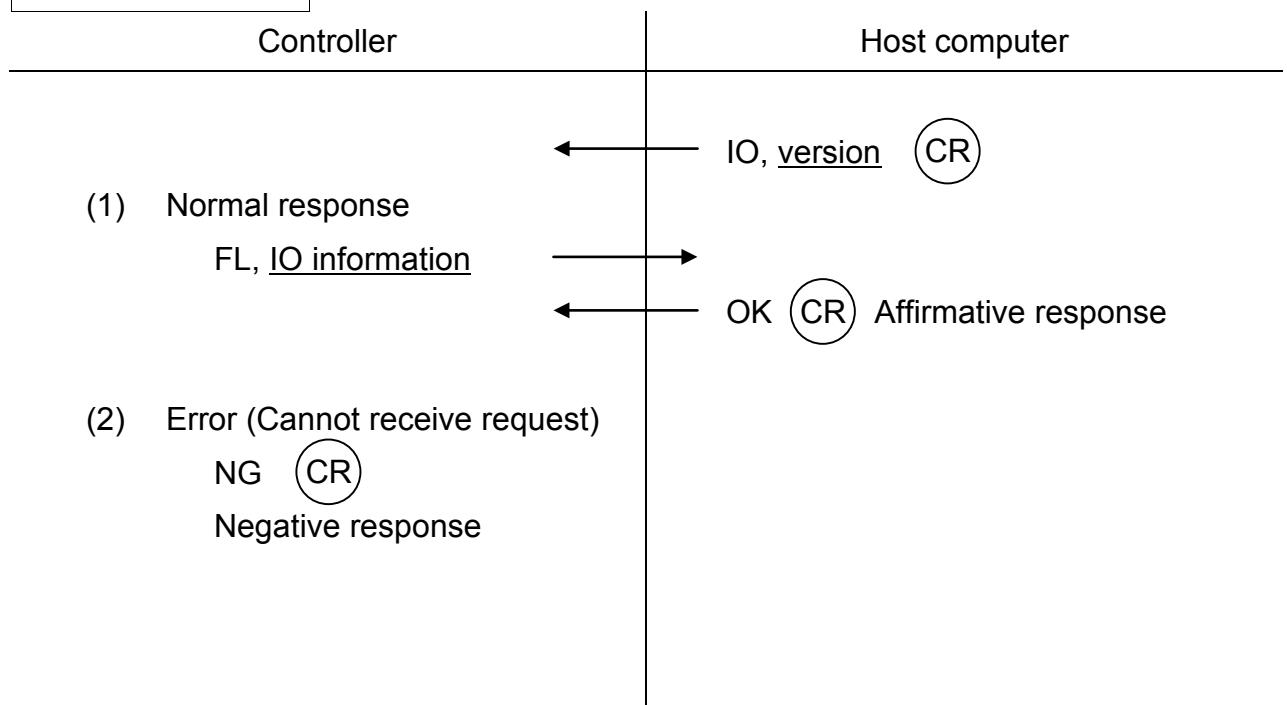
The IO command is used to transfer the specified version of IO information to the host computer.

For details on the contents of IO information, see the FL (File) command.

Specify the following number for the version.

1: TS3000 series robot controller

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Watchdog timer setting (Host computer → Controller) WD

Format

WD, timer valuealarm level (CR)

Description

The WD command is used to set the watchdog timer of the controller from the host computer.

Specify one of the following numbers for the timer value (msec).

0 to 99999

*However, if 0 is specified, it means that watchdog monitor is stopped.

Specify one of the following numbers for the alarm level.

0: No alarm is generated.

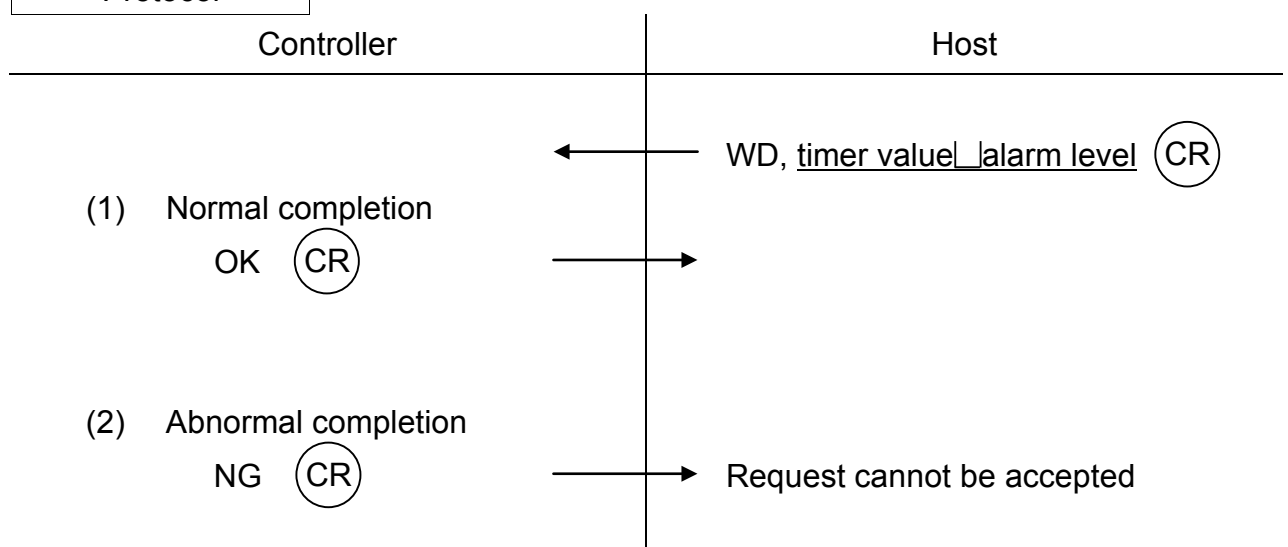
1: Level 1 alarm is generated. (001-200 Host Port Time Out)

2: Level 2 alarm is generated. (002-137 Host Port Time Out)

8: Level 8 alarm is generated. (008-352 Host Port Time Out)

*An alarm is generated during the set timer value (msec) when communication is discontinued.

Protocol



: Alarm history acquisition with message (Host computer → Controller) AH

Format

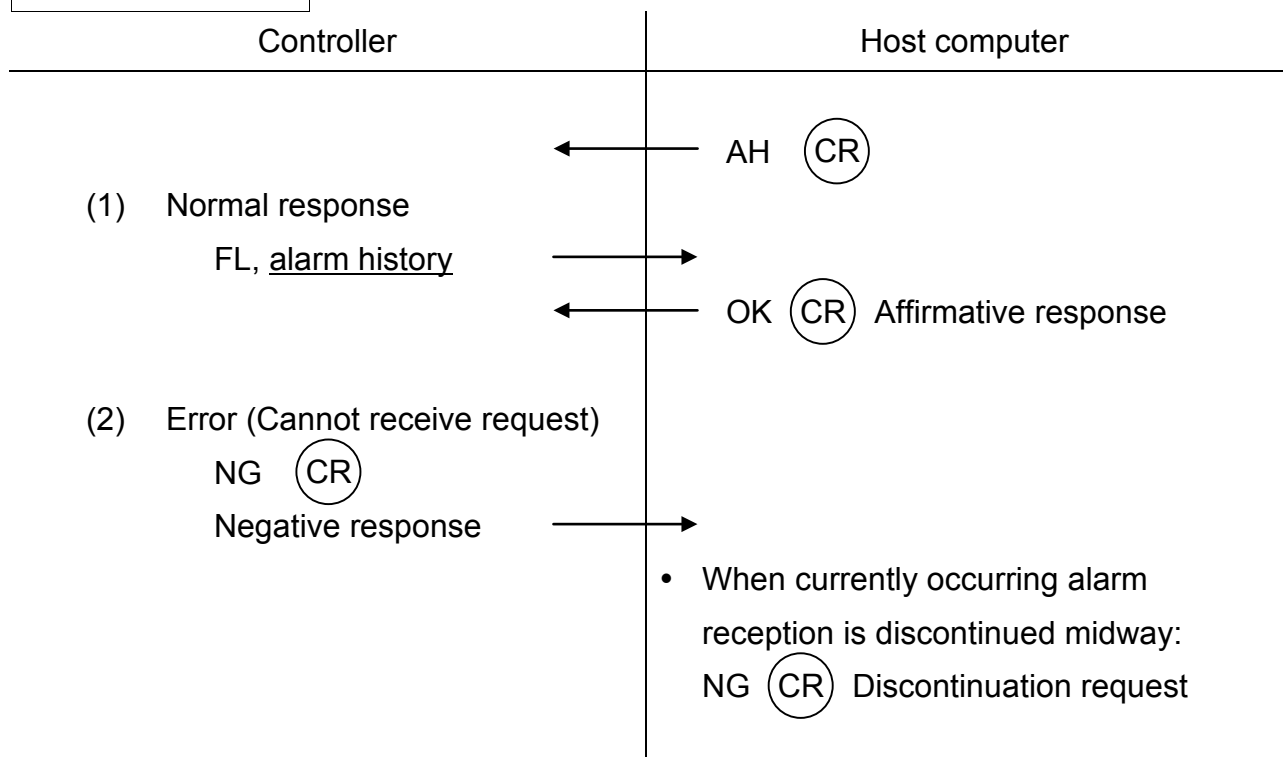
AH (CR)

Function

The AH command is used to transfer the alarm history with message of the controller to the host computer.

For details on the contents of the alarm history, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the currently occurring alarm.

If there are many alarm histories, all alarm histories may not be able to be received by one communication. To receive continuing alarm histories, send an affirmative response OK during each reception.

: Hexadecimal-coding and writing of output signal (Host computer → Controller)

HO

Format

HO, signal namesignal lengthExpression (CR)

Description

The HO command is used to direct the controller to hexadecimal-code and forcibly write up to 32 output signals from the host computer.

Specify one of the following numbers for the signal.

- 1 to 64
- 101 to 164
- 201 to 264
- 301 to 364
- 401 to 464

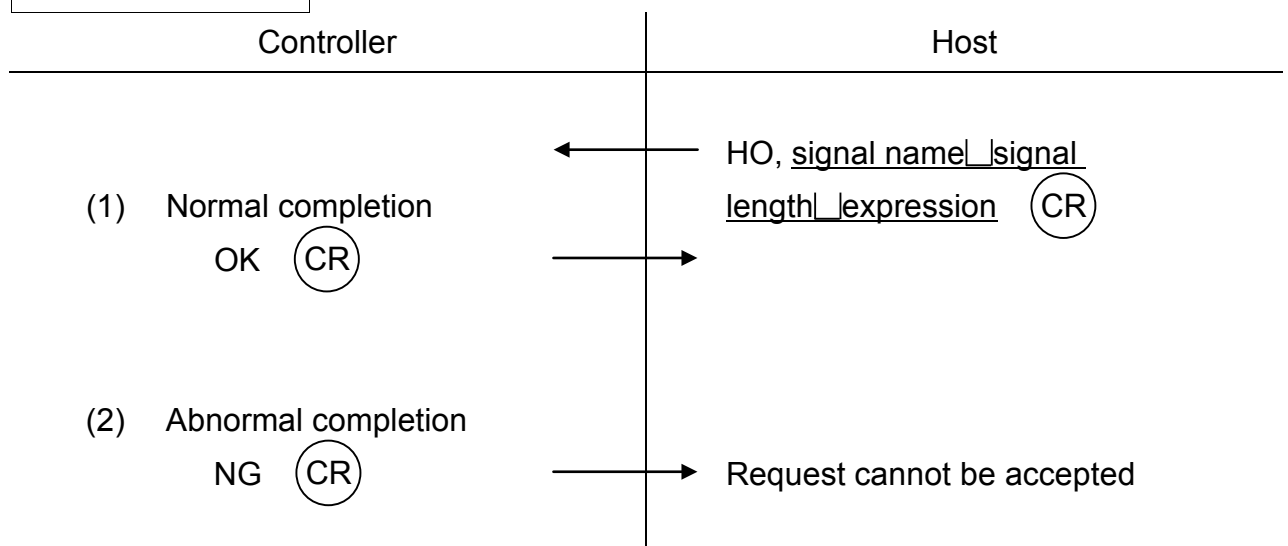
Specify one of the following numbers for the signal length.

- 1 to 32

Specify one of the following numbers for the expression (value).

- Decimal number of more than 0

Protocol



: Robot tip speed information acquisition (Host computer → Controller) VL

Format

VL, speed type (CR)

Description

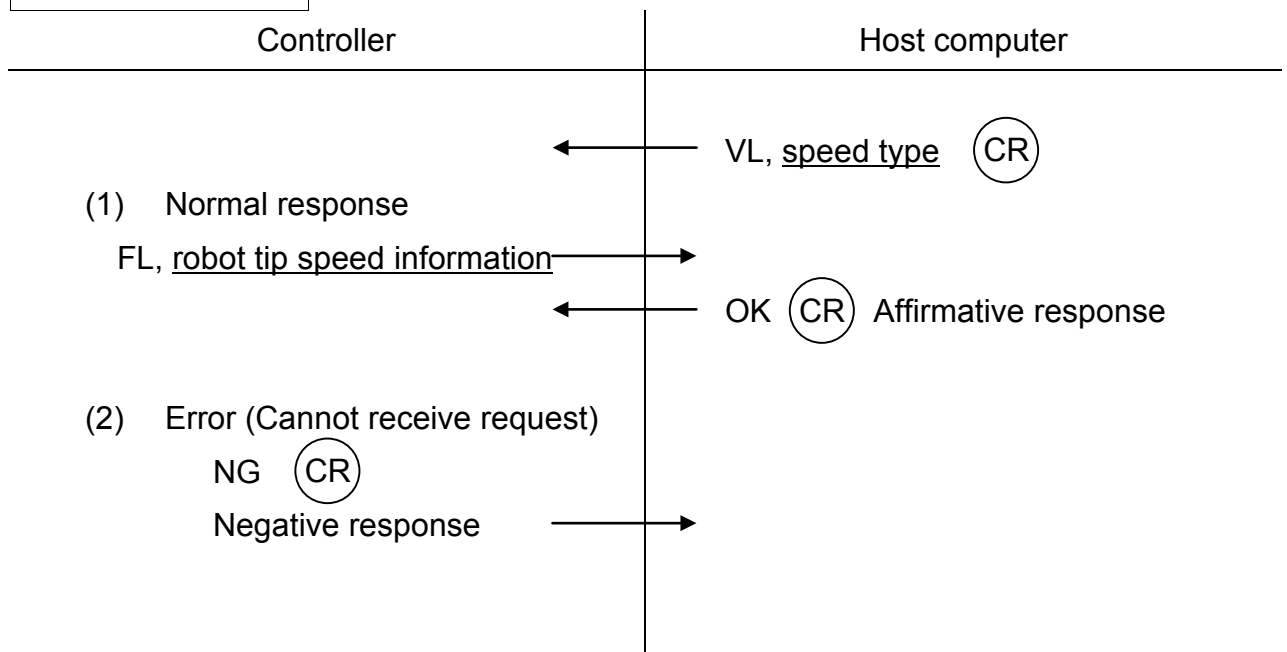
The VL command is used to transfer the specified robot tip speed information to the host computer.

For details on the contents of the specified robot tip speed information, see the FL (File) command.

Specify one of the following numbers for the speed type.

- 0: Robot tip speed (joint)
- 1: Robot tip speed (world)

Protocol



: Motion status request (Host computer → Controller) SM

Format

SM, version (CR)

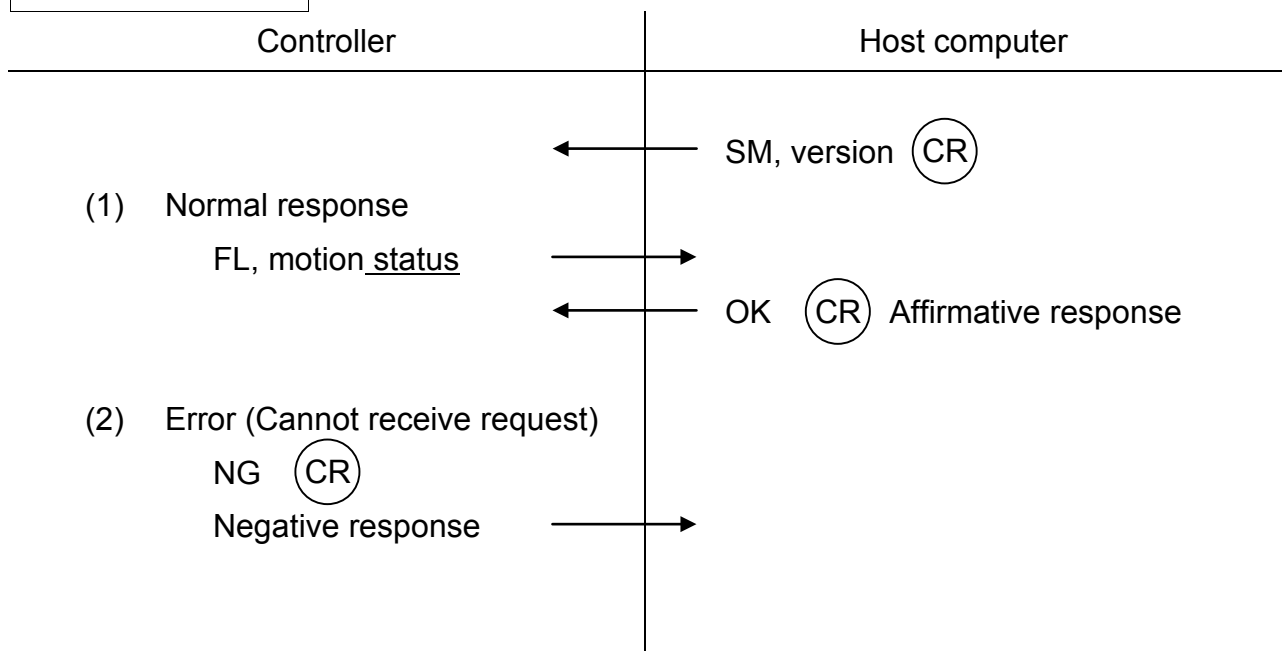
Function

The SM command is used to transfer the motion status of the controller to the host computer.

For details on the contents of the motion status, see the FL (File) command.
Specify the following number for the version.

1: TS3000 series robot controller

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the status.

: Hexadecimal-coding and reading of Input signal (Host computer → Controller)

HI

Format

HI, signal name_signal length (CR)

Description

The HI command is used to hexadecimal-code the input signal for transfer to the host computer.

For details on the contents of the expression, see the FL (File) command.

Specify one of the following numbers for the signal name.

1 to 64

101 to 164

201 to 264

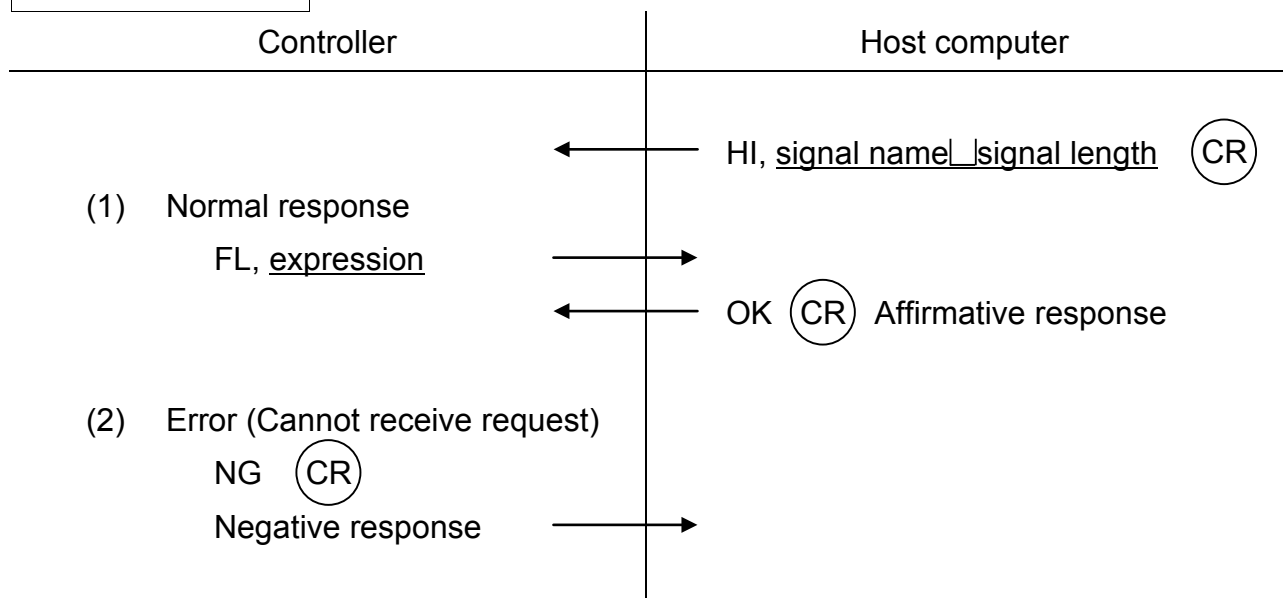
301 to 364

401 to 464

Specify one of the following numbers for the signal length.

1 to 32

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received IO information.

: 3-axis origin position change information acquisition (Host computer → Controller)
ZC

Format

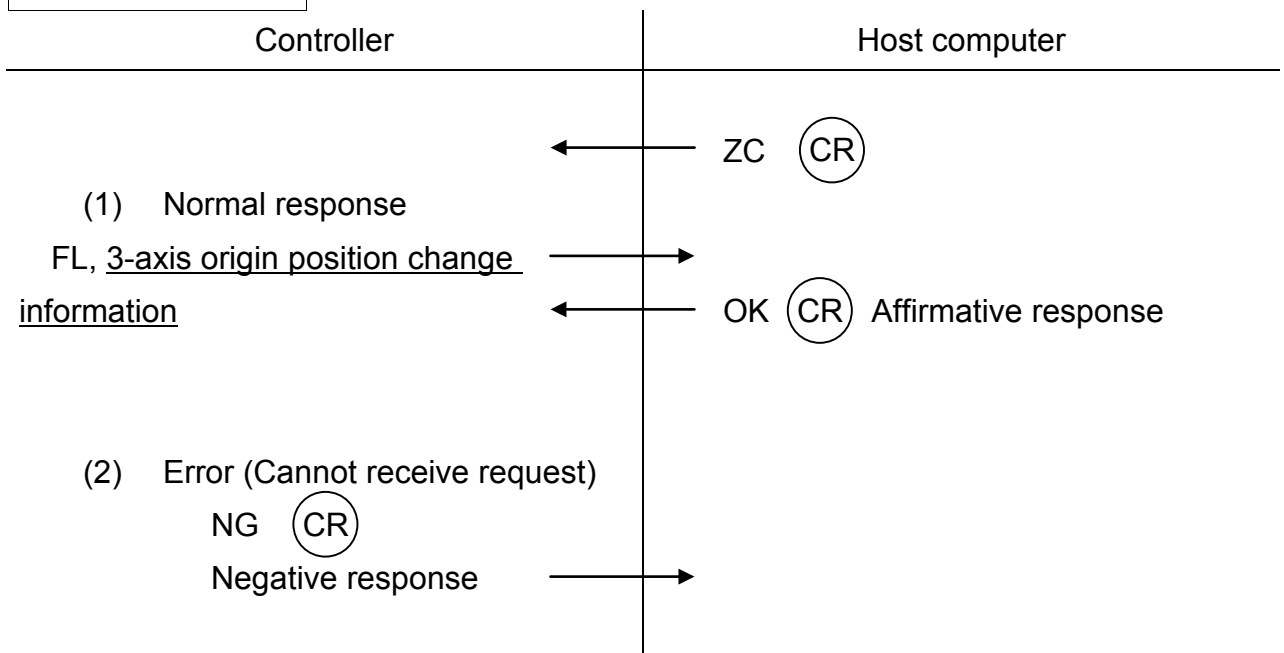
ZC (CR)

Description

The ZC command is used to transfer 3-axis origin position change information to the host computer.

For details on the contents of 3-axis origin position change information, see the FL (File) command.

Protocol



Note

Put in a delay of about 50 msec until the host computer sends an affirmative response OK to the controller after the host computer has received the file contents.

: Controller mode change request (Host computer → Controller) EM

Format

EM, EXTERNAL mode (CR)

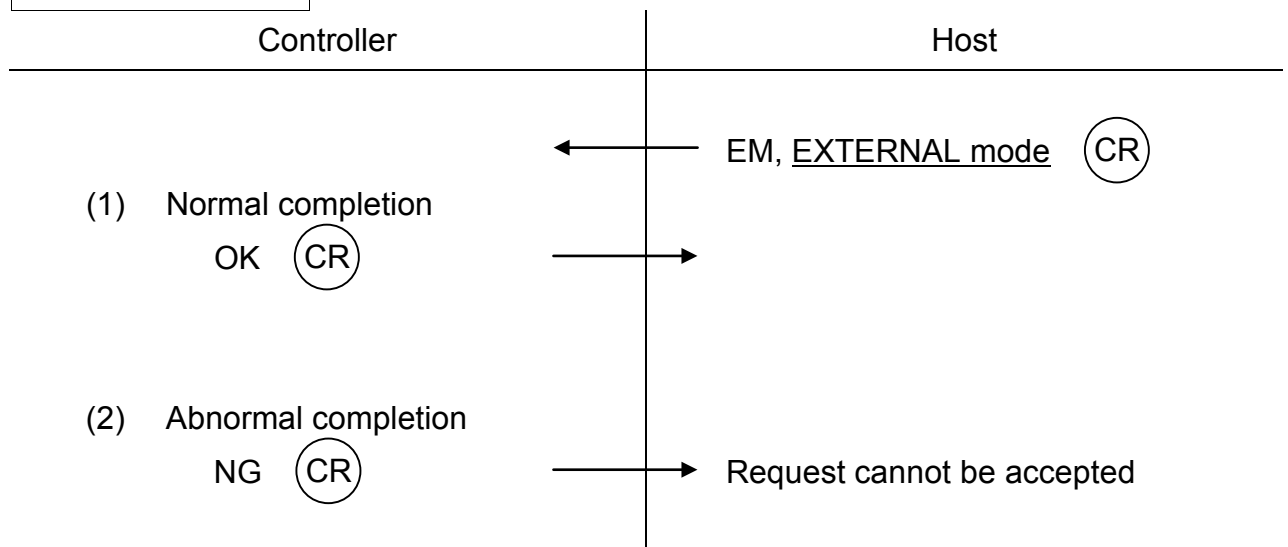
Function

The EM command is used to direct the controller to change the EXTERNAL mode from the host computer.

Specify one of the following numbers for the EXTERNAL mode.

- If the current mode is EXTERNAL RS232C:
 - 0: EXTERNAL SIG
 - 2: EXTERNAL ETHERNET
- If the current mode is EXTERNAL ETHERNET:
 - 0: EXTERNAL SIG
 - 1: EXTERNAL RS232C

Protocol



: File (Host computer ↔ Controller) FL

Format

FL, file contents EOF

- If the data exceeds one text field or 253 bytes, the data is transmitted in two or more texts, as shown below.

FL,	file contents 1	File contents 2	File contents 3	EOF
-----	-----------------	-----------------	-----------------	-----

- EOF (End of File): File end code (0x1A) — 1 byte
The STX, ETX, and SUM (if checksum exists) are attached to each text.

Descriptions

The FL command is used as a response command to the command given beforehand.

1. File types and contents

1.1 Types

Description	Corresponding command
RAM file	UL, DL
File directory	CA
Status	SU
Error history	EU
System total status	SF
Version information	VR
Memory read data	MR
Currently occurring alarm	AC
Current position information	PS
Individual current position information	PR
Maintenance information	MN
IO information	IO
Alarm history	AH

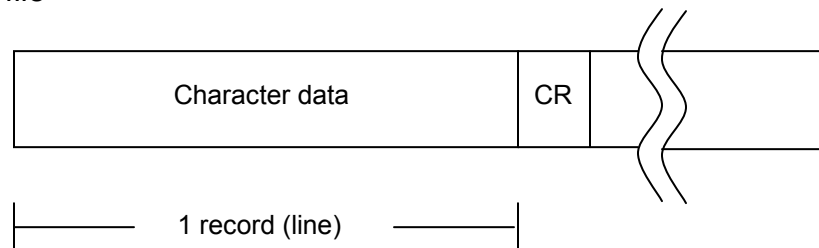
Robot tip speed information	VL
Motion status	SM
Hexadecimal-coding and reading of input signal	HI
3-axis origin position change information	ZC

1.2 File contents

A file is composed of one or more records. Different kinds of files are made up of different kinds of records, each of which is shown below.

Furthermore, in the tables below, the value in the Size column is the maximum size of the data section. "Fixed" means that the length of the data section is fixed; "Variable" means that the length is variable.

(1) RAM file



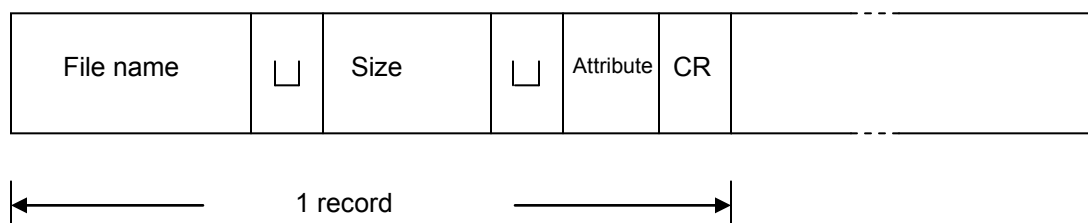
Record structure

No.	Name	Size(byte)	Description
1	Character data	Max 252 (Variable)	ASCII code alphanumeric characters and symbols
2	CR	1 (Fixed)	Record end code (0x0D)

This record corresponds to one line of a program or one line of positional data.

For information on file structure, see Para. 5.4, "RAM Files."

(2) User file directory



Record structure

No.	Name	Size(byte)	Description
1	File name	12(Variable)	File name in alphanumeric characters beginning with an alphabetic character.
2	Size	5(Variable)	Size of file in bytes.
3	Attribute	-(Variable)	Shows the attribution. The attribute is omitted if the attribution is not specified.
4	CR	1(Fixed)	Record end code (0x0D)

(3) Status file

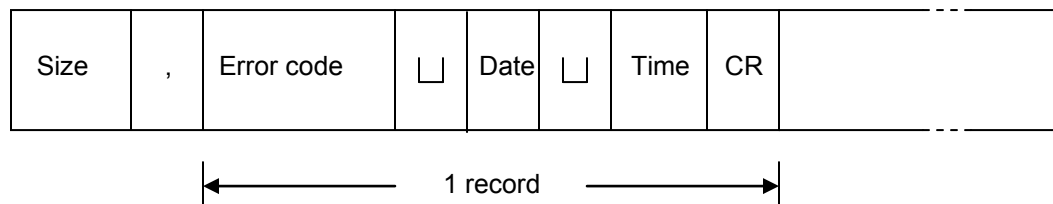
Mode	☐	File name	☐	Speed override	☐	Speed limit	☐
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Machine status	☐	Execution status
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No.	Name	Size(byte)	Description
1	Mode	30 (Variable)	The controller mode is shown in the following format: "MODE: (mode)/(operation mode)" (1) MODE : 5 byte fixed length (2) (mode) : Controller modes "external (sig)" : External Automatic Mode (External operation signal mode) "external (rs232C)" : External Automatic Mode (RS232C mode) "external (ethernet)" : External Automatic Mode (Ethernet mode) "internal" : Internal Operation Mode "teaching" : Test run mode (3) / : 1 byte fixed length (4) (operation mode) : Automatic Operation Modes "step" : Step Operation Mode "continuous" : Continuous Operation Mode "cycle" : Cycle Operation Mode "segment" : Segment operation mode
2	File name	17 (Variable)	File name presently selected is shown in the following format: "FILE : (file name)" (1) FILE : 5 byte fixed length (2) (file name) : Name of the file that has been selected.
3	Speed override	11 (Variable)	The speed override is shown in the following format: "OVRD : (override) %" (1) OVRD : 5-byte fixed length (2) (override) : Percentage for the speed override. (3) % : 1 byte fixed length

No.	Name	Size (byte)	Description
4	Re-served	11 (Variable)	Reserved
5	Machine status	12 (Variable)	The machine status is shown in the following format: "MACHINE : (status)" (1) MACHINE : 8 byte fixed length (2) (status) : Robot status "free" : Status in which machine lock has been released : (Mode in which the robot can move) "lock" : Status in which machine lock is in effect. (Status in which the robot cannot move)
6	Execution status	21 (Variable)	The operation status is shown in the following format: "STATUS : (execution status)" (1) STATUS : 7 byte fixed length (2) (execution status) : Operating state of the robot "running" : In Automatic Operation "stop (reset)" : In Stop (Initialization Mode) (Same status as that for program Reset) "stop (retry)" : In Stop (Retry Mode) (Restart operation from the interrupted movement) "stop (continue)" : In Stop (Continuous Mode) (Continue program from the present step)

(4) Error history file



No.	Name	Size(byte)	Description
1	Size	2 (Variable)	Number (in base 10) of error histories. Placed only at the beginning of the file.
2	Error code	7 (Fixed)	The error code of an error which occurred is shown in the following format. "XXX-YYY" XXX : Main code YYY : Sub code For more information on errors, see the "Operator's Manual."
3	Date	8 (Fixed)	The date on which the error occurred is shown in the following format. "YY-MM-DD" YY : Year (Last two numbers only) MM : Month DD : Day
4	Time	8 (Fixed)	The time on which the error occurred is shown in the following format. "HH:MM:SS" HH : Hours (In 24-hour"military time.") MM : Minutes SS : Seconds
5	CR	1 (Fixed)	Record end code (0x0D)

(5) System total status

The following information is transmitted as the system total status information in the binary notation (250 bytes).

Motion status	I/O information	Current value data
52 bytes	64 bytes	134 bytes

Detailed data of each group are tabled below.

l) Motion status

No.	Name	Size (byte)	Description
1	Servo power status	1	0: OFF, 1: ON
2	EMERGENCY stop switch status	1	0: OFF, 1: ON
3	Motion status	1	0: STOP (RESET) 1: RUN 2: STOP (RETRY) 3: STOP (CONT)
4	SU command request	1	0: Without request, 1: With request
5	Current alarm information	2 × 10 pcs.	Error of level 8: 0 to 367, 960 to 1471 Error of level 4: 368 to 511, 1472 to 1727 Error of level 2: 512 to 735, 1728 to 1983 Error of level 1: 736 to 959, 1984 to 2239 Other than above: No error
6	Program execution line	2	Line number during program execution
7	Program analysis line	2	Line number during program analysis
8	Program execution task	2	Task number during program execution
9	Program analysis task	2	Task number during program analysis
10	Feed hold status	2	0: OFF, 1: ON
11	Guidance coordinate system status	2	0: Joint, 1: Tool, 2: Work, 3: World
12	Guide rate status	2	0: Slow, 1: Mid, 2: Fast
13	Guide mode status	2	0: Jog, 1: Inching, 2: Free
14	Master mode status	2	0: TEACHING mode, 1: INTERNAL mode 2: EXT (SIG) mode, 3: EXT (HOST) mode
15	Dummy	2	Reserved
16	Power ON time	4	Unit: Min.
17	Program run time	4	Unit: Min.
Total		52	

II) I/O information

No.	Name	Size (byte)	Description																																																			
1	General input 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din1</td><td>General input</td></tr><tr><td>1</td><td>Din2</td><td>General input</td></tr><tr><td>2</td><td>Din3</td><td>General input</td></tr><tr><td>3</td><td>Din4</td><td>General input</td></tr><tr><td>4</td><td>Din5</td><td>General input</td></tr><tr><td>5</td><td>Din6</td><td>General input</td></tr><tr><td>6</td><td>Din7</td><td>General input</td></tr><tr><td>7</td><td>Din8</td><td>General input</td></tr><tr><td>8</td><td>Din9</td><td>General input</td></tr><tr><td>9</td><td>Din10</td><td>General input</td></tr><tr><td>10</td><td>Din11</td><td>General input</td></tr><tr><td>11</td><td>Din12</td><td>General input</td></tr><tr><td>12</td><td>Din13</td><td>General input</td></tr><tr><td>13</td><td>Din14</td><td>General input</td></tr><tr><td>14</td><td>Din15</td><td>General input</td></tr><tr><td>15</td><td>Din16</td><td>General input</td></tr></table>	Bit	Line No.	Signal name	0	Din1	General input	1	Din2	General input	2	Din3	General input	3	Din4	General input	4	Din5	General input	5	Din6	General input	6	Din7	General input	7	Din8	General input	8	Din9	General input	9	Din10	General input	10	Din11	General input	11	Din12	General input	12	Din13	General input	13	Din14	General input	14	Din15	General input	15	Din16	General input
Bit	Line No.	Signal name																																																				
0	Din1	General input																																																				
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2	General input 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din17</td><td>General input</td></tr><tr><td>1</td><td>Din18</td><td>General input</td></tr><tr><td>2</td><td>Din19</td><td>General input</td></tr><tr><td>3</td><td>Din20</td><td>General input</td></tr><tr><td>4</td><td>Din21</td><td>General input</td></tr><tr><td>5</td><td>Din22</td><td>General input</td></tr><tr><td>6</td><td>Din23</td><td>General input</td></tr><tr><td>7</td><td>Din24</td><td>General input</td></tr><tr><td>8</td><td>Din25</td><td>General input</td></tr><tr><td>9</td><td>Din26</td><td>General input</td></tr><tr><td>10</td><td>Din27</td><td>General input</td></tr><tr><td>11</td><td>Din28</td><td>General input</td></tr><tr><td>12</td><td>Din29</td><td>General input</td></tr><tr><td>13</td><td>Din30</td><td>General input</td></tr><tr><td>14</td><td>Din31</td><td>General input</td></tr><tr><td>15</td><td>Din32</td><td>General input</td></tr></table>	Bit	Line No.	Signal name	0	Din17	General input	1	Din18	General input	2	Din19	General input	3	Din20	General input	4	Din21	General input	5	Din22	General input	6	Din23	General input	7	Din24	General input	8	Din25	General input	9	Din26	General input	10	Din27	General input	11	Din28	General input	12	Din29	General input	13	Din30	General input	14	Din31	General input	15	Din32	General input
Bit	Line No.	Signal name																																																				
0	Din17	General input																																																				
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14	Din31	General input																																																				
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No.	Name	Size (byte)	Description																																																			
3	General input 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din33</td><td>General input</td></tr><tr><td>1</td><td>Din34</td><td>General input</td></tr><tr><td>2</td><td>Din35</td><td>General input</td></tr><tr><td>3</td><td>Din36</td><td>General input</td></tr><tr><td>4</td><td>Din37</td><td>General input</td></tr><tr><td>5</td><td>Din38</td><td>General input</td></tr><tr><td>6</td><td>Din39</td><td>General input</td></tr><tr><td>7</td><td>Din40</td><td>General input</td></tr><tr><td>8</td><td>Din41</td><td>General input</td></tr><tr><td>9</td><td>Din42</td><td>General input</td></tr><tr><td>10</td><td>Din43</td><td>General input</td></tr><tr><td>11</td><td>Din44</td><td>General input</td></tr><tr><td>12</td><td>Din45</td><td>General input</td></tr><tr><td>13</td><td>Din46</td><td>General input</td></tr><tr><td>14</td><td>Din47</td><td>General input</td></tr><tr><td>15</td><td>Din48</td><td>General input</td></tr></table>	Bit	Line No.	Signal name	0	Din33	General input	1	Din34	General input	2	Din35	General input	3	Din36	General input	4	Din37	General input	5	Din38	General input	6	Din39	General input	7	Din40	General input	8	Din41	General input	9	Din42	General input	10	Din43	General input	11	Din44	General input	12	Din45	General input	13	Din46	General input	14	Din47	General input	15	Din48	General input
Bit	Line No.	Signal name																																																				
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4	General input 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din49</td><td>General input</td></tr><tr><td>1</td><td>Din50</td><td>General input</td></tr><tr><td>2</td><td>Din51</td><td>General input</td></tr><tr><td>3</td><td>Din52</td><td>General input</td></tr><tr><td>4</td><td>Din53</td><td>General input</td></tr><tr><td>5</td><td>Din54</td><td>General input</td></tr><tr><td>6</td><td>Din55</td><td>General input</td></tr><tr><td>7</td><td>Din56</td><td>General input</td></tr><tr><td>8</td><td>Din57</td><td>General input</td></tr><tr><td>9</td><td>Din58</td><td>General input</td></tr><tr><td>10</td><td>Din59</td><td>General input</td></tr><tr><td>11</td><td>Din60</td><td>General input</td></tr><tr><td>12</td><td>Din61</td><td>General input</td></tr><tr><td>13</td><td>Din62</td><td>General input</td></tr><tr><td>14</td><td>Din63</td><td>General input</td></tr><tr><td>15</td><td>Din64</td><td>General input</td></tr></table>	Bit	Line No.	Signal name	0	Din49	General input	1	Din50	General input	2	Din51	General input	3	Din52	General input	4	Din53	General input	5	Din54	General input	6	Din55	General input	7	Din56	General input	8	Din57	General input	9	Din58	General input	10	Din59	General input	11	Din60	General input	12	Din61	General input	13	Din62	General input	14	Din63	General input	15	Din64	General input
Bit	Line No.	Signal name																																																				
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5	Extension input 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din101</td><td>Extension input</td></tr><tr><td>1</td><td>Din102</td><td>Extension input</td></tr><tr><td>2</td><td>Din103</td><td>Extension input</td></tr><tr><td>3</td><td>Din104</td><td>Extension input</td></tr><tr><td>4</td><td>Din105</td><td>Extension input</td></tr><tr><td>5</td><td>Din106</td><td>Extension input</td></tr><tr><td>6</td><td>Din107</td><td>Extension input</td></tr><tr><td>7</td><td>Din108</td><td>Extension input</td></tr><tr><td>8</td><td>Din109</td><td>Extension input</td></tr><tr><td>9</td><td>Din110</td><td>Extension input</td></tr><tr><td>10</td><td>Din111</td><td>Extension input</td></tr><tr><td>11</td><td>Din112</td><td>Extension input</td></tr><tr><td>12</td><td>Din113</td><td>Extension input</td></tr><tr><td>13</td><td>Din114</td><td>Extension input</td></tr><tr><td>14</td><td>Din115</td><td>Extension input</td></tr><tr><td>15</td><td>Din116</td><td>Extension input</td></tr></table>	Bit	Line No.	Signal name	0	Din101	Extension input	1	Din102	Extension input	2	Din103	Extension input	3	Din104	Extension input	4	Din105	Extension input	5	Din106	Extension input	6	Din107	Extension input	7	Din108	Extension input	8	Din109	Extension input	9	Din110	Extension input	10	Din111	Extension input	11	Din112	Extension input	12	Din113	Extension input	13	Din114	Extension input	14	Din115	Extension input	15	Din116	Extension input
Bit	Line No.	Signal name																																																				
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15	Din116	Extension input																																																				
6	Extension input 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din117</td><td>Extension input</td></tr><tr><td>1</td><td>Din118</td><td>Extension input</td></tr><tr><td>2</td><td>Din119</td><td>Extension input</td></tr><tr><td>3</td><td>Din120</td><td>Extension input</td></tr><tr><td>4</td><td>Din121</td><td>Extension input</td></tr><tr><td>5</td><td>Din122</td><td>Extension input</td></tr><tr><td>6</td><td>Din123</td><td>Extension input</td></tr><tr><td>7</td><td>Din124</td><td>Extension input</td></tr><tr><td>8</td><td>Din125</td><td>Extension input</td></tr><tr><td>9</td><td>Din126</td><td>Extension input</td></tr><tr><td>10</td><td>Din127</td><td>Extension input</td></tr><tr><td>11</td><td>Din128</td><td>Extension input</td></tr><tr><td>12</td><td>Din129</td><td>Extension input</td></tr><tr><td>13</td><td>Din130</td><td>Extension input</td></tr><tr><td>14</td><td>Din131</td><td>Extension input</td></tr><tr><td>15</td><td>Din132</td><td>Extension input</td></tr></table>	Bit	Line No.	Signal name	0	Din117	Extension input	1	Din118	Extension input	2	Din119	Extension input	3	Din120	Extension input	4	Din121	Extension input	5	Din122	Extension input	6	Din123	Extension input	7	Din124	Extension input	8	Din125	Extension input	9	Din126	Extension input	10	Din127	Extension input	11	Din128	Extension input	12	Din129	Extension input	13	Din130	Extension input	14	Din131	Extension input	15	Din132	Extension input
Bit	Line No.	Signal name																																																				
0	Din117	Extension input																																																				
1	Din118	Extension input																																																				
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9	Din126	Extension input																																																				
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12	Din129	Extension input																																																				
13	Din130	Extension input																																																				
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15	Din132	Extension input																																																				

No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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10	System input 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din217</td><td>Alarm of level 8</td></tr><tr><td>1</td><td>Din218</td><td>Alarm of level 8</td></tr><tr><td>2</td><td>Din219</td><td>Alarm of level 8</td></tr><tr><td>3</td><td>Din220</td><td>Alarm of level 8</td></tr><tr><td>4</td><td>Din221</td><td>Alarm of level 4</td></tr><tr><td>5</td><td>Din222</td><td>Alarm of level 4</td></tr><tr><td>6</td><td>Din223</td><td>Alarm of level 4</td></tr><tr><td>7</td><td>Din224</td><td>Alarm of level 4</td></tr><tr><td>8</td><td>Din225</td><td>Alarm of level 2</td></tr><tr><td>9</td><td>Din226</td><td>Alarm of level 2</td></tr><tr><td>10</td><td>Din227</td><td>Alarm of level 2</td></tr><tr><td>11</td><td>Din228</td><td>Alarm of level 2</td></tr><tr><td>12</td><td>Din229</td><td>Alarm of level 1</td></tr><tr><td>13</td><td>Din230</td><td>Alarm of level 1</td></tr><tr><td>14</td><td>Din231</td><td>Alarm of level 1</td></tr><tr><td>15</td><td>Din232</td><td>Alarm of level 1</td></tr></table>	Bit	Line No.	Signal name	0	Din217	Alarm of level 8	1	Din218	Alarm of level 8	2	Din219	Alarm of level 8	3	Din220	Alarm of level 8	4	Din221	Alarm of level 4	5	Din222	Alarm of level 4	6	Din223	Alarm of level 4	7	Din224	Alarm of level 4	8	Din225	Alarm of level 2	9	Din226	Alarm of level 2	10	Din227	Alarm of level 2	11	Din228	Alarm of level 2	12	Din229	Alarm of level 1	13	Din230	Alarm of level 1	14	Din231	Alarm of level 1	15	Din232	Alarm of level 1
Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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17	General output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout1</td><td>General output</td></tr><tr><td>1</td><td>Dout2</td><td>General output</td></tr><tr><td>2</td><td>Dout3</td><td>General output</td></tr><tr><td>3</td><td>Dout4</td><td>General output</td></tr><tr><td>4</td><td>Dout5</td><td>General output</td></tr><tr><td>5</td><td>Dout6</td><td>General output</td></tr><tr><td>6</td><td>Dout7</td><td>General output</td></tr><tr><td>7</td><td>Dout8</td><td>General output</td></tr><tr><td>8</td><td>Dout9</td><td>General output</td></tr><tr><td>9</td><td>Dout10</td><td>General output</td></tr><tr><td>10</td><td>Dout11</td><td>General output</td></tr><tr><td>11</td><td>Dout12</td><td>General output</td></tr><tr><td>12</td><td>Dout13</td><td>General output</td></tr><tr><td>13</td><td>Dout14</td><td>General output</td></tr><tr><td>14</td><td>Dout15</td><td>General output</td></tr><tr><td>15</td><td>Dout16</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout1	General output	1	Dout2	General output	2	Dout3	General output	3	Dout4	General output	4	Dout5	General output	5	Dout6	General output	6	Dout7	General output	7	Dout8	General output	8	Dout9	General output	9	Dout10	General output	10	Dout11	General output	11	Dout12	General output	12	Dout13	General output	13	Dout14	General output	14	Dout15	General output	15	Dout16	General output
			Bit	Line No.	Signal name																																																	
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			1	Dout2	General output																																																	
			2	Dout3	General output																																																	
			3	Dout4	General output																																																	
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			7	Dout8	General output																																																	
			8	Dout9	General output																																																	
			9	Dout10	General output																																																	
			10	Dout11	General output																																																	
			11	Dout12	General output																																																	
			12	Dout13	General output																																																	
			13	Dout14	General output																																																	
			14	Dout15	General output																																																	
15	Dout16	General output																																																				
18	General output 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout17</td><td>General output</td></tr><tr><td>1</td><td>Dout18</td><td>General output</td></tr><tr><td>2</td><td>Dout19</td><td>General output</td></tr><tr><td>3</td><td>Dout20</td><td>General output</td></tr><tr><td>4</td><td>Dout21</td><td>General output</td></tr><tr><td>5</td><td>Dout22</td><td>General output</td></tr><tr><td>6</td><td>Dout23</td><td>General output</td></tr><tr><td>7</td><td>Dout24</td><td>General output</td></tr><tr><td>8</td><td>Dout25</td><td>General output</td></tr><tr><td>9</td><td>Dout26</td><td>General output</td></tr><tr><td>10</td><td>Dout27</td><td>General output</td></tr><tr><td>11</td><td>Dout28</td><td>General output</td></tr><tr><td>12</td><td>Dout29</td><td>General output</td></tr><tr><td>13</td><td>Dout30</td><td>General output</td></tr><tr><td>14</td><td>Dout31</td><td>General output</td></tr><tr><td>15</td><td>Dout32</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout17	General output	1	Dout18	General output	2	Dout19	General output	3	Dout20	General output	4	Dout21	General output	5	Dout22	General output	6	Dout23	General output	7	Dout24	General output	8	Dout25	General output	9	Dout26	General output	10	Dout27	General output	11	Dout28	General output	12	Dout29	General output	13	Dout30	General output	14	Dout31	General output	15	Dout32	General output
			Bit	Line No.	Signal name																																																	
			0	Dout17	General output																																																	
			1	Dout18	General output																																																	
			2	Dout19	General output																																																	
			3	Dout20	General output																																																	
			4	Dout21	General output																																																	
			5	Dout22	General output																																																	
			6	Dout23	General output																																																	
			7	Dout24	General output																																																	
			8	Dout25	General output																																																	
			9	Dout26	General output																																																	
			10	Dout27	General output																																																	
			11	Dout28	General output																																																	
			12	Dout29	General output																																																	
			13	Dout30	General output																																																	
			14	Dout31	General output																																																	
15	Dout32	General output																																																				

No.	Name	Size (byte)	Description																																																			
19	General output 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout33</td><td>General output</td></tr><tr><td>1</td><td>Dout34</td><td>General output</td></tr><tr><td>2</td><td>Dout35</td><td>General output</td></tr><tr><td>3</td><td>Dout36</td><td>General output</td></tr><tr><td>4</td><td>Dout37</td><td>General output</td></tr><tr><td>5</td><td>Dout38</td><td>General output</td></tr><tr><td>6</td><td>Dout39</td><td>General output</td></tr><tr><td>7</td><td>Dout40</td><td>General output</td></tr><tr><td>8</td><td>Dout41</td><td>General output</td></tr><tr><td>9</td><td>Dout42</td><td>General output</td></tr><tr><td>10</td><td>Dout43</td><td>General output</td></tr><tr><td>11</td><td>Dout44</td><td>General output</td></tr><tr><td>12</td><td>Dout45</td><td>General output</td></tr><tr><td>13</td><td>Dout46</td><td>General output</td></tr><tr><td>14</td><td>Dout47</td><td>General output</td></tr><tr><td>15</td><td>Dout48</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout33	General output	1	Dout34	General output	2	Dout35	General output	3	Dout36	General output	4	Dout37	General output	5	Dout38	General output	6	Dout39	General output	7	Dout40	General output	8	Dout41	General output	9	Dout42	General output	10	Dout43	General output	11	Dout44	General output	12	Dout45	General output	13	Dout46	General output	14	Dout47	General output	15	Dout48	General output
			Bit	Line No.	Signal name																																																	
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			1	Dout34	General output																																																	
			2	Dout35	General output																																																	
			3	Dout36	General output																																																	
			4	Dout37	General output																																																	
			5	Dout38	General output																																																	
			6	Dout39	General output																																																	
			7	Dout40	General output																																																	
			8	Dout41	General output																																																	
			9	Dout42	General output																																																	
			10	Dout43	General output																																																	
			11	Dout44	General output																																																	
			12	Dout45	General output																																																	
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			14	Dout47	General output																																																	
15	Dout48	General output																																																				
20	General output 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout49</td><td>General output</td></tr><tr><td>1</td><td>Dout50</td><td>General output</td></tr><tr><td>2</td><td>Dout51</td><td>General output</td></tr><tr><td>3</td><td>Dout52</td><td>General output</td></tr><tr><td>4</td><td>Dout53</td><td>General output</td></tr><tr><td>5</td><td>Dout54</td><td>General output</td></tr><tr><td>6</td><td>Dout55</td><td>General output</td></tr><tr><td>7</td><td>Dout56</td><td>General output</td></tr><tr><td>8</td><td>Dout57</td><td>General output</td></tr><tr><td>9</td><td>Dout58</td><td>General output</td></tr><tr><td>10</td><td>Dout59</td><td>General output</td></tr><tr><td>11</td><td>Dout60</td><td>General output</td></tr><tr><td>12</td><td>Dout61</td><td>General output</td></tr><tr><td>13</td><td>Dout62</td><td>General output</td></tr><tr><td>14</td><td>Dout63</td><td>General output</td></tr><tr><td>15</td><td>Dout64</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout49	General output	1	Dout50	General output	2	Dout51	General output	3	Dout52	General output	4	Dout53	General output	5	Dout54	General output	6	Dout55	General output	7	Dout56	General output	8	Dout57	General output	9	Dout58	General output	10	Dout59	General output	11	Dout60	General output	12	Dout61	General output	13	Dout62	General output	14	Dout63	General output	15	Dout64	General output
			Bit	Line No.	Signal name																																																	
			0	Dout49	General output																																																	
			1	Dout50	General output																																																	
			2	Dout51	General output																																																	
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			9	Dout58	General output																																																	
			10	Dout59	General output																																																	
			11	Dout60	General output																																																	
			12	Dout61	General output																																																	
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			14	Dout63	General output																																																	
15	Dout64	General output																																																				

No.	Name	Size (byte)	Description																																																			
21	Extension output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout101</td><td>Extension output</td></tr><tr><td>1</td><td>Dout102</td><td>Extension output</td></tr><tr><td>2</td><td>Dout103</td><td>Extension output</td></tr><tr><td>3</td><td>Dout104</td><td>Extension output</td></tr><tr><td>4</td><td>Dout105</td><td>Extension output</td></tr><tr><td>5</td><td>Dout106</td><td>Extension output</td></tr><tr><td>6</td><td>Dout107</td><td>Extension output</td></tr><tr><td>7</td><td>Dout108</td><td>Extension output</td></tr><tr><td>8</td><td>Dout109</td><td>Extension output</td></tr><tr><td>9</td><td>Dout110</td><td>Extension output</td></tr><tr><td>10</td><td>Dout111</td><td>Extension output</td></tr><tr><td>11</td><td>Dout112</td><td>Extension output</td></tr><tr><td>12</td><td>Dout113</td><td>Extension output</td></tr><tr><td>13</td><td>Dout114</td><td>Extension output</td></tr><tr><td>14</td><td>Dout115</td><td>Extension output</td></tr><tr><td>15</td><td>Dout116</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout101	Extension output	1	Dout102	Extension output	2	Dout103	Extension output	3	Dout104	Extension output	4	Dout105	Extension output	5	Dout106	Extension output	6	Dout107	Extension output	7	Dout108	Extension output	8	Dout109	Extension output	9	Dout110	Extension output	10	Dout111	Extension output	11	Dout112	Extension output	12	Dout113	Extension output	13	Dout114	Extension output	14	Dout115	Extension output	15	Dout116	Extension output
Bit	Line No.	Signal name																																																				
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1	Dout102	Extension output																																																				
2	Dout103	Extension output																																																				
3	Dout104	Extension output																																																				
4	Dout105	Extension output																																																				
5	Dout106	Extension output																																																				
6	Dout107	Extension output																																																				
7	Dout108	Extension output																																																				
8	Dout109	Extension output																																																				
9	Dout110	Extension output																																																				
10	Dout111	Extension output																																																				
11	Dout112	Extension output																																																				
12	Dout113	Extension output																																																				
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14	Dout115	Extension output																																																				
15	Dout116	Extension output																																																				
22	Extension output 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout117</td><td>Extension output</td></tr><tr><td>1</td><td>Dout118</td><td>Extension output</td></tr><tr><td>2</td><td>Dout119</td><td>Extension output</td></tr><tr><td>3</td><td>Dout120</td><td>Extension output</td></tr><tr><td>4</td><td>Dout121</td><td>Extension output</td></tr><tr><td>5</td><td>Dout122</td><td>Extension output</td></tr><tr><td>6</td><td>Dout123</td><td>Extension output</td></tr><tr><td>7</td><td>Dout124</td><td>Extension output</td></tr><tr><td>8</td><td>Dout125</td><td>Extension output</td></tr><tr><td>9</td><td>Dout126</td><td>Extension output</td></tr><tr><td>10</td><td>Dout127</td><td>Extension output</td></tr><tr><td>11</td><td>Dout128</td><td>Extension output</td></tr><tr><td>12</td><td>Dout129</td><td>Extension output</td></tr><tr><td>13</td><td>Dout130</td><td>Extension output</td></tr><tr><td>14</td><td>Dout131</td><td>Extension output</td></tr><tr><td>15</td><td>Dout132</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout117	Extension output	1	Dout118	Extension output	2	Dout119	Extension output	3	Dout120	Extension output	4	Dout121	Extension output	5	Dout122	Extension output	6	Dout123	Extension output	7	Dout124	Extension output	8	Dout125	Extension output	9	Dout126	Extension output	10	Dout127	Extension output	11	Dout128	Extension output	12	Dout129	Extension output	13	Dout130	Extension output	14	Dout131	Extension output	15	Dout132	Extension output
Bit	Line No.	Signal name																																																				
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1	Dout118	Extension output																																																				
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7	Dout124	Extension output																																																				
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9	Dout126	Extension output																																																				
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12	Dout129	Extension output																																																				
13	Dout130	Extension output																																																				
14	Dout131	Extension output																																																				
15	Dout132	Extension output																																																				

No.	Name	Size (byte)	Description																																																			
23	Extension output 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout133</td><td>Extension output</td></tr><tr><td>1</td><td>Dout134</td><td>Extension output</td></tr><tr><td>2</td><td>Dout135</td><td>Extension output</td></tr><tr><td>3</td><td>Dout136</td><td>Extension output</td></tr><tr><td>4</td><td>Dout137</td><td>Extension output</td></tr><tr><td>5</td><td>Dout138</td><td>Extension output</td></tr><tr><td>6</td><td>Dout139</td><td>Extension output</td></tr><tr><td>7</td><td>Dout140</td><td>Extension output</td></tr><tr><td>8</td><td>Dout141</td><td>Extension output</td></tr><tr><td>9</td><td>Dout142</td><td>Extension output</td></tr><tr><td>10</td><td>Dout143</td><td>Extension output</td></tr><tr><td>11</td><td>Dout144</td><td>Extension output</td></tr><tr><td>12</td><td>Dout145</td><td>Extension output</td></tr><tr><td>13</td><td>Dout146</td><td>Extension output</td></tr><tr><td>14</td><td>Dout147</td><td>Extension output</td></tr><tr><td>15</td><td>Dout148</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout133	Extension output	1	Dout134	Extension output	2	Dout135	Extension output	3	Dout136	Extension output	4	Dout137	Extension output	5	Dout138	Extension output	6	Dout139	Extension output	7	Dout140	Extension output	8	Dout141	Extension output	9	Dout142	Extension output	10	Dout143	Extension output	11	Dout144	Extension output	12	Dout145	Extension output	13	Dout146	Extension output	14	Dout147	Extension output	15	Dout148	Extension output
Bit	Line No.	Signal name																																																				
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14	Dout147	Extension output																																																				
15	Dout148	Extension output																																																				
24	Extension output 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout149</td><td>Extension output</td></tr><tr><td>1</td><td>Dout150</td><td>Extension output</td></tr><tr><td>2</td><td>Dout151</td><td>Extension output</td></tr><tr><td>3</td><td>Dout152</td><td>Extension output</td></tr><tr><td>4</td><td>Dout153</td><td>Extension output</td></tr><tr><td>5</td><td>Dout154</td><td>Extension output</td></tr><tr><td>6</td><td>Dout155</td><td>Extension output</td></tr><tr><td>7</td><td>Dout156</td><td>Extension output</td></tr><tr><td>8</td><td>Dout157</td><td>Extension output</td></tr><tr><td>9</td><td>Dout158</td><td>Extension output</td></tr><tr><td>10</td><td>Dout159</td><td>Extension output</td></tr><tr><td>11</td><td>Dout160</td><td>Extension output</td></tr><tr><td>12</td><td>Dout161</td><td>Extension output</td></tr><tr><td>13</td><td>Dout162</td><td>Extension output</td></tr><tr><td>14</td><td>Dout163</td><td>Extension output</td></tr><tr><td>15</td><td>Dout164</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout149	Extension output	1	Dout150	Extension output	2	Dout151	Extension output	3	Dout152	Extension output	4	Dout153	Extension output	5	Dout154	Extension output	6	Dout155	Extension output	7	Dout156	Extension output	8	Dout157	Extension output	9	Dout158	Extension output	10	Dout159	Extension output	11	Dout160	Extension output	12	Dout161	Extension output	13	Dout162	Extension output	14	Dout163	Extension output	15	Dout164	Extension output
Bit	Line No.	Signal name																																																				
0	Dout149	Extension output																																																				
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13	Dout162	Extension output																																																				
14	Dout163	Extension output																																																				
15	Dout164	Extension output																																																				

No.	Name	Size (byte)	Description																																																			
25	System output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout201</td><td>Hand output</td></tr><tr><td>1</td><td>Dout202</td><td>Hand output</td></tr><tr><td>2</td><td>Dout203</td><td>Hand output</td></tr><tr><td>3</td><td>Dout204</td><td>Hand output</td></tr><tr><td>4</td><td>Dout205</td><td>Hand output</td></tr><tr><td>5</td><td>Dout206</td><td>Hand output</td></tr><tr><td>6</td><td>Dout207</td><td>Hand output</td></tr><tr><td>7</td><td>Dout208</td><td>Hand output</td></tr><tr><td>8</td><td>Dout209</td><td></td></tr><tr><td>9</td><td>Dout210</td><td></td></tr><tr><td>10</td><td>Dout211</td><td></td></tr><tr><td>11</td><td>Dout212</td><td></td></tr><tr><td>12</td><td>Dout213</td><td></td></tr><tr><td>13</td><td>Dout214</td><td></td></tr><tr><td>14</td><td>Dout215</td><td></td></tr><tr><td>15</td><td>Dout216</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout201	Hand output	1	Dout202	Hand output	2	Dout203	Hand output	3	Dout204	Hand output	4	Dout205	Hand output	5	Dout206	Hand output	6	Dout207	Hand output	7	Dout208	Hand output	8	Dout209		9	Dout210		10	Dout211		11	Dout212		12	Dout213		13	Dout214		14	Dout215		15	Dout216	
			Bit	Line No.	Signal name																																																	
			0	Dout201	Hand output																																																	
			1	Dout202	Hand output																																																	
			2	Dout203	Hand output																																																	
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			14	Dout215																																																		
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26	System output 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout217</td><td>Seq. parameter</td></tr><tr><td>1</td><td>Dout218</td><td>Seq. parameter</td></tr><tr><td>2</td><td>Dout219</td><td>Seq. parameter</td></tr><tr><td>3</td><td>Dout220</td><td>Seq. parameter</td></tr><tr><td>4</td><td>Dout221</td><td>Seq. parameter</td></tr><tr><td>5</td><td>Dout222</td><td>Seq. parameter</td></tr><tr><td>6</td><td>Dout223</td><td>Seq. parameter</td></tr><tr><td>7</td><td>Dout224</td><td>Seq. parameter</td></tr><tr><td>8</td><td>Dout225</td><td></td></tr><tr><td>9</td><td>Dout226</td><td></td></tr><tr><td>10</td><td>Dout227</td><td></td></tr><tr><td>11</td><td>Dout228</td><td></td></tr><tr><td>12</td><td>Dout229</td><td></td></tr><tr><td>13</td><td>Dout230</td><td></td></tr><tr><td>14</td><td>Dout231</td><td></td></tr><tr><td>15</td><td>Dout232</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout217	Seq. parameter	1	Dout218	Seq. parameter	2	Dout219	Seq. parameter	3	Dout220	Seq. parameter	4	Dout221	Seq. parameter	5	Dout222	Seq. parameter	6	Dout223	Seq. parameter	7	Dout224	Seq. parameter	8	Dout225		9	Dout226		10	Dout227		11	Dout228		12	Dout229		13	Dout230		14	Dout231		15	Dout232	
			Bit	Line No.	Signal name																																																	
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No.	Name	Size (byte)	Description																																																			
27	System output 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout233</td><td></td></tr><tr><td>1</td><td>Dout234</td><td></td></tr><tr><td>2</td><td>Dout235</td><td></td></tr><tr><td>3</td><td>Dout236</td><td></td></tr><tr><td>4</td><td>Dout237</td><td></td></tr><tr><td>5</td><td>Dout238</td><td></td></tr><tr><td>6</td><td>Dout239</td><td></td></tr><tr><td>7</td><td>Dout240</td><td></td></tr><tr><td>8</td><td>Dout241</td><td></td></tr><tr><td>9</td><td>Dout242</td><td></td></tr><tr><td>10</td><td>Dout243</td><td></td></tr><tr><td>11</td><td>Dout244</td><td></td></tr><tr><td>12</td><td>Dout245</td><td></td></tr><tr><td>13</td><td>Dout246</td><td></td></tr><tr><td>14</td><td>Dout247</td><td></td></tr><tr><td>15</td><td>Dout248</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout233		1	Dout234		2	Dout235		3	Dout236		4	Dout237		5	Dout238		6	Dout239		7	Dout240		8	Dout241		9	Dout242		10	Dout243		11	Dout244		12	Dout245		13	Dout246		14	Dout247		15	Dout248	
			Bit	Line No.	Signal name																																																	
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28	System output 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout249</td><td>EMG_ST</td></tr><tr><td>1</td><td>Dout250</td><td>SV_RDY</td></tr><tr><td>2</td><td>Dout251</td><td>ACK</td></tr><tr><td>3</td><td>Dout252</td><td>TEACH</td></tr><tr><td>4</td><td>Dout253</td><td>INT</td></tr><tr><td>5</td><td>Dout254</td><td>EXTSIG</td></tr><tr><td>6</td><td>Dout255</td><td>EXTHOST</td></tr><tr><td>7</td><td>Dout256</td><td>SYS_RDY</td></tr><tr><td>8</td><td>Dout257</td><td>AUTORUN</td></tr><tr><td>9</td><td>Dout258</td><td>CYC_END</td></tr><tr><td>10</td><td>Dout259</td><td>LOW_ST</td></tr><tr><td>11</td><td>Dout260</td><td>CYC_ST</td></tr><tr><td>12</td><td>Dout261</td><td>BT_ALM</td></tr><tr><td>13</td><td>Dout262</td><td>ALARM</td></tr><tr><td>14</td><td>Dout263</td><td></td></tr><tr><td>15</td><td>Dout264</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout249	EMG_ST	1	Dout250	SV_RDY	2	Dout251	ACK	3	Dout252	TEACH	4	Dout253	INT	5	Dout254	EXTSIG	6	Dout255	EXTHOST	7	Dout256	SYS_RDY	8	Dout257	AUTORUN	9	Dout258	CYC_END	10	Dout259	LOW_ST	11	Dout260	CYC_ST	12	Dout261	BT_ALM	13	Dout262	ALARM	14	Dout263		15	Dout264	
			Bit	Line No.	Signal name																																																	
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			8	Dout257	AUTORUN																																																	
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			13	Dout262	ALARM																																																	
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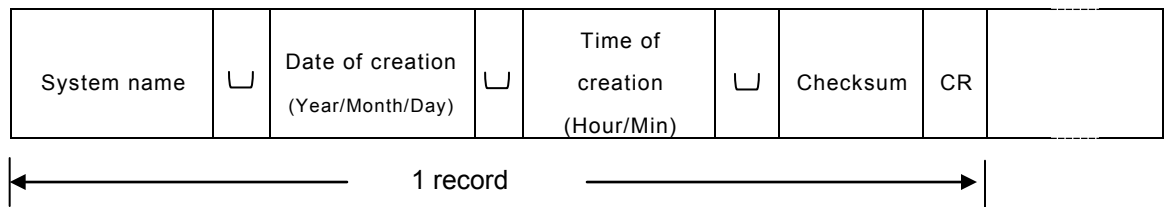
No.	Name	Size (byte)	Description																																																			
29	System reservation	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td></td><td></td></tr><tr><td>1</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td></tr><tr><td>7</td><td></td><td></td></tr><tr><td>8</td><td></td><td></td></tr><tr><td>9</td><td></td><td></td></tr><tr><td>10</td><td></td><td></td></tr><tr><td>11</td><td></td><td></td></tr><tr><td>12</td><td></td><td></td></tr><tr><td>13</td><td></td><td></td></tr><tr><td>14</td><td></td><td></td></tr><tr><td>15</td><td></td><td></td></tr></table>	Bit	Line No.	Signal name	0			1			2			3			4			5			6			7			8			9			10			11			12			13			14			15		
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No.	Name	Size (byte)	Description																																																			
31	System reservation	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td></td><td></td></tr><tr><td>1</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td></tr><tr><td>7</td><td></td><td></td></tr><tr><td>8</td><td></td><td></td></tr><tr><td>9</td><td></td><td></td></tr><tr><td>10</td><td></td><td></td></tr><tr><td>11</td><td></td><td></td></tr><tr><td>12</td><td></td><td></td></tr><tr><td>13</td><td></td><td></td></tr><tr><td>14</td><td></td><td></td></tr><tr><td>15</td><td></td><td></td></tr></table>	Bit	Line No.	Signal name	0			1			2			3			4			5			6			7			8			9			10			11			12			13			14			15		
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32	System reservation	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td></td><td></td></tr><tr><td>1</td><td></td><td></td></tr><tr><td>2</td><td></td><td></td></tr><tr><td>3</td><td></td><td></td></tr><tr><td>4</td><td></td><td></td></tr><tr><td>5</td><td></td><td></td></tr><tr><td>6</td><td></td><td></td></tr><tr><td>7</td><td></td><td></td></tr><tr><td>8</td><td></td><td></td></tr><tr><td>9</td><td></td><td></td></tr><tr><td>10</td><td></td><td></td></tr><tr><td>11</td><td></td><td></td></tr><tr><td>12</td><td></td><td></td></tr><tr><td>13</td><td></td><td></td></tr><tr><td>14</td><td></td><td></td></tr><tr><td>15</td><td></td><td></td></tr></table>	Bit	Line No.	Signal name	0			1			2			3			4			5			6			7			8			9			10			11			12			13			14			15		
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III) Current value data

No.	Name	Size (byte)	Description
1	Joint coordinate value	4 × 8 axes	As shown below, values are set in the order of axis 1 to axis 6. Axis 1 (float) Axis 2 (float) Axis 3 (float) Axis 4 (float) Axis 5 (float) Axis 6 (float)
2	World coordinate value	4 × 8 axes	
3	Work coordinate value	4 × 8 axes	
4	Work coordinate name	20	Name of work coordinate system
5	Tool coordinate name	20	Name of tool coordinate system
6	Base coordinate name	20	Name of base coordinate system
7	Reserved	2	
Total		134	

(6) Version information



Record structure

No.	Name	Size (byte)	Description
1	System name	10	
2	Date of creation (Year/Month/Day)	10	20**/**/**
3	Time of creation (Hour/Min)	5	**.**
4	Checksum	4	
5	CR	1	Record end code (0x0D)

(7) Variable read data

Read data 1	└─	Read data 2	└─	Read data 3	└─	...
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The variable read data differs as shown below with the type of variable.

Variable type data	Type	Read data
0	Integer type	1 pc. (long)
1	Real number type	1 pc. (float)
2	Load type	2 pcs. (float└float)
3	Coordinate type	4 pcs. (float└float└float└float)
4	Position type	6 pcs. (float└float└float└float└float└float)
5	Array integer type	1 pc. (long)
6	Array real number type	1 pc. (float)
7	Array load type	2 pcs. (float└float)
8	Array coordinate type	4 pcs. (float└float└float└float)
9	Array position type	6 pcs. (float└float└float└float└float└float)

The float data is expressed by a value calculated to three decimal places.

(8) Currently occurring alarm

Size	,	Alarm number	,	Message	,	Date	□	Time	CR	
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No.	Name	Size (byte)	Description
1	Size	2 (Variable)	Number of currently occurring alarms (decimal) This is added to only the beginning of the file.
2	Alarm number	7 (Fixed)	Shows the occurring alarm number in the following format. "XXX-YYY" XXX : Main code YYY : Sub code For details on the alarm contents, see the "Alarm Manual."
3	Alarm message	- (Variable)	Shows the message of occurring alarm.
4	Date	8 (Fixed)	The date on which the alarm occurred is shown in the following format. "YY-MM-DD" YY : Year (Last two numbers only) MM : Month DD : Day
5	Time	8 (Fixed)	The time on which the alarm occurred is shown in the following format. "HH:MM:SS" HH : Hours (In 24-hour"military time.") MM : Minutes SS : Seconds
6	CR	1 (Fixed)	Record end code (0x0D)

(9) Current position information

Motion status	☐	Program execution	☐	Axis 1 current position (integer part)	☐	Axis 2 current position (integer part)
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☐	Axis 3 current position (integer part)	☐	Axis 4 current position (integer part)	☐	Axis 5 current position (integer part)	☐	Axis 6 current position (integer part)
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☐	Axis 1 current position (real number)	☐	Axis 2 current position (real number)	☐	Axis 3 current position (real number)
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☐	Axis 4 current position (real number)	☐	Axis 5 current position (real number)	☐	Axis 6 current position (real number)
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☐	Axis 1 motor torque current value (real number)	☐	Axis 2 motor torque current value (real number)	☐	Axis 3 motor torque current value (real number)
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☐	Axis 4 motor torque current value (real number)	☐	Axis 5 motor torque current value (real number)	☐	Axis 6 motor torque current value (real number)
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No.	Name	Size (byte)	Description
1	Motion status	1 (Fixed)	0: STOP (RESET) 1: RUN 2: STOP (RETRY) 3: STOP (CONTINUE)
2	Program execution line number	- (Variable)	Line number during program execution
3 to 8	Current position	- (Variable)	Joint coordinate value of axis 1 to axis 6 However, the decimal point is deleted and only integer part is sent.

9 to 14	Current position	- (Variable)	Joint coordinate value of axis 1 to axis 6 The value is sent as a real number to three decimal places.
15 to 20	Motor torque current value	- (Variable)	Motor torque current value of axis 1 to axis 6 The value is sent as a real number to one decimal place.

(10) Individual current position information

Axis 1 (X) current position	☐	Axis 2 (Y) current position	☐	Axis 3 (Z) current position
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☐	Axis 4 (C) current position	☐	Axis 5 (T) current position	☐	Axis 6 current position
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☐	Current configuration
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No.	Name	Size (byte)	Description
1 to 7	Current position	- (Variable)	Current value of axis 1 to axis 6 and current configuration This value is sent as a real number to three decimal points. *The contents of the response varies depending on the request coordinate system. 0: Joint coordinate 1: World coordinate 2: Work coordinate 3: Joint feedback coordinate 4: World feedback coordinate 5: Work feedback coordinate

(11) Maintenance information

• Format 1

Maintenance information number	,	Axis 1 information	,	Axis 2 information
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,	Axis 3 information	,	Axis 4 information	,	Axis 5 information
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,	Axis 6 information
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No.	Name	Size (byte)	Description
1	Maintenance information number	- (Variable)	Maintenance information number of request
2 to 7	Axis information	- (Variable)	Axis 1 to axis 6 information This value is sent as a real number to one decimal point. *The contents of response varies depending on the maintenance information number. 1: Total rotational amount of motor 2: Movement amount of joint axis 3: Integrated amount of motor torque 4: Integrated amount of motor torque exceeding rating 5: Peak torque (cleared by communication request) 6: 3-second load factor 7: 1-minute load factor 8: 15-minute load factor 9: 2-hour load factor 17: Peak torque

- Format 2

Maintenance information number	,	Current information
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No.	Name	Size (byte)	Description
1	Maintenance information number	- (Variable)	Maintenance information number of request
2	Information	- (Variable)	Information This value is sent as a real number to one decimal point. *The contents of response varies depending on the maintenance information number. 10: Total time 11: Energizing time 12: Power OFF time 13: Motion time 14: Fault time 15: Servo information (servo OFF count) 16: Servo ON time

(12) IO information

No.	Name	Size (byte)	Description																																																			
1	General input 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din1</td><td>General input</td></tr><tr><td>1</td><td>Din2</td><td>General input</td></tr><tr><td>2</td><td>Din3</td><td>General input</td></tr><tr><td>3</td><td>Din4</td><td>General input</td></tr><tr><td>4</td><td>Din5</td><td>General input</td></tr><tr><td>5</td><td>Din6</td><td>General input</td></tr><tr><td>6</td><td>Din7</td><td>General input</td></tr><tr><td>7</td><td>Din8</td><td>General input</td></tr><tr><td>8</td><td>Din9</td><td>General input</td></tr><tr><td>9</td><td>Din10</td><td>General input</td></tr><tr><td>10</td><td>Din11</td><td>General input</td></tr><tr><td>11</td><td>Din12</td><td>General input</td></tr><tr><td>12</td><td>Din13</td><td>General input</td></tr><tr><td>13</td><td>Din14</td><td>General input</td></tr><tr><td>14</td><td>Din15</td><td>General input</td></tr><tr><td>15</td><td>Din16</td><td>General input</td></tr></table>	Bit	Line No.	Signal name	0	Din1	General input	1	Din2	General input	2	Din3	General input	3	Din4	General input	4	Din5	General input	5	Din6	General input	6	Din7	General input	7	Din8	General input	8	Din9	General input	9	Din10	General input	10	Din11	General input	11	Din12	General input	12	Din13	General input	13	Din14	General input	14	Din15	General input	15	Din16	General input
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14	Din15	General input																																																				
15	Din16	General input																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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15	Din64	General input																																																				

No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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6	Extension input 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din117</td><td>Extension input</td></tr><tr><td>1</td><td>Din118</td><td>Extension input</td></tr><tr><td>2</td><td>Din119</td><td>Extension input</td></tr><tr><td>3</td><td>Din120</td><td>Extension input</td></tr><tr><td>4</td><td>Din121</td><td>Extension input</td></tr><tr><td>5</td><td>Din122</td><td>Extension input</td></tr><tr><td>6</td><td>Din123</td><td>Extension input</td></tr><tr><td>7</td><td>Din124</td><td>Extension input</td></tr><tr><td>8</td><td>Din125</td><td>Extension input</td></tr><tr><td>9</td><td>Din126</td><td>Extension input</td></tr><tr><td>10</td><td>Din127</td><td>Extension input</td></tr><tr><td>11</td><td>Din128</td><td>Extension input</td></tr><tr><td>12</td><td>Din129</td><td>Extension input</td></tr><tr><td>13</td><td>Din130</td><td>Extension input</td></tr><tr><td>14</td><td>Din131</td><td>Extension input</td></tr><tr><td>15</td><td>Din132</td><td>Extension input</td></tr></table>	Bit	Line No.	Signal name	0	Din117	Extension input	1	Din118	Extension input	2	Din119	Extension input	3	Din120	Extension input	4	Din121	Extension input	5	Din122	Extension input	6	Din123	Extension input	7	Din124	Extension input	8	Din125	Extension input	9	Din126	Extension input	10	Din127	Extension input	11	Din128	Extension input	12	Din129	Extension input	13	Din130	Extension input	14	Din131	Extension input	15	Din132	Extension input
Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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10	System input 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Din217</td><td>Alarm of level 8</td></tr><tr><td>1</td><td>Din218</td><td>Alarm of level 8</td></tr><tr><td>2</td><td>Din219</td><td>Alarm of level 8</td></tr><tr><td>3</td><td>Din220</td><td>Alarm of level 8</td></tr><tr><td>4</td><td>Din221</td><td>Alarm of level 4</td></tr><tr><td>5</td><td>Din222</td><td>Alarm of level 4</td></tr><tr><td>6</td><td>Din223</td><td>Alarm of level 4</td></tr><tr><td>7</td><td>Din224</td><td>Alarm of level 4</td></tr><tr><td>8</td><td>Din225</td><td>Alarm of level 2</td></tr><tr><td>9</td><td>Din226</td><td>Alarm of level 2</td></tr><tr><td>10</td><td>Din227</td><td>Alarm of level 2</td></tr><tr><td>11</td><td>Din228</td><td>Alarm of level 2</td></tr><tr><td>12</td><td>Din229</td><td>Alarm of level 1</td></tr><tr><td>13</td><td>Din230</td><td>Alarm of level 1</td></tr><tr><td>14</td><td>Din231</td><td>Alarm of level 1</td></tr><tr><td>15</td><td>Din232</td><td>Alarm of level 1</td></tr></table>	Bit	Line No.	Signal name	0	Din217	Alarm of level 8	1	Din218	Alarm of level 8	2	Din219	Alarm of level 8	3	Din220	Alarm of level 8	4	Din221	Alarm of level 4	5	Din222	Alarm of level 4	6	Din223	Alarm of level 4	7	Din224	Alarm of level 4	8	Din225	Alarm of level 2	9	Din226	Alarm of level 2	10	Din227	Alarm of level 2	11	Din228	Alarm of level 2	12	Din229	Alarm of level 1	13	Din230	Alarm of level 1	14	Din231	Alarm of level 1	15	Din232	Alarm of level 1
Bit	Line No.	Signal name																																																				
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15	Din232	Alarm of level 1																																																				

No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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4	Din253	DO_RST																																																				
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7	Din256	EX_SVON																																																				
8	Din257	STOP																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
21	General output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout1</td><td>General output</td></tr><tr><td>1</td><td>Dout2</td><td>General output</td></tr><tr><td>2</td><td>Dout3</td><td>General output</td></tr><tr><td>3</td><td>Dout4</td><td>General output</td></tr><tr><td>4</td><td>Dout5</td><td>General output</td></tr><tr><td>5</td><td>Dout6</td><td>General output</td></tr><tr><td>6</td><td>Dout7</td><td>General output</td></tr><tr><td>7</td><td>Dout8</td><td>General output</td></tr><tr><td>8</td><td>Dout9</td><td>General output</td></tr><tr><td>9</td><td>Dout10</td><td>General output</td></tr><tr><td>10</td><td>Dout11</td><td>General output</td></tr><tr><td>11</td><td>Dout12</td><td>General output</td></tr><tr><td>12</td><td>Dout13</td><td>General output</td></tr><tr><td>13</td><td>Dout14</td><td>General output</td></tr><tr><td>14</td><td>Dout15</td><td>General output</td></tr><tr><td>15</td><td>Dout16</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout1	General output	1	Dout2	General output	2	Dout3	General output	3	Dout4	General output	4	Dout5	General output	5	Dout6	General output	6	Dout7	General output	7	Dout8	General output	8	Dout9	General output	9	Dout10	General output	10	Dout11	General output	11	Dout12	General output	12	Dout13	General output	13	Dout14	General output	14	Dout15	General output	15	Dout16	General output
Bit	Line No.	Signal name																																																				
0	Dout1	General output																																																				
1	Dout2	General output																																																				
2	Dout3	General output																																																				
3	Dout4	General output																																																				
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5	Dout6	General output																																																				
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8	Dout9	General output																																																				
9	Dout10	General output																																																				
10	Dout11	General output																																																				
11	Dout12	General output																																																				
12	Dout13	General output																																																				
13	Dout14	General output																																																				
14	Dout15	General output																																																				
15	Dout16	General output																																																				
22	General output 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout17</td><td>General output</td></tr><tr><td>1</td><td>Dout18</td><td>General output</td></tr><tr><td>2</td><td>Dout19</td><td>General output</td></tr><tr><td>3</td><td>Dout20</td><td>General output</td></tr><tr><td>4</td><td>Dout21</td><td>General output</td></tr><tr><td>5</td><td>Dout22</td><td>General output</td></tr><tr><td>6</td><td>Dout23</td><td>General output</td></tr><tr><td>7</td><td>Dout24</td><td>General output</td></tr><tr><td>8</td><td>Dout25</td><td>General output</td></tr><tr><td>9</td><td>Dout26</td><td>General output</td></tr><tr><td>10</td><td>Dout27</td><td>General output</td></tr><tr><td>11</td><td>Dout28</td><td>General output</td></tr><tr><td>12</td><td>Dout29</td><td>General output</td></tr><tr><td>13</td><td>Dout30</td><td>General output</td></tr><tr><td>14</td><td>Dout31</td><td>General output</td></tr><tr><td>15</td><td>Dout32</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout17	General output	1	Dout18	General output	2	Dout19	General output	3	Dout20	General output	4	Dout21	General output	5	Dout22	General output	6	Dout23	General output	7	Dout24	General output	8	Dout25	General output	9	Dout26	General output	10	Dout27	General output	11	Dout28	General output	12	Dout29	General output	13	Dout30	General output	14	Dout31	General output	15	Dout32	General output
Bit	Line No.	Signal name																																																				
0	Dout17	General output																																																				
1	Dout18	General output																																																				
2	Dout19	General output																																																				
3	Dout20	General output																																																				
4	Dout21	General output																																																				
5	Dout22	General output																																																				
6	Dout23	General output																																																				
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8	Dout25	General output																																																				
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11	Dout28	General output																																																				
12	Dout29	General output																																																				
13	Dout30	General output																																																				
14	Dout31	General output																																																				
15	Dout32	General output																																																				

No.	Name	Size (byte)	Description																																																			
23	General output 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout33</td><td>General output</td></tr><tr><td>1</td><td>Dout34</td><td>General output</td></tr><tr><td>2</td><td>Dout35</td><td>General output</td></tr><tr><td>3</td><td>Dout36</td><td>General output</td></tr><tr><td>4</td><td>Dout37</td><td>General output</td></tr><tr><td>5</td><td>Dout38</td><td>General output</td></tr><tr><td>6</td><td>Dout39</td><td>General output</td></tr><tr><td>7</td><td>Dout40</td><td>General output</td></tr><tr><td>8</td><td>Dout41</td><td>General output</td></tr><tr><td>9</td><td>Dout42</td><td>General output</td></tr><tr><td>10</td><td>Dout43</td><td>General output</td></tr><tr><td>11</td><td>Dout44</td><td>General output</td></tr><tr><td>12</td><td>Dout45</td><td>General output</td></tr><tr><td>13</td><td>Dout46</td><td>General output</td></tr><tr><td>14</td><td>Dout47</td><td>General output</td></tr><tr><td>15</td><td>Dout48</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout33	General output	1	Dout34	General output	2	Dout35	General output	3	Dout36	General output	4	Dout37	General output	5	Dout38	General output	6	Dout39	General output	7	Dout40	General output	8	Dout41	General output	9	Dout42	General output	10	Dout43	General output	11	Dout44	General output	12	Dout45	General output	13	Dout46	General output	14	Dout47	General output	15	Dout48	General output
			Bit	Line No.	Signal name																																																	
			0	Dout33	General output																																																	
			1	Dout34	General output																																																	
			2	Dout35	General output																																																	
			3	Dout36	General output																																																	
			4	Dout37	General output																																																	
			5	Dout38	General output																																																	
			6	Dout39	General output																																																	
			7	Dout40	General output																																																	
			8	Dout41	General output																																																	
			9	Dout42	General output																																																	
			10	Dout43	General output																																																	
			11	Dout44	General output																																																	
			12	Dout45	General output																																																	
			13	Dout46	General output																																																	
			14	Dout47	General output																																																	
15	Dout48	General output																																																				
24	General output 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout49</td><td>General output</td></tr><tr><td>1</td><td>Dout50</td><td>General output</td></tr><tr><td>2</td><td>Dout51</td><td>General output</td></tr><tr><td>3</td><td>Dout52</td><td>General output</td></tr><tr><td>4</td><td>Dout53</td><td>General output</td></tr><tr><td>5</td><td>Dout54</td><td>General output</td></tr><tr><td>6</td><td>Dout55</td><td>General output</td></tr><tr><td>7</td><td>Dout56</td><td>General output</td></tr><tr><td>8</td><td>Dout57</td><td>General output</td></tr><tr><td>9</td><td>Dout58</td><td>General output</td></tr><tr><td>10</td><td>Dout59</td><td>General output</td></tr><tr><td>11</td><td>Dout60</td><td>General output</td></tr><tr><td>12</td><td>Dout61</td><td>General output</td></tr><tr><td>13</td><td>Dout62</td><td>General output</td></tr><tr><td>14</td><td>Dout63</td><td>General output</td></tr><tr><td>15</td><td>Dout64</td><td>General output</td></tr></table>	Bit	Line No.	Signal name	0	Dout49	General output	1	Dout50	General output	2	Dout51	General output	3	Dout52	General output	4	Dout53	General output	5	Dout54	General output	6	Dout55	General output	7	Dout56	General output	8	Dout57	General output	9	Dout58	General output	10	Dout59	General output	11	Dout60	General output	12	Dout61	General output	13	Dout62	General output	14	Dout63	General output	15	Dout64	General output
			Bit	Line No.	Signal name																																																	
			0	Dout49	General output																																																	
			1	Dout50	General output																																																	
			2	Dout51	General output																																																	
			3	Dout52	General output																																																	
			4	Dout53	General output																																																	
			5	Dout54	General output																																																	
			6	Dout55	General output																																																	
			7	Dout56	General output																																																	
			8	Dout57	General output																																																	
			9	Dout58	General output																																																	
			10	Dout59	General output																																																	
			11	Dout60	General output																																																	
			12	Dout61	General output																																																	
			13	Dout62	General output																																																	
			14	Dout63	General output																																																	
15	Dout64	General output																																																				

No.	Name	Size (byte)	Description																																																			
25	Extension output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout101</td><td>Extension output</td></tr><tr><td>1</td><td>Dout102</td><td>Extension output</td></tr><tr><td>2</td><td>Dout103</td><td>Extension output</td></tr><tr><td>3</td><td>Dout104</td><td>Extension output</td></tr><tr><td>4</td><td>Dout105</td><td>Extension output</td></tr><tr><td>5</td><td>Dout106</td><td>Extension output</td></tr><tr><td>6</td><td>Dout107</td><td>Extension output</td></tr><tr><td>7</td><td>Dout108</td><td>Extension output</td></tr><tr><td>8</td><td>Dout109</td><td>Extension output</td></tr><tr><td>9</td><td>Dout110</td><td>Extension output</td></tr><tr><td>10</td><td>Dout111</td><td>Extension output</td></tr><tr><td>11</td><td>Dout112</td><td>Extension output</td></tr><tr><td>12</td><td>Dout113</td><td>Extension output</td></tr><tr><td>13</td><td>Dout114</td><td>Extension output</td></tr><tr><td>14</td><td>Dout115</td><td>Extension output</td></tr><tr><td>15</td><td>Dout116</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout101	Extension output	1	Dout102	Extension output	2	Dout103	Extension output	3	Dout104	Extension output	4	Dout105	Extension output	5	Dout106	Extension output	6	Dout107	Extension output	7	Dout108	Extension output	8	Dout109	Extension output	9	Dout110	Extension output	10	Dout111	Extension output	11	Dout112	Extension output	12	Dout113	Extension output	13	Dout114	Extension output	14	Dout115	Extension output	15	Dout116	Extension output
Bit	Line No.	Signal name																																																				
0	Dout101	Extension output																																																				
1	Dout102	Extension output																																																				
2	Dout103	Extension output																																																				
3	Dout104	Extension output																																																				
4	Dout105	Extension output																																																				
5	Dout106	Extension output																																																				
6	Dout107	Extension output																																																				
7	Dout108	Extension output																																																				
8	Dout109	Extension output																																																				
9	Dout110	Extension output																																																				
10	Dout111	Extension output																																																				
11	Dout112	Extension output																																																				
12	Dout113	Extension output																																																				
13	Dout114	Extension output																																																				
14	Dout115	Extension output																																																				
15	Dout116	Extension output																																																				
26	Extension output 2	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout117</td><td>Extension output</td></tr><tr><td>1</td><td>Dout118</td><td>Extension output</td></tr><tr><td>2</td><td>Dout119</td><td>Extension output</td></tr><tr><td>3</td><td>Dout120</td><td>Extension output</td></tr><tr><td>4</td><td>Dout121</td><td>Extension output</td></tr><tr><td>5</td><td>Dout122</td><td>Extension output</td></tr><tr><td>6</td><td>Dout123</td><td>Extension output</td></tr><tr><td>7</td><td>Dout124</td><td>Extension output</td></tr><tr><td>8</td><td>Dout125</td><td>Extension output</td></tr><tr><td>9</td><td>Dout126</td><td>Extension output</td></tr><tr><td>10</td><td>Dout127</td><td>Extension output</td></tr><tr><td>11</td><td>Dout128</td><td>Extension output</td></tr><tr><td>12</td><td>Dout129</td><td>Extension output</td></tr><tr><td>13</td><td>Dout130</td><td>Extension output</td></tr><tr><td>14</td><td>Dout131</td><td>Extension output</td></tr><tr><td>15</td><td>Dout132</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout117	Extension output	1	Dout118	Extension output	2	Dout119	Extension output	3	Dout120	Extension output	4	Dout121	Extension output	5	Dout122	Extension output	6	Dout123	Extension output	7	Dout124	Extension output	8	Dout125	Extension output	9	Dout126	Extension output	10	Dout127	Extension output	11	Dout128	Extension output	12	Dout129	Extension output	13	Dout130	Extension output	14	Dout131	Extension output	15	Dout132	Extension output
Bit	Line No.	Signal name																																																				
0	Dout117	Extension output																																																				
1	Dout118	Extension output																																																				
2	Dout119	Extension output																																																				
3	Dout120	Extension output																																																				
4	Dout121	Extension output																																																				
5	Dout122	Extension output																																																				
6	Dout123	Extension output																																																				
7	Dout124	Extension output																																																				
8	Dout125	Extension output																																																				
9	Dout126	Extension output																																																				
10	Dout127	Extension output																																																				
11	Dout128	Extension output																																																				
12	Dout129	Extension output																																																				
13	Dout130	Extension output																																																				
14	Dout131	Extension output																																																				
15	Dout132	Extension output																																																				

No.	Name	Size (byte)	Description																																																			
27	Extension output 3	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout133</td><td>Extension output</td></tr><tr><td>1</td><td>Dout134</td><td>Extension output</td></tr><tr><td>2</td><td>Dout135</td><td>Extension output</td></tr><tr><td>3</td><td>Dout136</td><td>Extension output</td></tr><tr><td>4</td><td>Dout137</td><td>Extension output</td></tr><tr><td>5</td><td>Dout138</td><td>Extension output</td></tr><tr><td>6</td><td>Dout139</td><td>Extension output</td></tr><tr><td>7</td><td>Dout140</td><td>Extension output</td></tr><tr><td>8</td><td>Dout141</td><td>Extension output</td></tr><tr><td>9</td><td>Dout142</td><td>Extension output</td></tr><tr><td>10</td><td>Dout143</td><td>Extension output</td></tr><tr><td>11</td><td>Dout144</td><td>Extension output</td></tr><tr><td>12</td><td>Dout145</td><td>Extension output</td></tr><tr><td>13</td><td>Dout146</td><td>Extension output</td></tr><tr><td>14</td><td>Dout147</td><td>Extension output</td></tr><tr><td>15</td><td>Dout148</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout133	Extension output	1	Dout134	Extension output	2	Dout135	Extension output	3	Dout136	Extension output	4	Dout137	Extension output	5	Dout138	Extension output	6	Dout139	Extension output	7	Dout140	Extension output	8	Dout141	Extension output	9	Dout142	Extension output	10	Dout143	Extension output	11	Dout144	Extension output	12	Dout145	Extension output	13	Dout146	Extension output	14	Dout147	Extension output	15	Dout148	Extension output
			Bit	Line No.	Signal name																																																	
			0	Dout133	Extension output																																																	
			1	Dout134	Extension output																																																	
			2	Dout135	Extension output																																																	
			3	Dout136	Extension output																																																	
			4	Dout137	Extension output																																																	
			5	Dout138	Extension output																																																	
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			7	Dout140	Extension output																																																	
			8	Dout141	Extension output																																																	
			9	Dout142	Extension output																																																	
			10	Dout143	Extension output																																																	
			11	Dout144	Extension output																																																	
			12	Dout145	Extension output																																																	
			13	Dout146	Extension output																																																	
			14	Dout147	Extension output																																																	
15	Dout148	Extension output																																																				
28	Extension output 4	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout149</td><td>Extension output</td></tr><tr><td>1</td><td>Dout150</td><td>Extension output</td></tr><tr><td>2</td><td>Dout151</td><td>Extension output</td></tr><tr><td>3</td><td>Dout152</td><td>Extension output</td></tr><tr><td>4</td><td>Dout153</td><td>Extension output</td></tr><tr><td>5</td><td>Dout154</td><td>Extension output</td></tr><tr><td>6</td><td>Dout155</td><td>Extension output</td></tr><tr><td>7</td><td>Dout156</td><td>Extension output</td></tr><tr><td>8</td><td>Dout157</td><td>Extension output</td></tr><tr><td>9</td><td>Dout158</td><td>Extension output</td></tr><tr><td>10</td><td>Dout159</td><td>Extension output</td></tr><tr><td>11</td><td>Dout160</td><td>Extension output</td></tr><tr><td>12</td><td>Dout161</td><td>Extension output</td></tr><tr><td>13</td><td>Dout162</td><td>Extension output</td></tr><tr><td>14</td><td>Dout163</td><td>Extension output</td></tr><tr><td>15</td><td>Dout164</td><td>Extension output</td></tr></table>	Bit	Line No.	Signal name	0	Dout149	Extension output	1	Dout150	Extension output	2	Dout151	Extension output	3	Dout152	Extension output	4	Dout153	Extension output	5	Dout154	Extension output	6	Dout155	Extension output	7	Dout156	Extension output	8	Dout157	Extension output	9	Dout158	Extension output	10	Dout159	Extension output	11	Dout160	Extension output	12	Dout161	Extension output	13	Dout162	Extension output	14	Dout163	Extension output	15	Dout164	Extension output
			Bit	Line No.	Signal name																																																	
			0	Dout149	Extension output																																																	
			1	Dout150	Extension output																																																	
			2	Dout151	Extension output																																																	
			3	Dout152	Extension output																																																	
			4	Dout153	Extension output																																																	
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			6	Dout155	Extension output																																																	
			7	Dout156	Extension output																																																	
			8	Dout157	Extension output																																																	
			9	Dout158	Extension output																																																	
			10	Dout159	Extension output																																																	
			11	Dout160	Extension output																																																	
			12	Dout161	Extension output																																																	
			13	Dout162	Extension output																																																	
			14	Dout163	Extension output																																																	
15	Dout164	Extension output																																																				

No.	Name	Size (byte)	Description																																																			
29	System output 1	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout201</td><td>Hand output</td></tr><tr><td>1</td><td>Dout202</td><td>Hand output</td></tr><tr><td>2</td><td>Dout203</td><td>Hand output</td></tr><tr><td>3</td><td>Dout204</td><td>Hand output</td></tr><tr><td>4</td><td>Dout205</td><td>Hand output</td></tr><tr><td>5</td><td>Dout206</td><td>Hand output</td></tr><tr><td>6</td><td>Dout207</td><td>Hand output</td></tr><tr><td>7</td><td>Dout208</td><td>Hand output</td></tr><tr><td>8</td><td>Dout209</td><td></td></tr><tr><td>9</td><td>Dout210</td><td></td></tr><tr><td>10</td><td>Dout211</td><td></td></tr><tr><td>11</td><td>Dout212</td><td></td></tr><tr><td>12</td><td>Dout213</td><td></td></tr><tr><td>13</td><td>Dout214</td><td></td></tr><tr><td>14</td><td>Dout215</td><td></td></tr><tr><td>15</td><td>Dout216</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout201	Hand output	1	Dout202	Hand output	2	Dout203	Hand output	3	Dout204	Hand output	4	Dout205	Hand output	5	Dout206	Hand output	6	Dout207	Hand output	7	Dout208	Hand output	8	Dout209		9	Dout210		10	Dout211		11	Dout212		12	Dout213		13	Dout214		14	Dout215		15	Dout216	
Bit	Line No.	Signal name																																																				
0	Dout201	Hand output																																																				
1	Dout202	Hand output																																																				
2	Dout203	Hand output																																																				
3	Dout204	Hand output																																																				
4	Dout205	Hand output																																																				
5	Dout206	Hand output																																																				
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Bit	Line No.	Signal name																																																				
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No.	Name	Size (byte)	Description																																																			
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No.	Name	Size (byte)	Description																																																			
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No.	Name	Size (byte)	Description																																																			
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40	Field bus output 8	2	<table><tr><th>Bit</th><th>Line No.</th><th>Signal name</th></tr><tr><td>0</td><td>Dout449</td><td></td></tr><tr><td>1</td><td>Dout450</td><td></td></tr><tr><td>2</td><td>Dout451</td><td></td></tr><tr><td>3</td><td>Dout452</td><td></td></tr><tr><td>4</td><td>Dout453</td><td></td></tr><tr><td>5</td><td>Dout454</td><td></td></tr><tr><td>6</td><td>Dout455</td><td></td></tr><tr><td>7</td><td>Dout456</td><td></td></tr><tr><td>8</td><td>Dout457</td><td></td></tr><tr><td>9</td><td>Dout458</td><td></td></tr><tr><td>10</td><td>Dout459</td><td></td></tr><tr><td>11</td><td>Dout460</td><td></td></tr><tr><td>12</td><td>Dout461</td><td></td></tr><tr><td>13</td><td>Dout462</td><td></td></tr><tr><td>14</td><td>Dout463</td><td></td></tr><tr><td>15</td><td>Dout464</td><td></td></tr></table>	Bit	Line No.	Signal name	0	Dout449		1	Dout450		2	Dout451		3	Dout452		4	Dout453		5	Dout454		6	Dout455		7	Dout456		8	Dout457		9	Dout458		10	Dout459		11	Dout460		12	Dout461		13	Dout462		14	Dout463		15	Dout464	
Bit	Line No.	Signal name																																																				
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14	Dout463																																																					
15	Dout464																																																					

(13) Alarm history

Size	,	Alarm number	,	Message	,	Date	□	Time	CR	
------	---	--------------	---	---------	---	------	---	------	----	--



No.	Name	Size (byte)	Description
1	Size	2 (Variable)	Number of alarm histories (decimal) This is added to only the beginning of the file.
2	Alarm number	7 (Fixed)	Shows the occurring alarm number in the following format. "XXX-YYY" XXX : Main code YYY : Sub code For details on the alarm contents, see the "Alarm Manual."
3	Alarm message	- (Variable)	Shows the message of occurring alarm.
4	Date	8 (Fixed)	The date on which the alarm occurred is shown in the following format. "YY-MM-DD" YY : Year (Last two numbers only) MM : Month DD : Day
5	Time	8 (Fixed)	The time on which the alarm occurred is shown in the following format. "HH:MM:SS" HH : Hours (In 24-hour"military time.") MM : Minutes SS : Seconds
6	CR	1 (Fixed)	Record end code (0x0D)

(14) Robot tip speed information

Axis 1 (X) tip speed information	,	Axis 2 (Y) tip speed information	,	Axis 3 (Z) tip speed information
-------------------------------------	---	-------------------------------------	---	-------------------------------------

,	Axis 4 (C) tip speed information	,	Axis 5 (T) tip speed information	,	Axis 6 tip speed information
---	-------------------------------------	---	-------------------------------------	---	---------------------------------

No.	Name	Size (byte)	Description
1 to 6	Axis tip speed information	- (Variable)	Axis 1 to axis 6 information This value is sent as a real number to three decimal points. *The speed type varies depending on the response contents. 0: Robot tip speed (joint) 1: Robot tip speed (world)

(15) Motion status

Data header		Data	□	Data header		Data	□	Data header		Data
E	E			S	E			S	C	

□	Data header		Data	□	Data header		Data	□	Data header		Data
	B	C			E	S			S	S	

□	Data header		Data	□	Data header		Data	□	Data header		Data
	S	V			M	M			R	M	

□	Data header		Data	□	Data header		Data	□	Data header		Data
	R	S			O	V			A	L	

□	Data header		Data	□	Data header		Data
	D	C			D	S	

Data header (2BYTE)		Data Size (byte)	Meaning	Data contents
E	E	1 (Fixed)	Emergency STOP Event	0: Without event 1: With event
S	E	1 (Fixed)	Safety SW Event	0: Without event 1: With event
S	C	1 (Fixed)	Stop command Event	0: Without event 1: With event
B	C	1 (Fixed)	Break command Event	0: Without event 1: With event
E	S	1 (Fixed)	Emergency Switch Status	0: Emergency stop OFF 1: Emergency stop ON
S	S	1 (Fixed)	Safety Switch Status	0: Safety Switch OFF 1: Safety Switch ON
S	V	1 (Fixed)	Servo Status	0: Servo OFF 1: Servo ON
M	M	1 (Fixed)	Master Mode Status	0: TEACHING 1: INTERNAL 2: EXT.SIG 4: EXT.RS232C 5: EXT.ETHER
R	M	1 (Fixed)	Run Mode Status	0: CONTINUOUS 1: CYCLE 2: STEP 3: SEGMENT
R	S	1 (Fixed)	Run Status	0: STOP(RESET) 1: RUN 2: STOP(RETRY) 3: STOP(CONT)
O	V	- (Variable)	Override	0 to 100
A	L	1 (Fixed)	Alarm *Alarm at the highest level among occurring alarms	0: No alarm 1: Level 1 alarm 2: Level 2 alarm 4: Level 4 alarm 8: Level 8 alarm
D	C	- (Variable)	Do Move Count	0 to 65536
D	S	1 (Fixed)	Do Move Status	0: Motion completion 1: Motion in progress 2: Stop end 3: Break end

(16) Motion status

Expression (value)

No.	Name	Size (byte)	Description
1	Expression (value)	- (Variable)	Hexadecimal-coded information of the specified input signal

(17) Motion status

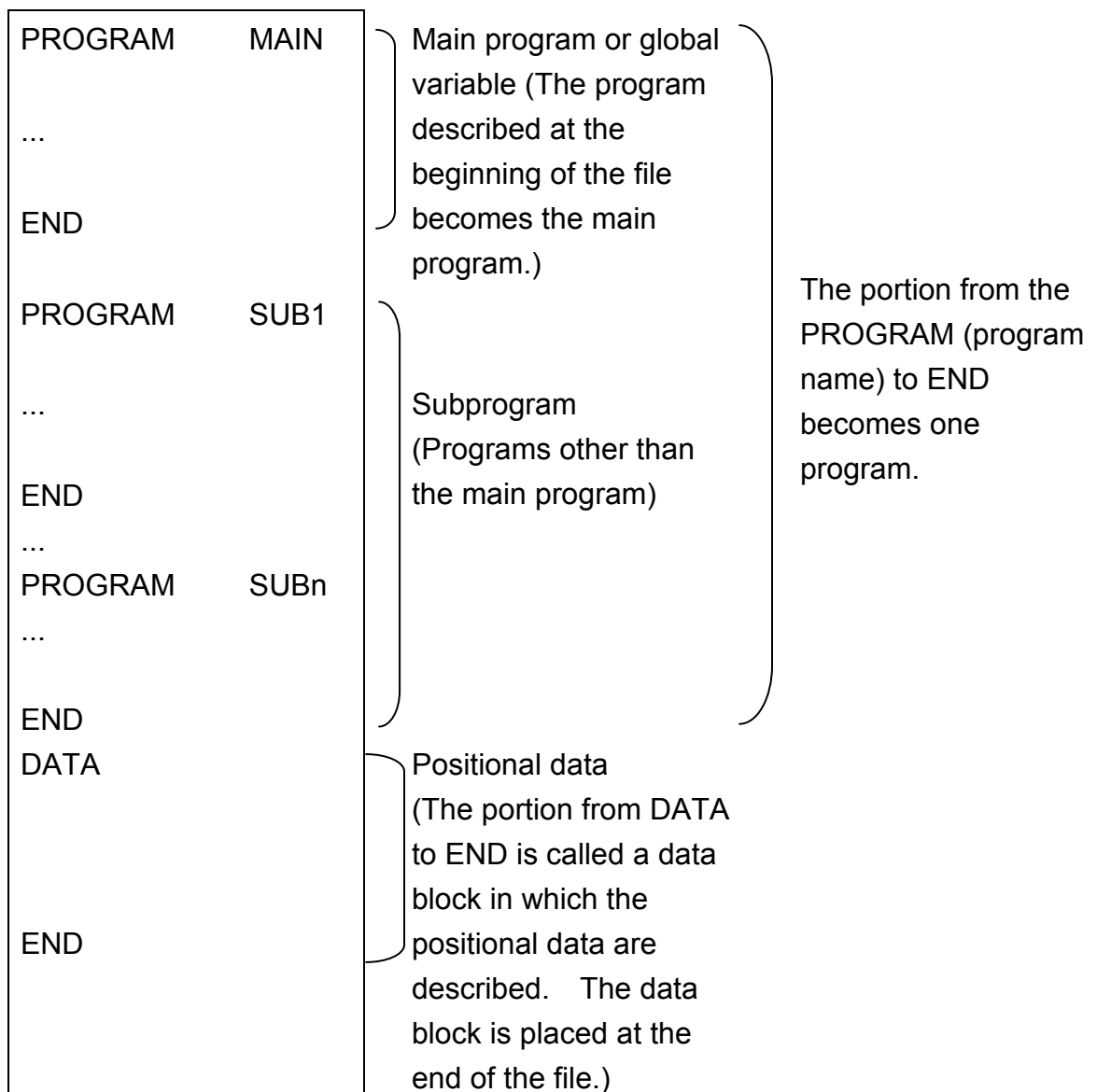
Function determination	☐	3-axis origin change value (integer)
------------------------	--------------------------	--------------------------------------

No.	Name	Size (byte)	Description
1	Function determination	1 (Fixed)	3-axis origin change function 0: Invalid 1: Valid
2	3-axis origin change value	- (Variable)	Offset value of 3-axis origin This value is sent as an integer value.

5.4 Ram Files

5.4.1 User Files (Work files)

The user files store some programs and positional data.



As shown in the above figure, the main program or global variable data at the beginning of the file, subprograms, and finally data block, in which the positional data are described, are placed in that order. The delimiting codes for file names and programs and the check sum are not placed. At the end of each line, a carriage return code is placed. At the end of the file, an EOF code is placed.

The characters to be used should be ASCII code alphanumeric characters and symbols.

Alphanumeric characters:

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9

Special symbols:

“ ‘ () + - * / , . < > = ! [] { } % ^ & ?

The robot controller does not distinguish between upper case and lower case characters. Do not use Chinese characters or double width characters. In particular, make sure that blanks are not double width characters. Moreover, do not place a tab code in the file.

(1) Global variable

The global variable area which can be referred from all programs are placed at the head of the program. If the global variable is not used, this area can be omitted in the program.

(2) Program

One program starts with the PROGRAM statement of the robot language and ends with the END statement. The program described at the beginning of the file becomes the main program. In the automatic operation mode, the program is executed from the beginning.

Each statement of the robot language is described on each line.

In one file, two or more programs can be described.

For details of the robot language, see the Robot Language Manual.

(3) Positional data

The positional data are described in the data block, following the program, at the end of the file.

The data block starts with a DATA statement and ends with an END statement.

The data block stores coordinate data and load data in addition to the positional data.

Each data is designated on each line.

The format of each data is as follows:

[1] Positional data

POINT <Position name>=<X>,<Y>,<Z>,C>,<T>/<configuration>

<Position name> : Name of positional data

<X>
<Y>
<Z> } : Coordinate values of X, Y, and Z (unit: mm)

<C> : Rotating angle of tool end (unit: deg)

<T> : Data of axis 5

<Configuration> : Configuration of the robot arm that rotates in horizontal direction

OmittedUndefined

LEFTY Left shoulder

RIGHTY Right shoulder

When the values of X, Y, Z, C, and T are omitted, they are treated as 0.

[2] Coordinate data

TRANS <Coordinate name>=<X>,<Y>,<Z>,<C>

<Coordinate name> : Name of coordinate data

<X>
<Y>
<Z> } : Coordinate values in directions of axes X, Y, and Z (unit: mm)

<C> : Rotating angle about Z axis (unit: deg)

When the values of X, Y, Z, and C are omitted, they are treated as 0.

[3] Load data

PAYLOAD <Load name>=<mass>, <center of gravity offset>

<Load name>: Name of load data

<mass>: Mass of the load on the end of the robot hand
(unit: kg)

<center of gravity offset>: Offset of the center of gravity of the load on the
end of the robot hand from the center of the tool
axis (unit: mm)

When the values of the mass and center-of-gravity offset are omitted, they are treated as 0.

[4] Designation of work coordinate system

The positional data represent the position in the work coordinate system. The positional data are designated by the work coordinate system.

When the work coordinate system is designated, the positional data described in the next line are treated as the position in the work coordinate system.

When the work coordinate system is not designated, it is assumed that the work coordinate system accords with the world coordinate system.

The work coordinate system is described in the following format.

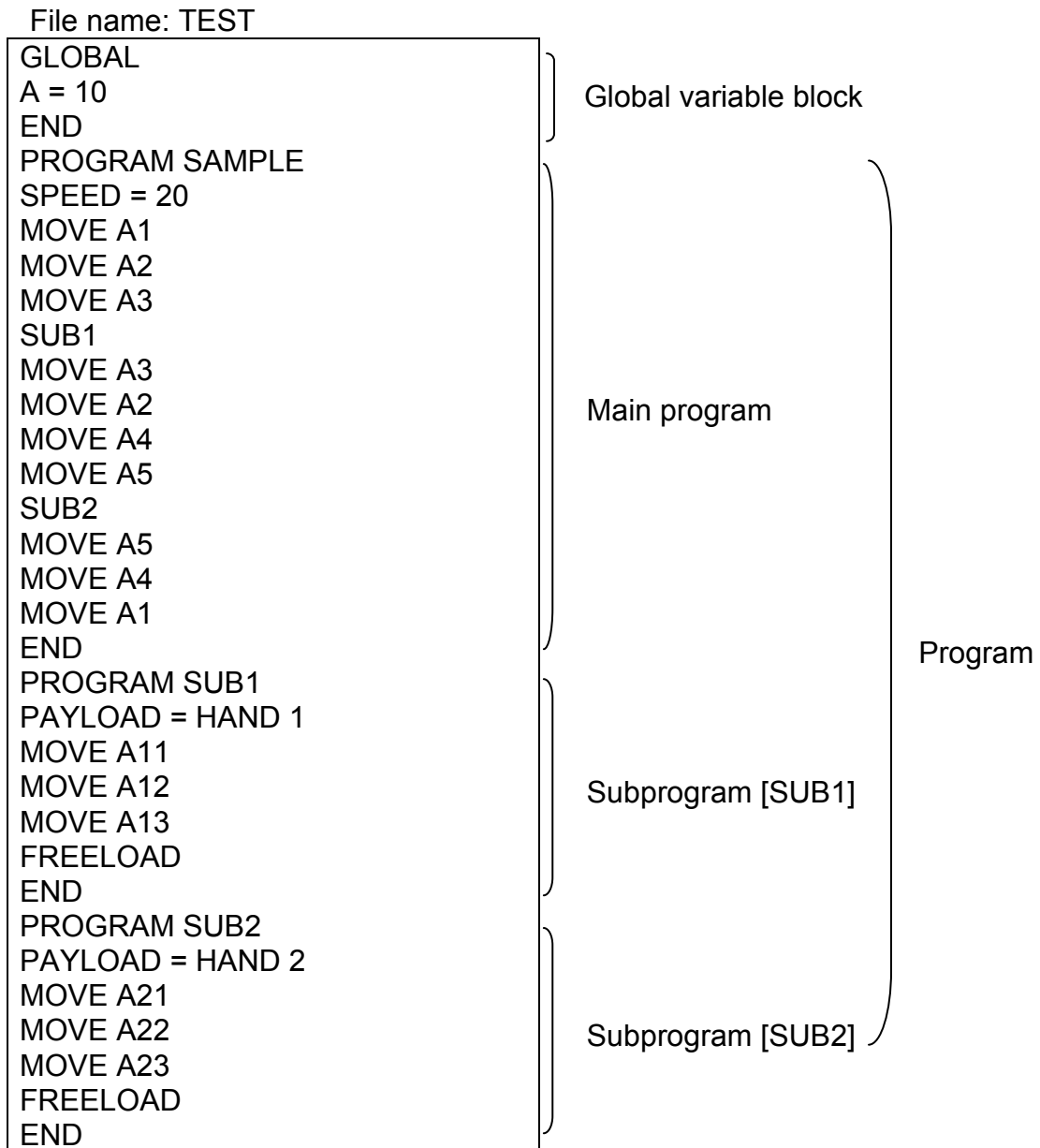
WORK <Coordinate name>

<Coordinate name>: Name of coordinate data used as work coordinate
system

(3) Example of user file

Examples of the user file is as follows:

A block end code (0x0A) is attached to the end of each line (or block).



DATA		
POINT A1=400,0,200,0,0/RIGHTY	Position in work coordinate system = world coordinate system	Data block
POINT A2=400,200,200,0,0/RIGHTY		
POINT A3=400,200,50,0,0/RIGHTY		
POINT A4=400,-200,200,0,0/LEFTY		
POINT A5=400,-200,50,0,0/LEFTY		
TRANS W1=400,200,50,90	Coordinate data W1	
TRANS W2=400,-200,50,0	Coordinate data W2	
PAYLOAD HAND1=3,100	Load data HAND1	
PAYLOAD HAND2=5,0	Load data HAND2	
WORK W1		
POINT A11=50,0,60,-45,0/RIGHTY	Position in work coordinate system = W1	
WORK W1		
POINT A12=0,50,70,0,0/RIGHTY		
WORK W1		
POINT A13=-50,0,60,45,0/RIGHTY		
WORK W2	Position in work coordinate system = W2	
POINT A21=-80,0,60,30,0/LEFTY		
WORK W2		
POINT A22=0,30,70,0,0/LEFTY		
WORK W2		
POINT A23=-80,0,60,-30,0/LEFTY		
END		

5.5 TSPC Settings

TSPC corresponding to the ethernet interface is Version 1.06 and subsequent versions.
Select Communication Settings from among the options.

Select TCP/IP Connection.

To connect to the controller through TCP/IP, enter IP address and port number ([E06]
Own port no {IP0}) specified for the controller.

For the operation, see TSCP Manual.

Set communication

Communication
☐ RS232C ☒ TCP/IP

Common
 Interval time: 1000

Setting RS232C
 Port: COM1
 Port Name:
 Baudrate: 9600
 Data length: 8BITS
 Parity bit: NONE
 Stop bit: 1BIT
 checksum: NONE

Setting TCP/IP
 IP address: 192 168 0 124
 PORT: 1000
 STX: NONE

Upload/Download directory

Ref ... Delete history

OK Cancel

Section 6

Robot Operation Sequence Using External Communication

Operating procedures for the robot in the External Control Mode are basically the same as those in the Internal Control Mode.

6.1 Basic Operation

Figure 6.1 presents a flow sequence in which a certain block is selected and executed repeatedly.

6.2 Program Download

Figure 6.2 presents a flow sequence in which a program is downloaded at the completion of each cycle and it is executed.

6.3 Reinitializing and Starting a Stopped Program

Figure 6.3 presents a flow sequence which reinitializes and starts a program which has been stopped.

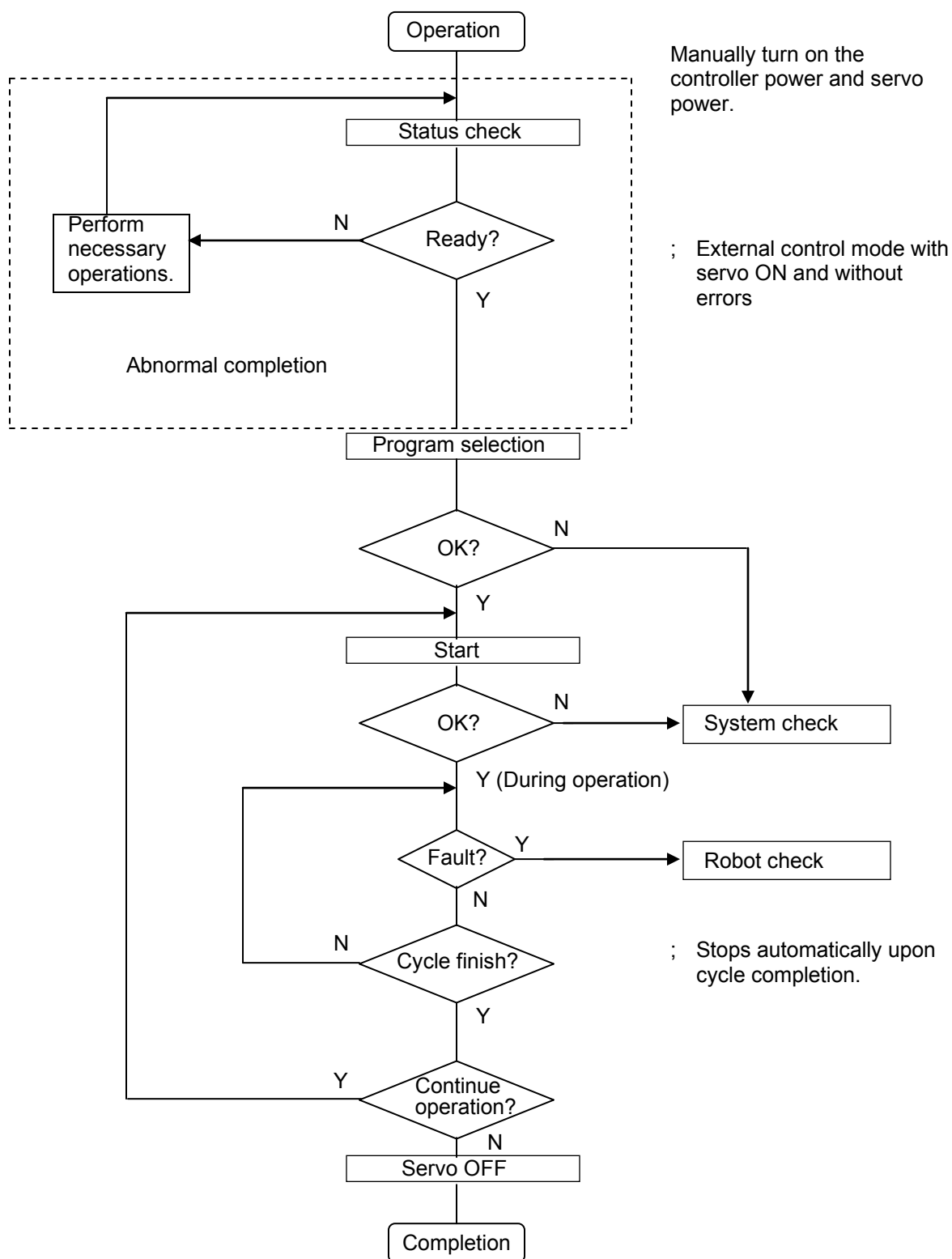


Fig. 6.1 Basic operation sequence

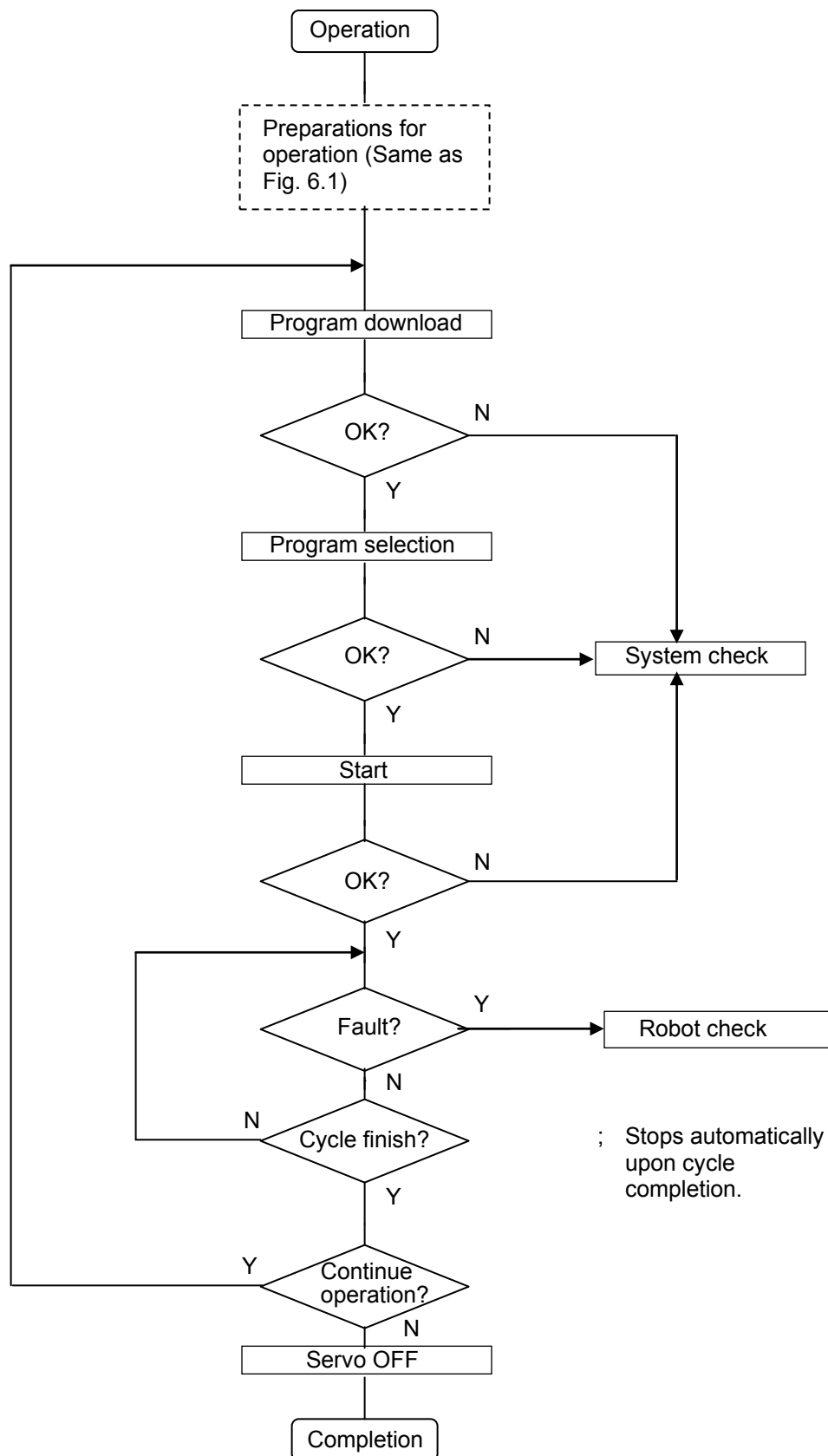


Fig. 6.2 Operation sequence including program download

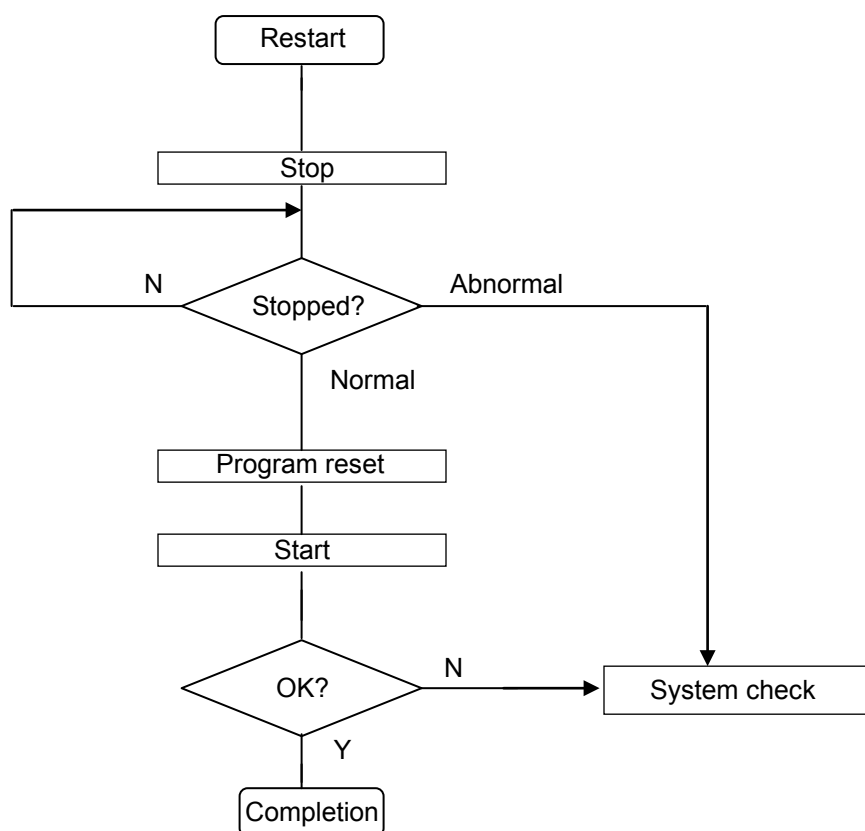


Fig. 6.3 Restart after program interruption

Section 7

Appendix

7.1 ASCII Code

		High-order 4 bits →								Hexadecimal number ←							
Low-order 4 bits ↓		0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
	0		D E		0	@	P	,	p								
	1	S H	D 1	!	1	A	Q	a	q								
	2	S X	D 2	“	2	B	R	b	r								
	3	E X	D 3	#	3	C	S	c	s								
	4	E T	D 4	\$	4	D	T	d	t								
	5	E Q	N K	%	5	E	U	e	u								
	6	A K	S N	&	6	F	V	f	v								
	7	B L	E B	'	7	G	W	g	w								
	8	B S	C N	(	8	H	X	h	x								
	9	H T	E M	)	9	I	Y	i	y								
	A	L F	S B	*	:	J	Z	j	z								
	B	H M	E C	+	;	K	[	k	(								
	C	C L	→	,	<	L	¥	l									
	D	C R	←	-	=	M	]	m	)								
	E	S O	↑	.	>	N	^	n	~								
	F	S I	↓	/	?	O	-	o									

↑ *
Hexadecimal number

- * Codes 00 to 1F correspond to control characters. These characters will either be interpreted as a space or as a code having a specific meaning. The meaning of these codes is shown in the above table.

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